

Improved Statistical Inference for Expectile-Based Risk Measures

An Optimal Pooling Approach in the Heavy-Tailed Regime

Filippo Gombac

[**TODO:** add supervisor / co-supervisor]

[**TODO:** add institution / department]

June 3, 2026

Abstract

[**TODO:** Write a ~ 200 -word abstract once [Chapter 3](#) is in shape. Outline of the abstract: (i) expectiles as the only coherent and elicitable law-invariant risk measure; (ii) the problem of combining tail-index and extreme-quantile information from several samples (distributed inference, multi-study evidence synthesis); (iii) our pooled extreme-expectile estimator, its joint CLT via plug-in into the expectile–quantile asymptotic equivalence, and the variance-/AMSE-optimal weights tailored to the expectile target; (iv) Weissman-style extrapolation to very-extreme expectile levels; (v) outlook on relaxing the iid assumption.]

Contents

1	Introduction	1
1.1	Motivation	1
1.2	Extreme expectile inference	2
1.3	Pooled inference across samples	2
1.4	Research question and scope	3
1.5	Contributions	3
1.6	Outline of the thesis	4
2	Background	5
2.1	Regular variation and the second-order condition	5
2.2	Tail-index and extreme-quantile estimation	6
2.2.1	The Hill estimator	6
2.2.2	Weissman extrapolation to extreme quantiles	7
2.3	Expectiles	7
2.3.1	Definition	8
2.3.2	Basic properties	8
2.3.3	Coherence and elicibility	9
2.4	Extreme expectile inference: the iid heavy-tailed baseline	9
2.4.1	The expectile–quantile asymptotic equivalence	9
2.4.2	Two families of estimators at intermediate levels	10
2.4.3	Weissman extrapolation to very-extreme expectile levels	11
2.4.4	The plug-in central limit theorem	12
2.5	The optimal pooling framework	13
2.5.1	Setup	13
2.5.2	The pooled Hill estimator	14
2.5.3	Variance- and AMSE-optimal weights	15
2.5.4	Weighted geometric pooling of extreme quantiles	16
2.5.5	Tail-homogeneity testing and confidence intervals	17
2.5.6	Distributed inference as a special case	17
3	Pooled intermediate expectiles	19
3.1	Setup and notation	19
3.2	The pooled QB expectile estimator	20

3.3	Joint asymptotic distribution of the pooled inputs	22
3.4	Asymptotic normality of the pooled expectile estimator	23
3.5	Variance- and AMSE-optimal weights	26
3.6	Estimated weights and confidence intervals	29
3.7	A root- k diagnostic for high-expectile homogeneity	32
4	Extrapolation to very-extreme expectile levels	35
4.1	Motivation	35
4.2	Extrapolation regime and input limit	36
4.3	The pooled extrapolated expectile estimator	38
4.4	Main extrapolation theorem	40
4.5	First-order optimal Hill weights for extrapolation	40
4.6	Bias-reduction	42
4.7	Asymptotic confidence intervals	43
5	Finite-sample performance	46
5.1	Purpose and scope	46
5.2	Simulation design	46
5.3	Estimators and metrics	47
5.4	Reproducibility	48
5.5	Generated summaries	48
6	Discussion and future work	52
6.1	Summary of contributions	52
6.2	Connections to Expected Shortfall	52
6.3	Limitations	52
6.4	Future work	53
A	Proofs	54
A.1	Proof of Proposition 2.14	54
A.2	Proof of Proposition 3.1	55
A.3	Proof of Proposition 3.2	58
A.4	Proof of Theorem 3.3	62
A.5	Proof of Propositions 3.4 and 3.5	65
A.6	Proof of Proposition 3.6	67
A.7	Proof of Corollary 3.7	69
A.8	Proof of Theorem 3.9 and Corollary 3.10	71
A.9	Proof of Lemma 4.1	75
A.10	Proof of Proposition 4.2	76
A.11	Proof of Theorem 4.4	77
A.12	Proof of Proposition 4.5	78
A.13	Proof of Corollary 4.6	78
A.14	Proof of Corollary 4.7	79

<i>CONTENTS</i>	iv
B Additional finite-sample diagnostics	80
Bibliography	83

Chapter 1

Introduction

1.1 Motivation

Quantitative risk management routinely needs to summarise the tail of a loss distribution by a single number. The industry-standard summary has long been the Value-at-Risk (VaR), defined as a high quantile of the loss distribution. VaR is intuitive and easy to compute, but it suffers from two well-known shortcomings that have shaped the search for alternatives. First, VaR is not subadditive: it can deem a diversified portfolio more risky than its components, contradicting the diversification principle and disqualifying VaR from the class of *coherent* risk measures in the sense of Artzner et al. (1999). Expected Shortfall (ES) corrects this defect and has progressively replaced VaR in regulatory practice. However, ES is not *elicitable* (Gneiting 2011): no scoring function has its expectation uniquely minimised at the true ES, so ES forecasts cannot be ranked or backtested directly.

Expectiles are the only law-invariant risk measure that is simultaneously coherent (for confidence levels $\tau \geq 1/2$) and elicitable (Bellini et al. 2014). Originally introduced by Newey and Powell (1987), the τ -expectile of a random loss X is defined as the minimiser of an asymmetric squared loss:

$$\xi_\tau(X) = \arg \min_{\theta \in \mathbb{R}} \mathbb{E} \left[|\tau - \mathbb{1}\{X \leq \theta\}| (X - \theta)^2 \right].$$

Setting $\tau = 1/2$ recovers the mean; setting $\tau \rightarrow 1$ moves the expectile into the right tail. Because the asymmetric loss is squared rather than absolute, expectiles — unlike quantiles — are sensitive to the *magnitude* of tail losses, not just to their frequency. Ehm et al. (2016) place expectiles within a broader theory of consistent scoring functions and Choquet representations.

This thesis addresses inference for expectiles at levels τ close to one, in two situations that are increasingly relevant in practice:

- (i) the loss distribution is heavy-tailed, in the sense of having an extreme-value index $\gamma > 0$, so that the empirical asymmetric-least-squares estimator becomes unstable as τ approaches one and extreme-value theory becomes the natural framework;

- (ii) the data are spread across several independent samples — whether because of distributed computation, regulatory data-sharing restrictions, or natural heterogeneity between studies — and inference must be conducted by communicating only summary statistics between sources.

1.2 Extreme expectile inference

The central bridge between expectile estimation and classical extreme-value theory is the *expectile-quantile asymptotic equivalence*. Under heavy-tailed assumptions with extreme-value index $\gamma \in (0, 1)$ and finite first moment,

$$\frac{\xi_\tau(X)}{Q_X(\tau)} \xrightarrow{\tau \uparrow 1} \psi(\gamma) := (\gamma^{-1} - 1)^{-\gamma}, \quad (1.1)$$

where $Q_X(\tau)$ is the τ -quantile of X . Two families of estimators arise naturally from (1.1). The *asymmetric-least-squares* (LAWS) estimator minimises the empirical asymmetric criterion over the data; the *quantile-based* (QB) estimator substitutes consistent estimators of γ and of an extreme quantile of X into the asymptotic-equivalence formula. Their asymptotic theory at intermediate levels and via Weissman-type extrapolation to very-extreme levels was developed for iid heavy-tailed data by Daouia, Girard, et al. (2018) and subsequent work, and extended to the dependent setting by Davison et al. (2023); to the multivariate setting by Padoan and Stupfler (2022). The relevant machinery is reviewed in Chapter 2.

1.3 Pooled inference across samples

When m samples are available, with heterogeneous sizes and possibly heterogeneous tails, the natural question is how best to combine them. Daouia, Padoan, et al. (2024) develop an optimal pooling framework for the tail index γ and for extreme quantiles. Their estimators take the form of weighted arithmetic combinations (for the tail index) and weighted *geometric* combinations (for extreme quantiles), the latter specifically adapted to the multiplicative-power structure of the Weissman extrapolation. They derive the joint asymptotic normality of the per-sample estimators across m samples, allowing for asymptotic dependence between samples driven by a tail-copula, and identify the variance-optimal and AMSE-optimal pooling weights in closed form. In the important special case of *distributed inference* — where data are iid both within and across machines — the cross-sample tail-copula vanishes, off-diagonal entries of the joint asymptotic covariance collapse to zero, and the pooled estimators recover the asymptotic efficiency of the unfeasible Hill / Weissman estimator built from the combined sample.

The framework of Daouia, Padoan, et al. (2024) is the methodological foundation of this thesis. The framework does not address expectiles; the question of how to combine m pairs $(\hat{\gamma}_j, \hat{Q}_j(p_j))$ into a coherent inference procedure for an extreme expectile ξ_{τ_n} has not been studied.

1.4 Research question and scope

The thesis answers the following question:

Given m independent samples of iid heavy-tailed observations, with each sample j contributing point estimates $(\hat{\gamma}_j, \hat{Q}_j(p_j))$ together with their joint asymptotic covariance, how should one combine these summary statistics to perform inference and Weissman-style extrapolation for an extreme expectile of the common underlying distribution?

The scope is restricted as follows:

- the data within each sample are univariate iid (no time-series dependence), matching the independent common-distribution distributed setting of Daouia, Padoan, et al. (2024);
- the loss distribution is heavy-tailed with extreme-value index $\gamma \in (0, 1)$ and finite first moment, the natural range for the quantile-based expectile–quantile bridge used by the thesis; the stronger finite-variance range $\gamma \in (0, 1/2)$ is invoked only when direct LAWS expectile estimators are discussed for comparison;
- the input to the pooling procedure is summary statistics, not raw data, in line with practical distributed-inference constraints.

1.5 Contributions

1. We define a *pooled extreme-expectile estimator* at intermediate levels by feeding the pooled tail-index estimator $\hat{\gamma}_n(\omega_\gamma)$ and the pooled weighted-geometric Weissman extreme-quantile estimator $\hat{q}_n^*(\cdot; \omega_q)$ of Daouia, Padoan, et al. (2024) into the asymptotic-equivalence formula (1.1).
2. We derive the joint asymptotic distribution of the pooled inputs, accounting for the cross-source structure (which is diagonal because the samples are independent) and for the within-source dependence between $\hat{\gamma}_j$ and $\hat{Q}_j$ induced by the use of overlapping top order statistics.
3. Applying the delta method through ψ to this joint pooled CLT, we obtain the asymptotic normality of the pooled intermediate-level expectile estimator together with explicit forms of its asymptotic bias and variance.
4. We characterise the *variance-optimal* and *AMSE-optimal* pooling weights tailored to the pooled expectile estimator. These need not coincide with the weights optimal for the tail index or for the extreme quantile alone, owing to the non-trivial dependence between the two pieces of information.
5. We extend the construction to very-extreme levels $\tau'_n \uparrow 1$ with $n(1 - \tau'_n) \rightarrow c < \infty$ via a Weissman-style extrapolation of the pooled intermediate-level estimator, and derive the corresponding central limit theorem by feeding the joint intermediate-level limit of the

previous contribution into the plug-in CLT of Davison et al. (2023, Proposition C.6). As in the single-sample case, the extrapolated estimator inherits the asymptotic variance of the pooled tail-index estimator.

6. We construct asymptotic confidence intervals for extreme expectiles and a target-level diagnostic for *expectile-tail-homogeneity* across the m samples, modelled on the tail-quantile-homogeneity test of Daouia, Padoan, et al. (2024) and on the GLS-weighted equality-of-means test of Padoan and Stupfler (2022, Section 3.3). This diagnostic is not a test of equality of the full marginal distributions.

1.6 Outline of the thesis

Chapter 2 collects the technical background: regular variation and the second-order condition, the Hill and Weissman estimators, expectiles and their characterisation, single-sample iid extreme-expectile inference, and the optimal pooling framework of Daouia, Padoan, et al. (2024). Chapter 3 develops the pooled estimator at intermediate expectile levels and the associated asymptotic theory. Chapter 4 treats the Weissman-style extrapolation to very-extreme expectile levels. Chapter 6 discusses limitations, the link to Expected Shortfall, and avenues for future work, including extensions to time-series dependence and multivariate observations. Detailed proofs are gathered in Appendix A.

Chapter 2

Background

2.1 Regular variation and the second-order condition

Throughout the thesis we work with a real-valued random loss X that has continuous distribution function F and survival function $\bar{F} = 1 - F$. Given an iid sample $X_1, \dots, X_n$ from F , the corresponding order statistics are denoted by $X_{1:n} \leq X_{2:n} \leq \dots \leq X_{n:n}$.

Heavy-tailed asymptotics are governed by the language of regular variation; the reference is Haan and Ferreira (2006, Appendix B.1).

Definition 2.1 (Regular variation). A Lebesgue measurable function $f : (0, \infty) \rightarrow \mathbb{R}$ that is eventually positive is *regularly varying at infinity with index* $\alpha \in \mathbb{R}$, written $f \in \text{RV}_\alpha$, if for every $y > 0$,

$$\lim_{t \rightarrow \infty} \frac{f(ty)}{f(t)} = y^\alpha.$$

The functions in RV_0 are called *slowly varying*.

The first-order condition under which all of our asymptotics are taken asserts that F has a Pareto-type tail.

Condition 2.2 (First-order condition). There exists $\gamma > 0$ such that $\bar{F} \in \text{RV}_{-\gamma}$, or equivalently the *tail quantile function*

$$U(t) := \inf \left\{ x \in \mathbb{R} : \frac{1}{\bar{F}(x)} \geq t \right\}, \quad t > 1,$$

satisfies $U \in \text{RV}_\gamma$.

Throughout, for a continuous distribution function F , we write

$$Q_X(\tau) := F^{\leftarrow}(\tau) = \inf \{ x \in \mathbb{R} : F(x) \geq \tau \}, \quad 0 < \tau < 1.$$

With the tail quantile $U(t) = F^{\leftarrow}(1 - 1/t)$, $t > 1$, this gives

$$Q_X(\tau) = U((1 - \tau)^{-1}), \quad Q_X(1 - p) = U(1/p).$$

Condition 2.2 is the Fréchet max-domain condition with extreme-value index γ (Haan and Ferreira 2006, Theorem 1.2.1); standard examples include Pareto, Burr, Student- t , Fréchet and F -distribution tails (Haan and Ferreira 2006, §2.6). For normal limits we use the standard second-order refinement.

Condition 2.3 (Second-order condition $\mathcal{C}_2(\gamma, \rho, A)$). There exist $\gamma > 0$, a second-order parameter $\rho \leq 0$, and a measurable auxiliary function A of constant sign with $A(t) \rightarrow 0$ as $t \rightarrow \infty$, such that for every $y > 0$,

$$\lim_{t \rightarrow \infty} \frac{1}{A(t)} \left[\frac{U(ty)}{U(t)} - y^\gamma \right] = y^\gamma \frac{y^\rho - 1}{\rho},$$

where the right-hand side is read as $y^\gamma \log y$ when $\rho = 0$.

The function $|A|$ is itself regularly varying with index ρ (Haan and Ferreira 2006, Theorem 2.3.9). Since A has constant sign eventually, signed ratios $A(ty)/A(t)$ converge to y^ρ ; the rate A is the source of the second-order bias terms below.

Condition 2.4 (Finite first-moment condition). The extreme-value index satisfies $\gamma \in (0, 1)$ and the negative part $X^- = \max(-X, 0)$ satisfies

$$\mathbb{E} X^- < \infty.$$

Under **Condition 2.2**, $\gamma < 1$ gives $\mathbb{E} X^+ < \infty$; together with the displayed lower-tail condition this is the finite-mean regime for the quantile-based expectile approximation.

Condition 2.5 (LAWS lower-tail moment condition). For some $\delta > 0$, the negative part $X^- = \max(-X, 0)$ satisfies

$$\mathbb{E}(X^-)^{2+\delta} < \infty.$$

This stronger finite-variance-type condition is used only for direct LAWS expectile estimation. It is not part of the quantile-based plug-in route used in the contribution chapters.

2.2 Tail-index and extreme-quantile estimation

We collect the two classical estimators used throughout the thesis: the Hill estimator of γ and the Weissman extrapolation for extreme quantiles. The textbook reference is Haan and Ferreira (2006, Ch. 3–4).

2.2.1 The Hill estimator

Given an iid sample $X_1, \dots, X_n$ from F and an integer $k = k_n \in \{1, \dots, n-1\}$, the *Hill estimator* (Hill 1975) of γ based on the $k+1$ top order statistics is

$$\hat{\gamma}_n(k) = \frac{1}{k} \sum_{i=1}^k \log \left(\frac{X_{n-i+1:n}}{X_{n-k:n}} \right). \quad (2.1)$$

Since $U(t) \rightarrow \infty$, $X_{n-k:n} > 0$ with probability tending to one for intermediate k ; log-based estimators are understood on this event. For tail-index estimation one may also shift the data, but shifts must be undone for quantile or expectile estimation. The consistency result below is Haan and Ferreira (2006, Theorem 3.2.2); the normal limit is Haan and Ferreira (2006, Theorem 3.2.5) when $\rho < 0$, with the $\rho = 0$ endpoint covered by the $m = 1$ specialisation of Daouia, Padoan, et al. (2024, Theorem 1).

Theorem 2.6. *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma > 0$. Let $k = k_n \rightarrow \infty$ and $k/n \rightarrow 0$ as $n \rightarrow \infty$. Then $\hat{\gamma}_n(k) \xrightarrow{\mathbb{P}} \gamma$. If, in addition, $\sqrt{k} A(n/k) \rightarrow \lambda \in \mathbb{R}$, then*

$$\sqrt{k} (\hat{\gamma}_n(k) - \gamma) \xrightarrow{d} \mathcal{N}\left(\frac{\lambda}{1-\rho}, \gamma^2\right).$$

The choice of k controls the usual bias–variance trade-off; here it is treated as fixed by the data analyst.

2.2.2 Weissman extrapolation to extreme quantiles

The *Weissman extrapolation* (Weissman 1978) extrapolates from the intermediate-order statistic $X_{n-k:n}$ — which estimates the quantile $Q_X(1 - k/n)$ at the intermediate level $1 - k/n$ — to a quantile at a very-extreme level $1 - p$ with $p = p_n \rightarrow 0$ as $n \rightarrow \infty$. Plugging the Hill estimator into the first-order identity $U(ty)/U(t) \rightarrow y^\gamma$ gives the Weissman estimator

$$\hat{q}_n^*(1 - p \mid k) = \left(\frac{k}{np}\right)^{\hat{\gamma}_n(k)} X_{n-k:n}. \quad (2.2)$$

The theorem below, the Hill-estimator specialisation of Haan and Ferreira (2006, Theorem 4.3.8), covers the far-tail regime $np = o(k)$; the case $np = O(1)$ is very-extreme in the sense that only finitely many observations are expected beyond the target.

Theorem 2.7. *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma > 0$ and $\rho < 0$. Let $k = k_n \rightarrow \infty$, $k/n \rightarrow 0$, $p = p_n \rightarrow 0$ with $k/(np) \rightarrow \infty$ and $\sqrt{k}/\log(k/(np)) \rightarrow \infty$. Suppose $\sqrt{k} A(n/k) \rightarrow \lambda \in \mathbb{R}$. Then*

$$\frac{\sqrt{k}}{\log(k/(np))} \left(\frac{\hat{q}_n^*(1 - p \mid k)}{Q_X(1 - p)} - 1 \right) \xrightarrow{d} \mathcal{N}\left(\frac{\lambda}{1-\rho}, \gamma^2\right).$$

The log-denominator is the cost of extrapolation. Up to that factor the limit is the Hill limit, which is why Daouia, Padoan, et al. (2024, §2.3) pool Weissman estimators geometrically and why Chapter 4 again reduces to pooled tail-index inference at first order.

2.3 Expectiles

Expectiles were introduced by Newey and Powell (1987) as asymmetric-least-squares analogues of quantiles. Their role as coherent and elicitable risk measures is developed in Bellini et al. (2014), Ziegel (2016), Gneiting (2011), and Ehm et al. (2016).

2.3.1 Definition

Let X be a real-valued random variable with $\mathbb{E}|X| < \infty$, and fix a level $\tau \in (0, 1)$. The τ -*expectile* of X is defined as

$$\xi_\tau(X) := \arg \min_{\theta \in \mathbb{R}} \mathbb{E}[\eta_\tau(X - \theta) - \eta_\tau(X)], \quad \eta_\tau(u) := |\tau - \mathbb{1}\{u \leq 0\}| u^2. \quad (2.3)$$

The subtraction of $\eta_\tau(X)$ makes the criterion finite under $\mathbb{E}|X| < \infty$ and does not affect the minimiser; at $\tau = 1/2$ one recovers the mean.

The convex objective (2.3) has a unique minimiser, and its first-order condition reduces to the implicit equation

$$\tau \mathbb{E}[(X - \xi_\tau(X))_+] = (1 - \tau) \mathbb{E}[(\xi_\tau(X) - X)_+], \quad (2.4)$$

where $u_+ = \max(u, 0)$. Thus expectiles balance upper and lower first partial moments, unlike quantiles, which balance tail probabilities.

2.3.2 Basic properties

The following properties are immediate from (2.3) or (2.4); we collect them without proof and refer to Newey and Powell (1987, §2) and Bellini et al. (2014, Sections 2–3) for details.

Proposition 2.8. *Let X and Y be real-valued random variables with $\mathbb{E}|X|, \mathbb{E}|Y| < \infty$. For each $\tau \in (0, 1)$:*

- (i) **Existence and uniqueness.** $\xi_\tau(X)$ exists and is uniquely defined.
- (ii) **Law invariance.** If X and Y have the same distribution, then $\xi_\tau(X) = \xi_\tau(Y)$.
- (iii) **Mean recovery.** $\xi_{1/2}(X) = \mathbb{E}X$.
- (iv) **Location equivariance.** $\xi_\tau(X + c) = \xi_\tau(X) + c$ for $c \in \mathbb{R}$.
- (v) **Positive homogeneity.** $\xi_\tau(\lambda X) = \lambda \xi_\tau(X)$ for $\lambda \geq 0$.
- (vi) **Sign reflection.** $\xi_\tau(-X) = -\xi_{1-\tau}(X)$.
- (vii) **Monotonicity in τ .** The map $\tau \mapsto \xi_\tau(X)$ is nondecreasing on $(0, 1)$, and is strictly increasing when X is nondegenerate, with the boundary limits $\xi_\tau(X) \rightarrow \text{ess inf}(X)$ as $\tau \rightarrow 0$ and $\xi_\tau(X) \rightarrow \text{ess sup}(X)$ as $\tau \rightarrow 1$ (infinite limits allowed).
- (viii) **Monotonicity in the argument.** If $X \leq Y$ almost surely, then $\xi_\tau(X) \leq \xi_\tau(Y)$.

For losses, expectiles are translation equivariant, positively homogeneous and monotone. The remaining coherence issue is subadditivity.

2.3.3 Coherence and elicibility

In the rest of the thesis X denotes a loss. The coherence literature is often written for financial positions, so in this subsection only write Y for a position and define the capital requirement

$$\rho_\tau(Y) = \xi_\tau(-Y).$$

Theorem 2.9 (Bellini et al. (2014)). *The functional $\rho_\tau(Y) = \xi_\tau(-Y)$ is a coherent risk measure if and only if $\tau \geq 1/2$.*

Recall that a functional is *elicitable* if it is the unique minimiser of an expected strictly consistent score (Gneiting 2011). Expectiles are elicitable on classes with finite first moment; the asymmetric squared score is strictly consistent under finite second moments, and the general scoring class has Bregman-type and Choquet-type representations (Gneiting 2011; Ehm et al. 2016). This scoring structure motivates forecast comparison, but the asymptotic theory below uses only (2.4) and the extreme expectile–quantile equivalence.

2.4 Extreme expectile inference: the iid heavy-tailed baseline

We now consider high expectile levels for a heavy-tailed loss with $\gamma \in (0, 1)$. Two regimes are used throughout:

- *intermediate-level*: $\tau_n \rightarrow 1$ with $n(1 - \tau_n) \rightarrow \infty$. There are still many observations above $Q_X(\tau_n)$, and central limit theorems hold at the rate $\sqrt{n(1 - \tau_n)}$.
- *extreme-level*: $\tau'_n \rightarrow 1$ with $n(1 - \tau'_n) \rightarrow c \in [0, \infty)$. There are at most finitely many observations above $Q_X(\tau'_n)$, and direct estimation is infeasible; extrapolation is required.

The iid results are due to Daouia, Girard, et al. (2018), Daouia, Girard, et al. (2019), and Daouia, Girard, et al. (2020); the plug-in formulation used later is adapted from the time-series extension of Davison et al. (2023).

2.4.1 The expectile–quantile asymptotic equivalence

The single most important fact for extreme expectile inference is that, in the heavy-tailed regime, the expectile is asymptotically proportional to the quantile of the same level.

Proposition 2.10 (Asymptotic equivalence; Bellini et al. (2014), Daouia, Girard, et al. (2019), Corollary 1 with $p = 2$). *Suppose X satisfies Condition 2.2 with index $\gamma \in (0, 1)$ and $\mathbb{E} X^- < \infty$; equivalently, suppose Condition 2.4 holds. Then*

$$\frac{\xi_\tau(X)}{Q_X(\tau)} \xrightarrow{\tau \uparrow 1} \psi(\gamma) := (\gamma^{-1} - 1)^{-\gamma}. \quad (2.5)$$

The bridge requires only the finite-mean range $\gamma < 1$ and lower-tail integrability; the stronger $\gamma < 1/2$ restriction enters direct LAWS CLTs, not the quantile-based route. The function ψ is

continuous on $(0, 1)$, with $\psi(\gamma) \rightarrow 1$ as $\gamma \downarrow 0$ and $\psi(\gamma) \rightarrow \infty$ as $\gamma \uparrow 1$. It is *not* monotone: its logarithmic derivative

$$m(\gamma) := \frac{d}{d\gamma} \log \psi(\gamma) = \frac{1}{1-\gamma} - \log(\gamma^{-1} - 1) \quad (2.6)$$

changes sign at $\gamma^* \approx 0.2178$, where ψ reaches its minimum. This non-monotonicity is relevant later when $m(\gamma) = 0$ makes some weight formulas degenerate.

Under the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0, 1)$ and $\mathbb{E} X^- < \infty$, the first-order equivalence (2.5) sharpens to the second-order expansion (Daouia, Girard, et al. 2020, Proposition 1(i))

$$\begin{aligned} \frac{\xi_\tau(X)}{Q_X(\tau)} &= \psi(\gamma) \left[1 + c(\gamma, \rho) A((1-\tau)^{-1}) + \gamma(\gamma^{-1} - 1)^\gamma \frac{\mathbb{E} X}{Q_X(\tau)} \right. \\ &\quad \left. + o(A((1-\tau)^{-1})) + o(Q_X(\tau)^{-1}) \right], \\ c(\gamma, \rho) &:= \frac{(\gamma^{-1} - 1)^{-\rho}}{1 - \gamma - \rho} + \frac{(\gamma^{-1} - 1)^{-\rho} - 1}{\rho}. \end{aligned} \quad (2.7)$$

At $\rho = 0$ the second term in $c(\gamma, \rho)$ is read by continuity, so $c(\gamma, 0) = m(\gamma)$. The A -term is the second-order expectile-quantile bias used later; the displayed first-moment term is $O(Q_X(\tau)^{-1})$ and is negligible under the moment-rate condition imposed in the QB CLT below.

2.4.2 Two families of estimators at intermediate levels

Two natural estimators of ξ_{τ_n} at an intermediate level τ_n arise from the asymmetric-LS definition (2.3) on the one hand, and from the asymptotic equivalence (2.5) on the other.

LAWS direct estimator. The *asymmetric-least-squares* (LAWS) estimator replaces the population minimisation (2.3) with its empirical analogue:

$$\tilde{\xi}_{\tau_n} = \arg \min_{\theta \in \mathbb{R}} \sum_{i=1}^n \eta_{\tau_n}(X_i - \theta). \quad (2.8)$$

Theorem 2.11 (Intermediate LAWS CLT; Daouia, Girard, et al. (2018), Theorem 2). *Assume Condition 2.2 with $\gamma \in (0, 1/2)$, and assume Condition 2.5. Let $\tau_n \rightarrow 1$ with $n(1 - \tau_n) \rightarrow \infty$. Then*

$$\sqrt{n(1 - \tau_n)} \begin{pmatrix} \tilde{\xi}_{\tau_n} \\ \xi_{\tau_n} \end{pmatrix} - 1 \xrightarrow{d} \mathcal{N}\left(0, \frac{2\gamma^3}{1 - 2\gamma}\right).$$

Quantile-based plug-in estimator. A second family substitutes estimators of γ and of the intermediate quantile $Q_X(\tau_n)$ into the asymptotic-equivalence formula (2.5):

$$\hat{\xi}_{\tau_n} = \psi(\hat{\gamma}_n) \hat{Q}(\tau_n), \quad (2.9)$$

where $\hat{Q}(\tau_n) = X_{n - \lfloor n(1 - \tau_n) \rfloor : n}$ is the empirical quantile (an intermediate-order statistic) and $\hat{\gamma}_n$ is a tail-index estimator admitting the required $\sqrt{n(1 - \tau_n)}$ limit theory. The natural choice is the Hill estimator of (2.1) with effective sample size $k = \lfloor n(1 - \tau_n) \rfloor$.

Theorem 2.12 (Intermediate QB CLT; Daouia, Girard, et al. (2018), Corollary 2; Daouia, Girard, et al. (2020), Proposition 1). *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0, 1)$ and assume $\mathbb{E} X^- < \infty$. Let $\tau_n \rightarrow 1$ with $n(1 - \tau_n) \rightarrow \infty$ and suppose*

$$\sqrt{n(1 - \tau_n)} A((1 - \tau_n)^{-1}) \rightarrow \lambda \in \mathbb{R}, \quad \frac{\sqrt{n(1 - \tau_n)}}{Q_X(\tau_n)} \rightarrow 0.$$

With $\hat{\gamma}_n$ the Hill estimator at effective sample size $k = \lfloor n(1 - \tau_n) \rfloor$,

$$\sqrt{n(1 - \tau_n)} \left(\frac{\hat{\xi}_{\tau_n}}{\xi_{\tau_n}} - 1 \right) \xrightarrow{d} \mathcal{N} \left(\left(\frac{m(\gamma)}{1 - \rho} - c(\gamma, \rho) \right) \lambda, \gamma^2 (1 + m(\gamma)^2) \right),$$

with $m(\gamma)$ as in (2.6) and $c(\gamma, \rho)$ the second-order expectile-quantile constant of (2.7). The asymptotic bias has two pieces: the Hill bias $\lambda/(1 - \rho)$ propagated through the logarithmic derivative $m(\gamma)$ of ψ , and the deterministic expectile-quantile gap $-c(\gamma, \rho)\lambda$. The empirical quantile $\hat{Q}(\tau_n)$ contributes no asymptotic mean term at the $\sqrt{n(1 - \tau_n)}$ scale when centered at $Q_X(\tau_n)$; the condition $\sqrt{n(1 - \tau_n)}/Q_X(\tau_n) \rightarrow 0$ inherited from the theorem ensures that the moment-term contribution to the expectile-quantile gap is negligible.

This is the zero-first-moment-bias specialisation of Daouia, Girard, et al. (2018, Corollary 2), combined with Daouia, Girard, et al. (2020, Proposition 1). The imposed $\sqrt{n(1 - \tau_n)}/Q_X(\tau_n) \rightarrow 0$ removes the additional first-moment bias term present in the more general statement and is the finite-mean QB regime used in the contribution chapters.

Comparison. On the direct-LAWS range $\gamma \in (0, 1/2)$ neither LAWS nor QB uniformly dominates: their asymptotic variances cross at $\gamma \approx 0.2623$, and the LAWS variance diverges as $\gamma \uparrow 1/2$. The thesis uses QB because it composes with the pooled Hill and Weissman quantile primitives of Daouia, Padoan, et al. (2024), not because it is uniformly more efficient.

2.4.3 Weissman extrapolation to very-extreme expectile levels

For levels τ'_n so close to one that $n(1 - \tau'_n) \rightarrow c \in [0, \infty)$, neither $\tilde{\xi}_{\tau'_n}$ nor $\hat{\xi}_{\tau'_n}$ provides a consistent direct estimator of $\xi_{\tau'_n}$. Extrapolation from an intermediate level τ_n to the extreme level τ'_n is required.

The asymptotic equivalence (2.5) and the first-order identity $U(ty)/U(t) \rightarrow y^\gamma$ together yield the heuristic

$$\frac{\xi_{\tau'_n}}{\xi_{\tau_n}} \sim \frac{Q_X(\tau'_n)}{Q_X(\tau_n)} \sim \left(\frac{1 - \tau'_n}{1 - \tau_n} \right)^{-\gamma},$$

which suggests the *Weissman-type* extrapolated estimator

$$\bar{\xi}_{\tau'_n}^* = \left(\frac{1 - \tau'_n}{1 - \tau_n} \right)^{-\hat{\gamma}_n} \bar{\xi}_{\tau_n}, \quad (2.10)$$

where $\bar{\xi}_{\tau_n}$ is any consistent intermediate-level estimator satisfying a $\sqrt{n(1 - \tau_n)}$ -CLT, such as $\tilde{\xi}_{\tau_n}$ or $\hat{\xi}_{\tau_n}$ under the conditions above, and $\hat{\gamma}_n$ is a $\sqrt{n(1 - \tau_n)}$ -consistent tail-index estimator. We write $\tilde{\xi}^*$ and $\hat{\xi}^*$ for the LAWS-anchored and QB-anchored extrapolations, respectively.

Theorem 2.13 (Extreme-level CLT; Daouia, Girard, et al. (2018), Corollaries 3–4). *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0, 1)$ and $\rho < 0$, and assume $\mathbb{E} X^- < \infty$. Let $\tau_n \rightarrow 1$ satisfy $n(1 - \tau_n) \rightarrow \infty$ and let $\tau'_n \rightarrow 1$ satisfy $n(1 - \tau'_n) \rightarrow c \in [0, \infty)$, $(1 - \tau'_n)/(1 - \tau_n) \rightarrow 0$ and $\sqrt{n(1 - \tau_n)} / \log((1 - \tau_n)/(1 - \tau'_n)) \rightarrow \infty$. Suppose moreover that*

$$\sqrt{n(1 - \tau_n)} A((1 - \tau_n)^{-1}) \rightarrow \lambda \in \mathbb{R}, \quad \frac{\sqrt{n(1 - \tau_n)}}{Q_X(\tau_n)} \rightarrow 0.$$

If $\hat{\gamma}_n$ is the Hill estimator and $\bar{\xi}_{\tau_n}$ is the QB intermediate estimator of Theorem 2.12, then

$$\frac{\sqrt{n(1 - \tau_n)}}{\log((1 - \tau_n)/(1 - \tau'_n))} \left(\frac{\bar{\xi}_{\tau'_n}^*}{\bar{\xi}_{\tau'_n}} - 1 \right) \xrightarrow{d} \mathcal{N}\left(\frac{\lambda}{1 - \rho}, \gamma^2\right).$$

The same conclusion holds with $\bar{\xi}_{\tau_n}$ equal to the LAWS estimator of Theorem 2.11 under the additional direct-LAWS assumptions $\gamma \in (0, 1/2)$ and Condition 2.5.

The log-rate is the classical Weissman rate. At this scale the anchor estimator is first-order invisible, so the limit variance is γ^2 for both LAWS and QB anchors.

2.4.4 The plug-in central limit theorem

For pooled inference we need a version that treats $(\hat{\gamma}_n, \bar{\xi}_{\tau_n})$ as generic inputs. The following thesis-level lemma is adapted from Davison et al. (2023, Supplement, Proposition C.6); it uses a generic rate a_n and makes the deterministic extrapolation error explicit.

Proposition 2.14 (Plug-in CLT for extrapolated expectiles; adapted from Davison et al. (2023), Prop. C.6). *Assume $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0, 1)$ and $\rho < 0$, and assume $\mathbb{E} X^- < \infty$. Let $a_n \rightarrow \infty$, let $\hat{\gamma}_n$ be an estimator of γ , and let $\hat{\xi}_{\tau_n}^I$ be an intermediate-level estimator of ξ_{τ_n} satisfying the joint asymptotic representation*

$$a_n \begin{pmatrix} \hat{\gamma}_n - \gamma \\ \hat{\xi}_{\tau_n}^I / \xi_{\tau_n} - 1 \end{pmatrix} \xrightarrow{d} \begin{pmatrix} \Gamma \\ \Delta \end{pmatrix}, \quad (2.11)$$

for some random vector $(\Gamma, \Delta)^\top$. Let $\tau_n, \tau'_n \rightarrow 1$ with $n(1 - \tau_n) \rightarrow \infty$, $n(1 - \tau'_n) \rightarrow c \in [0, \infty)$, $(1 - \tau'_n)/(1 - \tau_n) \rightarrow 0$, put

$$\ell_n := \log\left(\frac{1 - \tau_n}{1 - \tau'_n}\right),$$

and suppose

$$\frac{a_n}{\ell_n} \rightarrow \infty, \quad a_n A((1 - \tau_n)^{-1}) \rightarrow \lambda_1 \in \mathbb{R}, \quad \frac{a_n}{Q_X(\tau_n)} \rightarrow \lambda_2 \in \mathbb{R}.$$

Assume moreover that $\mathbb{P}(\hat{\xi}_{\tau_n}^I > 0) \rightarrow 1$ and that the deterministic extrapolation error satisfies

$$\frac{a_n}{\ell_n} \{\log \xi_{\tau'_n} - \log \xi_{\tau_n} - \gamma \ell_n\} \rightarrow 0.$$

Then the Weissman-extrapolated expectile

$$\hat{\xi}_{\tau'_n}^* = \left(\frac{1 - \tau'_n}{1 - \tau_n} \right)^{-\hat{\gamma}_n} \hat{\xi}_{\tau_n}^I$$

satisfies

$$\frac{a_n}{\log((1 - \tau_n)/(1 - \tau'_n))} \left(\frac{\hat{\xi}_{\tau'_n}^*}{\hat{\xi}_{\tau_n}^I} - 1 \right) \xrightarrow{d} \Gamma.$$

Proof. See [Appendix A.1](#). □

Remark 2.15 (Deterministic ratio control). The deterministic log-ratio condition in [Proposition 2.14](#) is the high-level extrapolation-bias control

$$\frac{a_n}{\ell_n} \{ \log \xi_{\tau'_n} - \log \xi_{\tau_n} - \gamma \ell_n \} \longrightarrow 0.$$

In the iid heavy-tailed setting this control is supplied by the very-extreme expectile extrapolation argument of Daouia, Girard, et al. (2020, Theorem 5), invoked in Davison et al. (2023, Proposition C.6). It uses the sequential Weissman expansion along $y_n = (1 - \tau_n)/(1 - \tau'_n) \rightarrow \infty$ together with (2.7).

[Proposition 2.14](#) is the technical workhorse for [Chapter 4](#): after division by the diverging log-rate, the extreme-level law is governed only by the tail-index component Γ , while the intermediate expectile component washes out.

2.5 The optimal pooling framework

We now recall the pooling framework of Daouia, Padoan, et al. (2024), which the contribution chapters extend from quantiles to expectiles. The framework pools tail-index and extreme-quantile estimators from m samples, allowing heterogeneous sizes, second-order properties and cross-sample tail dependence.

2.5.1 Setup

Throughout [Section 2.5](#), m is fixed while $n = \sum_{j=1}^m n_j \rightarrow \infty$. Let $\mathbf{X}_i = (X_{i,1}, \dots, X_{i,m})^\top$, $i \geq 1$, be iid copies of $\mathbf{X} = (X_1, \dots, X_m)^\top$. Margin j is observed for $i = 1, \dots, n_j$ and has continuous right-heavy-tailed distribution F_j , tail quantile function $U_j = (1/\bar{F}_j)^\leftarrow$ and extreme-value index $\gamma_j > 0$; its order statistics are denoted $X_{1:n_j,j} \leq \dots \leq X_{n_j:n_j,j}$.

Tail inference in sample j uses its top $k_j + 1$ order statistics, where the effective sample size $k_j = k_j(n)$ satisfies $k_j \rightarrow \infty$ and $k_j/n_j \rightarrow 0$ as $n \rightarrow \infty$. The total effective sample size is $k = \sum_{j=1}^m k_j$. Assume the asymptotic-proportionality conditions

$$\frac{n_1}{n_j} \rightarrow b_j \in (0, \infty), \quad \frac{k_1}{k_j} \rightarrow c_j \in (0, \infty), \quad (2.12)$$

with $b_1 = c_1 = 1$. The marginal F_j satisfies $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$ (Condition 2.3). The thesis works under the *common-tail-index* assumption

$$(H_\gamma) \quad \gamma_1 = \dots = \gamma_m = \gamma.$$

Under this assumption, all common- γ vector limits below are expressed on the aggregate scale $\sqrt{k}$, with the constants c_j translating the local $\sqrt{k_j}$ rates.

The cross-sample asymptotic dependence is captured by a pairwise *tail-copula* structure (Daouia, Padoan, et al. 2024, §2.1).

Condition $\mathcal{J}(\mathbf{R})$ (tail-copula condition). For each pair (j, ℓ) with $j \neq \ell$, there exists a function $R_{j,\ell} : [0, \infty]^2 \setminus \{(\infty, \infty)\} \rightarrow [0, \infty)$ such that, for every $(x_j, x_\ell) \in [0, \infty]^2 \setminus \{(\infty, \infty)\}$,

$$\lim_{s \rightarrow \infty} s \mathbb{P}(1 - F_j(X_j) \leq x_j/s, 1 - F_\ell(X_\ell) \leq x_\ell/s) = R_{j,\ell}(x_j, x_\ell).$$

The case $R_{j,\ell} \equiv 0$ is asymptotic tail independence, implied by the independent samples used later in the thesis.

2.5.2 The pooled Hill estimator

Each sample contributes a Hill estimator $\hat{\gamma}_j(k_j)$ as in (2.1); we collect these into the random vector

$$\hat{\gamma}_n = (\hat{\gamma}_1(k_1), \dots, \hat{\gamma}_m(k_m))^\top.$$

Given a weight vector $\omega \in \mathbb{R}^m$ with $\omega^\top \mathbf{1} = 1$, the *pooled Hill estimator* is the weighted arithmetic combination

$$\hat{\gamma}_n(\omega) := \omega^\top \hat{\gamma}_n = \sum_{j=1}^m \omega_j \hat{\gamma}_j(k_j). \quad (2.13)$$

The affine constraint $\omega^\top \mathbf{1} = 1$ ensures consistency for the common γ under (H_γ) .

The fundamental asymptotic result is the joint CLT for the marginal Hill estimators.

Theorem 2.16 (Daouia, Padoan, et al. (2024), Theorem 1). *Assume $\mathcal{C}_2(\gamma_j, \rho_j, A_j)$ for every j , the tail-copula condition $\mathcal{J}(\mathbf{R})$, the proportionality conditions (2.12), and $\sqrt{k_j} A_j(n_j/k_j) \rightarrow \lambda_j \in \mathbb{R}$ for every j . Then*

$$(\sqrt{k_1}(\hat{\gamma}_1(k_1) - \gamma_1), \dots, \sqrt{k_m}(\hat{\gamma}_m(k_m) - \gamma_m))^\top \xrightarrow{d} \mathcal{N}(\mathbf{B}, \mathbf{V}),$$

with $\mathbf{B} = (\lambda_j/(1 - \rho_j))_{j=1}^m$ and $\mathbf{V}$ the symmetric matrix whose entries are

$$V_{j,\ell} = \begin{cases} \gamma_j^2, & j = \ell, \\ \frac{\gamma_j \gamma_\ell}{\max(b_j, b_\ell) \sqrt{c_j c_\ell}} R_{j,\ell}(b_j c_\ell, b_\ell c_j), & j < \ell. \end{cases}$$

The matrix is symmetric, so $V_{\ell,j} = V_{j,\ell}$ for $\ell > j$. Suppose, in addition, the common- γ condition

(H_γ) . Then the same marginal estimators, now viewed on the common $\sqrt{k}$ scale, satisfy

$$\sqrt{k}(\hat{\gamma}_n - \gamma \mathbf{1}) \xrightarrow{d} \mathcal{N}(\mathbf{B}_c, \mathbf{V}_c),$$

where

$$(\mathbf{B}_c)_j = \sqrt{c_j \sum_{i=1}^m c_i^{-1} \frac{\lambda_j}{1 - \rho_j}} \quad j = 1, \dots, m.$$

The covariance matrix $\mathbf{V}_c$ is given by

$$(\mathbf{V}_c)_{j,\ell} = \left(\sum_{i=1}^m c_i^{-1} \right) \begin{cases} \gamma^2 c_j, & j = \ell, \\ \frac{\gamma^2}{\max(b_j, b_\ell)} R_{j,\ell}(b_j c_\ell, b_\ell c_j), & j < \ell, \end{cases}$$

with $(\mathbf{V}_c)_{\ell,j} = (\mathbf{V}_c)_{j,\ell}$ for $\ell > j$. Consequently, for any $\boldsymbol{\omega}$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$,

$$\sqrt{k}(\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma) \xrightarrow{d} \mathcal{N}(\boldsymbol{\omega}^\top \mathbf{B}_c, \boldsymbol{\omega}^\top \mathbf{V}_c \boldsymbol{\omega}).$$

For independent samples, all off-diagonal entries vanish and $\mathbf{V}_c$ becomes the diagonal matrix

$$\mathbf{V}_c = \gamma^2 \left(\sum_{i=1}^m c_i^{-1} \right) \text{diag}(c_1, \dots, c_m). \quad (2.14)$$

Equation (2.14) is the covariance form used in Chapters 3 and 4.

2.5.3 Variance- and AMSE-optimal weights

The relevant weights solve quadratic programmes under the affine constraint $\boldsymbol{\omega}^\top \mathbf{1} = 1$; unless stated otherwise, they need not be nonnegative.

Proposition 2.17 (Optimal weights; Daouia, Padoan, et al. (2024), Theorem 1; estimated-weight implementation in §2.2 and Corollary 1). *Suppose $\mathbf{V}_c$ is positive definite. Under the common- γ condition (H_γ) :*

(i) *The variance-optimal weights, minimising $\boldsymbol{\omega}^\top \mathbf{V}_c \boldsymbol{\omega}$ subject to $\boldsymbol{\omega}^\top \mathbf{1} = 1$, are*

$$\boldsymbol{\omega}^{\text{Var}} = \frac{\mathbf{V}_c^{-1} \mathbf{1}}{\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}},$$

with corresponding asymptotic variance $\boldsymbol{\omega}^{\text{Var},\top} \mathbf{V}_c \boldsymbol{\omega}^{\text{Var}} = (\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1})^{-1}$.

(ii) *The AMSE-optimal weights, minimising $(\boldsymbol{\omega}^\top \mathbf{B}_c)^2 + \boldsymbol{\omega}^\top \mathbf{V}_c \boldsymbol{\omega}$ subject to $\boldsymbol{\omega}^\top \mathbf{1} = 1$, are*

$$\boldsymbol{\omega}^{\text{AMSE}} = \frac{(1 + \mathbf{B}_c^\top \mathbf{V}_c^{-1} \mathbf{B}_c) \mathbf{V}_c^{-1} \mathbf{1} - (\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{B}_c) \mathbf{V}_c^{-1} \mathbf{B}_c}{(1 + \mathbf{B}_c^\top \mathbf{V}_c^{-1} \mathbf{B}_c)(\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}) - (\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{B}_c)^2}.$$

In the independent-samples case, $\mathbf{V}_c$ is diagonal (Equation (2.14)) and the variance-optimal

weights reduce to

$$\omega_j^{\text{Var}} = \left(\sum_{i=1}^m c_i^{-1} \right)^{-1} c_j^{-1} \sim \frac{k_j}{k}, \quad (2.15)$$

where the equivalence uses $c_i^{-1} = \lim k_i/k_1$. The finite-sample distributed version is

$$\tilde{\omega}_n^{\text{Var}} = (k_1/k, \dots, k_m/k)^\top,$$

which is exactly normalised and converges to ω^{Var} .

Feasible weights replace $\mathbf{V}_c$ and $\mathbf{B}_c$ by consistent plug-ins (Daouia, Padoan, et al. 2024, §2.2). If $\hat{\omega}^\top \mathbf{1} = 1$ exactly and $\hat{\omega} \xrightarrow{\mathbb{P}} \omega$, then $\hat{\gamma}_n(\hat{\omega})$ is $\sqrt{k}$ -equivalent to $\hat{\gamma}_n(\omega)$; the difficult practical step is stable estimation of $\mathbf{V}_c$ and the bias quantities.

2.5.4 Weighted geometric pooling of extreme quantiles

For estimation of an extreme quantile $q(1-p)$ with $p = p(n) \rightarrow 0$, each sample contributes a Weissman estimator

$$\hat{q}_j^*(1-p \mid k_j) = \left(\frac{k_j}{n_j p} \right)^{\hat{\gamma}_j(k_j)} X_{n_j - k_j : n_j, j}.$$

Because Weissman extrapolation is multiplicative, pooling is geometric on the original scale, equivalently linear on the log scale. It requires asymptotic equivalence of the marginal extreme quantiles:

Condition 2.18 (Tail homoskedasticity ($\mathcal{H}$)). For every $j \neq \ell$, $U_j(t)/U_\ell(t) \rightarrow 1$ as $t \rightarrow \infty$.

Equivalently, $Q_j(1-p)/Q_\ell(1-p) \rightarrow 1$ as $p \downarrow 0$. The pooled extreme-quantile estimator of Daouia, Padoan, et al. (2024, §2.3) is

$$\hat{q}_n^*(1-p \mid \omega) := \prod_{j=1}^m [\hat{q}_j^*(1-p \mid k_j)]^{\omega_j}. \quad (2.16)$$

Taking logs reduces pooling to weighted sums of the local identities $\log \hat{q}_j^*(1-p \mid k_j) = \log(k_j/(n_j p)) \hat{\gamma}_j(k_j) + \log X_{n_j - k_j : n_j, j}$; so the Hill part is handled by Theorem 2.16; (2.12) gives $\log(k_j/(n_j p)) = \log(k/(np)) + O(1)$.

Theorem 2.19 (Daouia, Padoan, et al. (2024), Theorem 2). Assume the hypotheses of Theorem 2.16, the common- γ condition (H_γ), the tail-homoskedasticity condition ($\mathcal{H}$), and, in addition, $\rho_j < 0$ for every $j = 1, \dots, m$. Let $p = p(n) \rightarrow 0$ with $k/(np) \rightarrow \infty$ and $\sqrt{k}/\log(k/(np)) \rightarrow \infty$. Then for any ω with $\omega^\top \mathbf{1} = 1$ and any $j \in \{1, \dots, m\}$,

$$\frac{\sqrt{k}}{\log(k/(np))} \left(\frac{\hat{q}_n^*(1-p \mid \omega)}{Q_j(1-p)} - 1 \right) \xrightarrow{d} \mathcal{N}(\omega^\top \mathbf{B}_c, \omega^\top \mathbf{V}_c \omega).$$

The strict $\rho_j < 0$ assumption is part of the Weissman theorem. It also supplies the rate needed to recenter by any marginal $Q_j(1-p)$ under ($\mathcal{H}$) (Daouia, Padoan, et al. 2024, Supplement,

Lemma B.2): for any j, ℓ ,

$$\log Q_j(1-p) - \log Q_\ell(1-p) = O(|A_j(1/p)|) + O(|A_\ell(1/p)|) = o(k^{-1/2}),$$

along the Weissman sequence.

Thus the pooled Weissman and pooled Hill limits share the same mean, variance and optimal weights up to the log-rate factor. This is the multi-sample analogue of the tail-index dominance in [Theorem 2.13](#) and [Proposition 2.14](#).

2.5.5 Tail-homogeneity testing and confidence intervals

The covariance structure $\mathbf{V}_c$ also yields a Wald-type chi-square test for the common- γ hypothesis (H_γ), sometimes called *tail homogeneity* across samples.

Theorem 2.20 (Tail-homogeneity test; Daouia, Padoan, et al. (2024), Corollary 3). *Assume the hypotheses of [Theorem 2.16](#), the null $H_0 : \gamma_1 = \dots = \gamma_m$, and the bias-free calibration $\lambda_j = 0$ for every j . The statistic*

$$\Lambda_n = k(\hat{\gamma}_n - \bar{\mu}_n \mathbf{1})^\top \widehat{\mathbf{V}}_c^{-1}(\hat{\gamma}_n - \bar{\mu}_n \mathbf{1}), \quad \bar{\mu}_n = \frac{\mathbf{1}^\top \widehat{\mathbf{V}}_c^{-1} \hat{\gamma}_n}{\mathbf{1}^\top \widehat{\mathbf{V}}_c^{-1} \mathbf{1}},$$

where $\widehat{\mathbf{V}}_c$ is a consistent estimator of the covariance matrix of $\sqrt{k}(\hat{\gamma}_n - \gamma \mathbf{1})$ under H_0 . Then $\Lambda_n \xrightarrow{d} \chi_{m-1}^2$ under H_0 and $\Lambda_n \xrightarrow{\mathbb{P}} \infty$ under fixed alternatives. A test of asymptotic size α rejects H_0 when $\Lambda_n > \chi_{m-1, 1-\alpha}^2$.

Daouia, Padoan, et al. (2024) state the bias-free version above. As a direct Gaussian consequence, rather than as an additional result stated in that paper, the same GLS calculation also shows that, if the asymptotic bias is retained rather than removed by undersmoothing or bias correction, the deviance has the noncentral limit

$$\Lambda_n \xrightarrow{d} \chi_{m-1}^2(\delta_\gamma), \quad \delta_\gamma = \mathbf{B}_c^\top \mathbf{V}_c^{-1} \mathbf{B}_c - \frac{(\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{B}_c)^2}{\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}}.$$

Thus the central chi-square limit holds whenever $\mathbf{B}_c \in \text{span}\{\mathbf{1}\}$, in particular under $\lambda_j = 0$ for every j . Under contiguous alternatives of order $k^{-1/2}$, the limit is likewise noncentral rather than divergent.

The same Gaussian limits give confidence intervals for the common γ and log-scale intervals for pooled Weissman quantiles; retained bias must be removed or included in the centering.

2.5.6 Distributed inference as a special case

In the independent common-distribution distributed setting, the samples are iid subdivisions of one common population. Then:

- the common- γ condition (H_γ) and the tail-homoskedasticity condition ($\mathcal{H}$) are trivially satisfied;

- the tail copula $R_{j,\ell}$ between any pair of distinct samples vanishes identically (independence implies tail independence);
- $\mathbf{V}_c$ in [Theorem 2.16](#) becomes the diagonal matrix [\(2.14\)](#);
- the variance-optimal weights of [Proposition 2.17](#) reduce to $\omega_j^{\text{Var}} \propto k_j$ as in [\(2.15\)](#).

Corollary 2.21 (Distributed oracle equivalence; Daouia, Padoan, et al. [\(2024\)](#), Corollary 6, Theorem 3 and Corollary 8). *Under the distributed-inference assumptions (iid across machines) of Corollary 5 in Daouia, Padoan, et al. [\(2024\)](#), suppose the sample fractions are asymptotically equal in the stronger sense $k_j/n_j = (k/n)(1+O(k^{-1/2}))$ for every j . Then, under the conditions of their Theorem 3, the variance-optimally-pooled Hill estimator $\hat{\gamma}_n(\tilde{\omega}_n^{\text{Var}})$ is $\sqrt{k}$ -asymptotically equivalent to the unfeasible benchmark Hill estimator $\hat{\gamma}_n^{(\text{Hill})}(k)$ built from the entire combined sample. If, in addition, $\rho < 0$ and $p = p(n)$ satisfies $k/(np) \rightarrow \infty$ and $\sqrt{k}/\log(k/(np)) \rightarrow \infty$, then the variance-optimally-pooled Weissman estimator $\hat{q}_n^*(1-p \mid \tilde{\omega}_n^{\text{Var}})$ is asymptotically equivalent to the unfeasible benchmark Weissman estimator built from the combined sample.*

[Corollary 2.21](#) is the efficiency benchmark the pooled-expectile constructions in [Chapters 3 and 4](#) seek to preserve.

Chapter 3

Pooled inference for extreme expectiles at intermediate levels

3.1 Setup and notation

This chapter specialises the pooling framework of [Section 2.5](#) to independent samples with a common marginal distribution, the distributed setting of [Section 2.5.6](#). This is stronger than common tail index alone, but it gives one target ξ_{τ_n} and one tail quantile Q_X to pool.

The equal effective sample fraction condition $k_j/n_j = (k/n)(1 + o(1))$ is not part of the standing setup; it is invoked only in [Sections 3.5](#) to [3.7](#). Oracle comparisons use the stronger $k_j/n_j = (k/n)(1 + O(k^{-1/2}))$ condition of [Corollary 2.21](#). The pooled-Hill CLT is imported from [Theorem 2.16](#); the finite-log Weissman limit needed here is derived in this chapter.

Samples and pooling primitives. There are m independent samples $\{X_{i,j}\}_{i=1,\dots,n_j}$, $j = 1, \dots, m$, with common marginal distribution F and common extreme-value index γ . Let $n = \sum_{j=1}^m n_j$. All limits are finite- m triangular-array limits with $n_j \rightarrow \infty$, $k_j \rightarrow \infty$ and $k_j/n_j \rightarrow 0$. Independence makes all pairwise tail-copula functions vanish:

$$R_{j,\ell} \equiv 0 \quad \text{for every } j \neq \ell. \quad (3.1)$$

The proportionality conditions of [\(2.12\)](#) are in force:

$$\frac{n_1}{n_j} \rightarrow b_j \in (0, \infty), \quad \frac{k_1}{k_j} \rightarrow c_j \in (0, \infty), \quad j = 1, \dots, m,$$

with $b_1 = c_1 = 1$, and $k = \sum_{j=1}^m k_j$. Hence $k_j \asymp k$ uniformly in j and, with $S = \sum_{i=1}^m c_i^{-1}$, $k/k_j \rightarrow c_j S$. Each sample supplies the Paper-3 primitives: the local Hill estimator [\(2.1\)](#), the local Weissman estimator [\(2.2\)](#), and their joint asymptotic covariance. At any fixed target level this is one quantile scalar per sample; no raw observation and no expectile-specific summary statistic crosses between machines.

Heavy-tail regime. The common marginal F satisfies the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ of [Condition 2.3](#) with

$$\gamma \in (0, 1), \quad \mathbb{E} X^- < \infty, \quad (3.2)$$

the per-sample bias-rate condition $\sqrt{k_j} A(n_j/k_j) \rightarrow \lambda_j \in \mathbb{R}$ holds for every j , and the moment-negligibility condition

$$\frac{\sqrt{k}}{Q_X(\tau_n)} \rightarrow 0 \quad (3.3)$$

is in force. The finite-mean regime (3.2) is enough for the QB route; the stronger $\gamma < 1/2$ restriction belongs only to the direct LAWS CLT. Condition (3.3) kills the $O(Q_X(\tau_n)^{-1})$ first-moment term in (2.7) at the $\sqrt{k}$ scale. Under the equal-fraction scaling $1 - \tau_n \asymp k/n$, and suppressing slow variation, it requires $k^{1/2+\gamma}/n^\gamma \rightarrow 0$. The endpoint $\rho = 0$ is allowed here; the stricter $\rho < 0$ re-enters only in [Chapter 4](#).

The target. We target the common population expectile ξ_{τ_n} at an intermediate level $\tau_n \rightarrow 1$ with $n(1 - \tau_n) \rightarrow \infty$. The per-sample log-rate factor

$$L_j(\tau_n) := \log \frac{k_j}{n_j(1 - \tau_n)}$$

has a finite limit $L_j^* \in \mathbb{R}$ for every j . This is the finite-log regime used in [Proposition 3.2](#); the very-extreme level is deferred to [Chapter 4](#).

Per-sample QB plug-in: a forced consequence of the pooling protocol. Because each sample shares only the Paper-3 primitives and no expectile-specific summary statistic, the available local intermediate expectile construction is the QB plug-in (2.9):

$$\hat{\xi}_{\tau,j} = \psi(\hat{\gamma}_j(k_j)) \hat{q}_j^*(\tau | k_j).$$

The LAWS estimator (2.8) would require communicating local expectiles, which is outside [Section 2.5](#). Thus QB is used because it is forced by the communication protocol, not because it uniformly dominates LAWS on the direct-LAWS range $\gamma \in (0, 1/2)$.

3.2 The pooled QB expectile estimator

The shared primitives admit two natural single-weight constructions. In (A), Hill and Weissman inputs are pooled separately and the QB plug-in is applied once; in (B), local QB expectiles are computed first and then geometrically pooled. At matched weights, [Proposition 3.1](#) makes the difference asymptotically negligible. We adopt (A) because it is the direct pooled analogue of the iid QB pipeline and it exposes the two primitive weight vectors $(\omega_\gamma, \omega_q)$ used in [Section 3.5](#).

Pooled Hill and pooled Weissman primitives. For weight vectors $\omega_\gamma, \omega_q \in \mathbb{R}^m$ satisfying $\omega_\gamma^\top \mathbf{1} = \omega_q^\top \mathbf{1} = 1$, the linear pooled Hill estimator (2.13) and the weighted-geometric pooled

Weissman estimator (2.16) read

$$\hat{\gamma}_n(\omega_\gamma) = \omega_\gamma^\top \hat{\gamma}_n = \sum_{j=1}^m \omega_{\gamma,j} \hat{\gamma}_j(k_j), \quad (3.4)$$

$$\hat{q}_n^*(\tau_n \mid \omega_q) = \prod_{j=1}^m [\hat{q}_j^*(\tau_n \mid k_j)]^{\omega_{q,j}}, \quad (3.5)$$

both evaluated at τ_n . The weights are affine and need not be nonnegative. Geometric products are interpreted on the log scale, on the high-probability event where the Weissman factors are positive and the Hill estimators entering $\log \psi$ lie in $(0, 1)$.

The two candidate estimators. The first construction, which we adopt as canonical, pools the primitives separately and applies the QB plug-in once at the centre,

$$\tilde{\xi}_{\tau_n}^{(A)}(\omega_\gamma, \omega_q) := \psi(\hat{\gamma}_n(\omega_\gamma)) \hat{q}_n^*(\tau_n \mid \omega_q). \quad (3.6)$$

The second computes a per-sample QB expectile on each machine and geometric-pools the resulting m local estimates,

$$\tilde{\xi}_{\tau_n}^{(B)}(\omega) := \prod_{j=1}^m [\psi(\hat{\gamma}_j(k_j)) \hat{q}_j^*(\tau_n \mid k_j)]^{\omega_j} = \left(\prod_{j=1}^m \psi(\hat{\gamma}_j(k_j))^{\omega_j} \right) \cdot \hat{q}_n^*(\tau_n \mid \omega). \quad (3.7)$$

At matched weights the Weissman factor is identical in (3.6) and (3.7); only the placement of the nonlinearity ψ differs.

Proposition 3.1 ($\sqrt{k}$ -asymptotic equivalence of (A) and (B)). *Under the assumptions of Section 3.1, for fixed $m \in \mathbb{N}$ and every deterministic weight vector $\omega \in \mathbb{R}^m$ with $\omega^\top \mathbf{1} = 1$,*

$$\sqrt{k} \log \frac{\tilde{\xi}_{\tau_n}^{(A)}(\omega, \omega)}{\tilde{\xi}_{\tau_n}^{(B)}(\omega)} \xrightarrow{\mathbb{P}} 0.$$

In fact the unscaled log-ratio is itself $O_p(k^{-1})$; multiplying by $\sqrt{k}$ therefore gives $O_p(k^{-1/2}) = o_p(1)$, so the discrepancy is negligible at the CLT scale. Consequently, when construction (A) is evaluated at matched weights $(\omega_\gamma, \omega_q) = (\omega, \omega)$, the two estimators share the same $\sqrt{k}$ -asymptotic distribution and Theorem 3.3 applies to either without modification. The same conclusion extends to estimated weights $\hat{\omega}$ satisfying $\hat{\omega}^\top \mathbf{1} = 1$ exactly and $\hat{\omega} \xrightarrow{\mathbb{P}} \omega$. If exact normalisation is relaxed to $\sqrt{k} \left| \hat{\omega}^\top \mathbf{1} - 1 \right| \xrightarrow{\mathbb{P}} 0$, the $\sqrt{k}$ -equivalence still holds, although the unscaled $O_p(k^{-1})$ refinement need not.

Proof. See Appendix A.2. □

Canonical choice. We adopt

$$\hat{\xi}_{\tau_n}^{\text{pool}}(\omega_\gamma, \omega_q) := \psi(\hat{\gamma}_n(\omega_\gamma)) \hat{q}_n^*(\tau_n \mid \omega_q), \quad (3.8)$$

identical to (3.6), as the canonical pooled QB intermediate-expectile estimator. It reproduces the iid QB pipeline of Theorem 2.12 at the pooled level, while Proposition 3.1 shows that the local-QB alternative adds no first-order information at matched weights.

Two weight vectors, not one. Definition (3.8) carries separate Hill and Weissman weights. This is needed because the delta method through ψ creates the cross term displayed later in (3.21); Section 3.5 then decides when the two optima coincide.

Why the pooled Weissman at the intermediate level. At the intermediate level, $\hat{q}_j^*(\tau_n | k_j)$ moves sample j from its local level $1 - k_j/n_j$ to the common target τ_n . If these levels coincide, the Weissman power is one and the estimator is the intermediate order statistic. Otherwise the finite-log-rate CLT in Proposition 3.2 controls the interpolation or mild extrapolation. Using this same Paper-3 primitive also keeps Chapter 4 on the same notation.

3.3 Joint asymptotic distribution of the pooled inputs

This section establishes the joint $\sqrt{k}$ -Gaussian limit of $(\hat{\gamma}_n(\omega_\gamma), \log \hat{q}_n^*(\tau_n | \omega_q))$ at intermediate level τ_n . The result is the input that Section 3.4 feeds through the delta method to obtain the asymptotic distribution of the pooled QB expectile estimator (3.8).

Within-sample decomposition. For each sample j , the log of the Weissman estimator decomposes exactly as

$$\log \hat{q}_j^*(\tau_n | k_j) = \log X_{n_j - k_j : n_j, j} + L_j(\tau_n) \hat{\gamma}_j(k_j), \quad L_j(\tau_n) := \log \frac{k_j}{n_j(1 - \tau_n)}. \quad (3.9)$$

By the tail-empirical-process representation (Haan and Ferreira 2006, Theorem 2.4.8 and §5.1), the local Hill estimator and intermediate log-order statistic are jointly Gaussian on the $\sqrt{k_j}$ scale, asymptotically independent, and each has variance γ^2 . Combined with (3.9) and the deterministic second-order centering, this yields

$$\sqrt{k_j} \begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log \hat{q}_j^*(\tau_n | k_j) - \log Q_X(\tau_n) \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(\mu_j^*, \gamma^2 \begin{pmatrix} 1 & L_j^* \\ L_j^* & 1 + (L_j^*)^2 \end{pmatrix} \right), \quad (3.10)$$

under the common-marginal assumption of Section 3.1 and the condition $L_j(\tau_n) \rightarrow L_j^* \in \mathbb{R}$. With Ψ_ρ the second-order kernel defined in (3.14), the limiting mean is

$$\mu_j^* = \begin{pmatrix} \lambda_j/(1 - \rho) \\ \lambda_j \{ L_j^*/(1 - \rho) - \Psi_\rho(e^{L_j^*}) \} \end{pmatrix}.$$

The second mean component combines Hill bias transported through L_j^* with the deterministic second-order Weissman remainder; the off-diagonal covariance $\gamma^2 L_j^*$ is induced by the same log-rate factor.

Pooling under independence. Independence makes the cross-sample covariance blocks vanish. Linear pooling then combines the within-sample blocks of (3.10), each rescaled to the common $\sqrt{k}$ scale by (2.12).

Proposition 3.2 (Joint pooled CLT at intermediate level). *Under the assumptions of Section 3.1 and the convergence $L_j(\tau_n) \rightarrow L_j^* \in \mathbb{R}$ for every j , for any weight vectors $\omega_\gamma, \omega_q \in \mathbb{R}^m$ satisfying $\omega_\gamma^\top \mathbf{1} = \omega_q^\top \mathbf{1} = 1$,*

$$\sqrt{k} \begin{pmatrix} \hat{\gamma}_n(\omega_\gamma) - \gamma \\ \log \hat{q}_n^*(\tau_n | \omega_q) - \log Q_X(\tau_n) \end{pmatrix} \xrightarrow{d} \mathcal{N}(\mu(\omega_\gamma, \omega_q), \Sigma(\omega_\gamma, \omega_q)), \quad (3.11)$$

with covariance

$$\begin{aligned} D^* &:= \text{diag}(L^*), \\ \Sigma(\omega_\gamma, \omega_q) &= \begin{pmatrix} \omega_\gamma^\top V_c \omega_\gamma & \omega_\gamma^\top D^* V_c \omega_q \\ \omega_q^\top V_c D^* \omega_\gamma & \omega_q^\top (V_c + D^* V_c D^*) \omega_q \end{pmatrix}. \end{aligned} \quad (3.12)$$

Here V_c is the diagonal matrix (2.14), so $D^* V_c D^* = (D^*)^2 V_c$, and $L^* = (L_1^*, \dots, L_m^*)^\top$. The mean

$$\mu(\omega_\gamma, \omega_q) = \begin{pmatrix} \omega_\gamma^\top B_c \\ \omega_q^\top B_c^* \end{pmatrix}, \quad (3.13)$$

where B_c is the pooled-Hill bias vector of Theorem 2.16 and the limiting pooled-Weissman bias vector at the intermediate level has entries

$$(B_c^*)_j = \sqrt{c_j \sum_{i=1}^m c_i^{-1}} \lambda_j \left[\frac{L_j^*}{1 - \rho} - \Psi_\rho(e^{L_j^*}) \right], \quad \Psi_\rho(y) = \begin{cases} (y^\rho - 1)/\rho, & \rho < 0, \\ \log y, & \rho = 0, \end{cases} \quad (3.14)$$

combining Hill bias, weighted by L_j^* , with the limiting second-order Weissman remainder. The intermediate log-order statistic contributes no $\sqrt{k_j}$ -bias term of its own, and the covariance (3.12) does not depend on B_c^* . The same limit holds for estimated weights satisfying $\hat{\omega}_\gamma^\top \mathbf{1} = \hat{\omega}_q^\top \mathbf{1} = 1$ exactly and $\hat{\omega}_\gamma \xrightarrow{\mathbb{P}} \omega_\gamma$, $\hat{\omega}_q \xrightarrow{\mathbb{P}} \omega_q$; this estimated-weight lift is deferred to Section 3.6.

Proof. See Appendix A.3. □

Proposition 3.2 is the finite-log counterpart of the pooled-Hill and geometric-Weissman machinery in Section 2.5. Its new expectile-relevant feature is the cross-covariance $\omega_\gamma^\top \text{diag}(L^*) V_c \omega_q$, which drives the weight analysis below.

3.4 Asymptotic normality of the pooled expectile estimator

The main result of this chapter is the central limit theorem for the pooled QB intermediate expectile estimator (3.8). The result is obtained by feeding the joint pooled CLT of Proposition 3.2 through the smooth log-linear map $(g, x) \mapsto \log \psi(g) + x$ via a first-order Taylor expansion,

combined with the second-order expectile–quantile expansion that controls the deterministic gap between $\log \xi_{\tau_n}$ and $\log \psi(\gamma) + \log Q_X(\tau_n)$.

Derivative of $\log \psi$. A short calculation gives

$$\frac{d}{dg} \log \psi(g) = \frac{1}{1-g} - \log(g^{-1} - 1) =: m(g), \quad (3.15)$$

which is the same $m(\gamma)$ as in [Theorem 2.12](#). It vanishes at $\gamma^* \approx 0.2178$, a value relevant for the weight formulas.

The deterministic expectile–quantile gap. The first-order asymptotic equivalence of [Proposition 2.10](#),

$$\xi_\tau / Q_X(\tau) \rightarrow \psi(\gamma), \quad \tau \uparrow 1,$$

sharpens, under [Condition 2.3](#), through the second-order expansion [\(2.7\)](#) (Daouia, Girard, et al. [2020](#), Proposition 1(i)),

$$\begin{aligned} \log \xi_\tau - \log \psi(\gamma) - \log Q_X(\tau) &= c(\gamma, \rho) A((1-\tau)^{-1}) + o(A((1-\tau)^{-1})) \\ &\quad + O(Q_X(\tau)^{-1}), \quad \tau \uparrow 1. \end{aligned} \quad (3.16)$$

Here $c(\gamma, \rho)$ is defined in [\(2.7\)](#). The final term is not generally $o(A)$, but [\(3.3\)](#) makes it negligible after multiplication by $\sqrt{k}$, while the bias-rate assumptions give $\sqrt{k}A((1-\tau_n)^{-1}) = O(1)$. Hence

$$\sqrt{k} [\log \xi_{\tau_n} - \log \psi(\gamma) - \log Q_X(\tau_n)] = c(\gamma, \rho) \sqrt{k} A((1-\tau_n)^{-1}) + o(1). \quad (3.17)$$

The setup already implies the finite limit

$$\Lambda^\bullet := \lim_{n \rightarrow \infty} \sqrt{k} A((1-\tau_n)^{-1}) \in \mathbb{R}, \quad (3.18)$$

with no extra hypothesis. Equivalently, with $S = \sum_i c_i^{-1}$, the introduced constants obey

$$\sqrt{c_j S} \lambda_j e^{\rho L_j^*} = \sqrt{c_\ell S} \lambda_\ell e^{\rho L_\ell^*} \quad \text{for all } j, \ell,$$

because both sides are limits of $\sqrt{k}A((1-\tau_n)^{-1})$. The finite asymptotic constant

$$b^{\xi/Q} := \lim_{n \rightarrow \infty} \sqrt{k} [\log \psi(\gamma) + \log Q_X(\tau_n) - \log \xi_{\tau_n}] = -c(\gamma, \rho) \Lambda^\bullet \quad (3.19)$$

follows. The sign convention makes $b^{\xi/Q}$ enter [\(3.22\)](#) with a plus sign; it is the expectile-specific deterministic bias contribution, absent from the quantile-target pooling theorem.

Theorem 3.3 (Pooled QB intermediate expectile CLT). *Under the setup and heavy-tail regime of [Section 3.1](#) (including [Equations \(3.2\)](#) and [\(3.3\)](#)), the positivity/log-domain convention of [Section 3.2](#), the assumptions of [Proposition 3.2](#), and the finite expectile–quantile bias limit [\(3.18\)](#)*

implied by these conditions, for any weight vectors $\omega_\gamma, \omega_q \in \mathbb{R}^m$ satisfying $\omega_\gamma^\top \mathbf{1} = \omega_q^\top \mathbf{1} = 1$,

$$\sqrt{k} \left(\frac{\hat{\xi}_{\tau_n}^{\text{pool}}(\omega_\gamma, \omega_q)}{\xi_{\tau_n}} - 1 \right) \xrightarrow{d} \mathcal{N}(\mu^\xi(\omega_\gamma, \omega_q), \sigma^\xi(\omega_\gamma, \omega_q)^2), \quad (3.20)$$

with asymptotic variance

$$\sigma^\xi(\omega_\gamma, \omega_q)^2 = m(\gamma)^2 \omega_\gamma^\top \mathbf{V}_c \omega_\gamma + 2m(\gamma) \omega_\gamma^\top \text{diag}(\mathbf{L}^*) \mathbf{V}_c \omega_q + \omega_q^\top (\mathbf{V}_c + \text{diag}(\mathbf{L}^*)^2 \mathbf{V}_c) \omega_q, \quad (3.21)$$

and asymptotic bias

$$\mu^\xi(\omega_\gamma, \omega_q) = m(\gamma) \omega_\gamma^\top \mathbf{B}_c + \omega_q^\top \mathbf{B}_c^* + b^{\xi/Q}, \quad (3.22)$$

where $\mathbf{B}_c$ and $\mathbf{B}_c^*$ are as in [Proposition 3.2](#) (the pooled-Hill and limiting intermediate-level pooled-Weissman bias vectors) and $b^{\xi/Q}$ is the asymptotic expectile-quantile bias contribution of [\(3.19\)](#).

Proof. See [Appendix A.4](#). □

Writing $\mathbf{D}^* = \text{diag}(\mathbf{L}^*)$, the variance in [\(3.21\)](#) factorises as

$$\sigma^\xi(\omega_\gamma, \omega_q)^2 = (m(\gamma) \omega_\gamma + \mathbf{D}^* \omega_q)^\top \mathbf{V}_c (m(\gamma) \omega_\gamma + \mathbf{D}^* \omega_q) + \omega_q^\top \mathbf{V}_c \omega_q.$$

This factorisation is the starting point for the weight optimisation.

Reduction to the iid baseline. [Theorem 3.3](#) contains [Theorem 2.12](#) as the aligned $m = 1$ special case. Then $L_1^* = 0$, $k = k_1$, and

$$\mathbf{V}_c = \gamma^2, \quad \mathbf{B}_c = \frac{\lambda}{1 - \rho}, \quad \Lambda^\bullet = \lambda, \quad b^{\xi/Q} = -c(\gamma, \rho)\lambda.$$

Also $\mathbf{B}_c^* = 0$, so [\(3.21\)](#) becomes $\gamma^2(1 + m(\gamma)^2)$ and [\(3.22\)](#) becomes

$$\mu^\xi = m(\gamma) \mathbf{B}_c + \mathbf{B}_c^* + b^{\xi/Q} = \frac{m(\gamma)}{1 - \rho} \lambda - c(\gamma, \rho) \lambda = \left(\frac{m(\gamma)}{1 - \rho} - c(\gamma, \rho) \right) \lambda, \quad (3.23)$$

in exact agreement with [Theorem 2.12](#).

Equal-fraction distributed simplification. Under $k_j/n_j = (k/n)(1 + o(1))$, the log-rate factor satisfies

$$L_j(\tau_n) = \log \frac{k}{n(1 - \tau_n)} + o(1).$$

Thus all finite limits equal a common L^* and [\(3.21\)](#) simplifies to

$$\sigma^\xi(\omega_\gamma, \omega_q)^2 = m(\gamma)^2 \omega_\gamma^\top \mathbf{V}_c \omega_\gamma + 2m(\gamma) L^* \omega_\gamma^\top \mathbf{V}_c \omega_q + (1 + (L^*)^2) \omega_q^\top \mathbf{V}_c \omega_q. \quad (3.24)$$

With matched weights and $1 - \tau_n = (k/n)(1 + o(1))$, $L^* = 0$ and the variance becomes $(1 + m(\gamma)^2) \omega^\top \mathbf{V}_c \omega$.

Why two weight vectors. The cross term in (3.21) is why the two-weight definition is useful. Section 3.5 shows that this extra freedom collapses in the equal-fraction distributed case: if $m(\gamma) \neq 0$, the variance-optimal pair is matched at ω^{Var} ; if $m(\gamma) = 0$, the Hill weight is variance-unidentified.

3.5 Variance- and AMSE-optimal weights

We now identify the weight vectors $(\omega_\gamma, \omega_q)$ that minimise the asymptotic variance (3.21) and the AMSE $(\mu^\xi)^2 + \sigma^{\xi,2}$ of the pooled QB expectile estimator, subject to the affine sum-to-one constraints $\omega_\gamma^\top \mathbf{1} = \omega_q^\top \mathbf{1} = 1$. The qualitative question driving this section is whether the expectile-optimal weights coincide with those for the quantile target of Proposition 2.17, or whether the expectile target requires its own weight scheme. The answer has an important collapse case: under equal effective fractions, the limiting bias is common across samples, so the AMSE criterion reduces to the variance criterion up to an additive constant; target-specific AMSE weights arise only away from that regime.

Variance-optimal weights

The variance (3.21) admits the identity

$$\sigma^\xi(\omega_\gamma, \omega_q)^2 = \mathbf{u}^\top \mathbf{V}_c \mathbf{u} + \omega_q^\top \mathbf{V}_c \omega_q, \quad \mathbf{u} := m(\gamma) \omega_\gamma + \text{diag}(\mathbf{L}^*) \omega_q, \quad (3.25)$$

which one verifies by direct expansion using the diagonality of $\mathbf{V}_c$ (and hence its commutativity with $\text{diag}(\mathbf{L}^*)$). For a quadratic form $\mathbf{x}^\top \mathbf{V}_c \mathbf{x}$ under a constraint $\mathbf{x}^\top \mathbf{1} = d$, the minimiser is proportional to $\mathbf{V}_c^{-1} \mathbf{1}$. Whether both terms in (3.25) can be minimised simultaneously by one pair $(\omega_\gamma, \omega_q)$ depends on whether the linear constraint induced on $\mathbf{u}$ by the sum-to-one constraints is sample-free. This happens when $\mathbf{L}^*$ is a scalar multiple of $\mathbf{1}$; in the independent common-distribution distributed specialisation of Section 3.1, this algebraic collapse is induced by the equal-fraction condition $k_j/n_j = (k/n)(1 + o(1))$.

Proposition 3.4 (Variance-optimal weights, equal-fraction distributed inference). *Under the assumptions of Theorem 3.3 and the equal-effective-sample-fraction condition $k_j/n_j = (k/n)(1 + o(1))$ described in Section 3.1, the variance-optimal weights for the pooled QB expectile estimator satisfy*

$$\omega_q^{\xi, \text{Var}} = \omega^{\text{Var}} = \frac{\mathbf{V}_c^{-1} \mathbf{1}}{\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}}, \quad (3.26)$$

with corresponding asymptotic variance

$$\sigma^\xi(\omega_\gamma^{\xi, \text{Var}}, \omega_q^{\xi, \text{Var}})^2 = [(m(\gamma) + L^*)^2 + 1] (\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1})^{-1}. \quad (3.27)$$

If $m(\gamma) \neq 0$, the variance-optimal pair is unique and $\omega_\gamma^{\xi, \text{Var}} = \omega^{\text{Var}}$ as well, so both pooled inputs receive the quantile-pooling weights of Daouia, Padoan, et al. (2024). If $m(\gamma) = 0$, the asymptotic variance does not depend on ω_γ : any admissible vector is then variance-optimal, and the matched representative $\omega_\gamma^{\xi, \text{Var}} = \omega^{\text{Var}}$ is one canonical choice.

Proof. The proposition is the specialisation $\mathbf{L}^* = L^* \mathbf{1}$ of [Proposition 3.5](#); see [Appendix A.5](#), which discharges both the $m(\gamma) \neq 0$ closed form and the reduced $m(\gamma) = 0$ case. $\square$

[Proposition 3.4](#) is the central variance-efficiency statement: under equal fractions, changing the target from quantile to expectile does not change the variance-optimal fusion weights. Only the variance multiplier changes. The AMSE criterion needs separate bookkeeping because its bias term is target-specific, but it also collapses to the variance criterion in the equal-fraction regime.

Beyond the equal-fraction setting. When k_j/n_j varies across samples, $\text{diag}(\mathbf{L}^*)$ is no longer scalar and the decoupling behind [Proposition 3.4](#) breaks. Set $\mathbf{D} = \text{diag}(\mathbf{L}^*)$ and define

$$a := \mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}, \quad b := \mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{L}^*, \quad c := (\mathbf{L}^*)^\top \mathbf{V}_c^{-1} \mathbf{L}^*, \quad \Delta := a(a + c) - b^2. \quad (3.28)$$

Proposition 3.5 (Variance-optimal weights, general case). *Under the assumptions of [Theorem 3.3](#) and $m(\gamma) \neq 0$, the Cauchy–Schwarz inequality in the $\mathbf{V}_c^{-1}$ -inner product gives $ac - b^2 \geq 0$, with equality iff $\mathbf{L}^*$ is a scalar multiple of $\mathbf{1}$. This is the limiting collapse induced by the equal-effective-sample-fraction condition in the distributed specialisation. Consequently*

$$\Delta = a^2 + (ac - b^2) \geq a^2 > 0,$$

with $\Delta = a^2$ exactly under the equal-fraction distributed specialisation of [Proposition 3.4](#). The variance-optimal pooling weights for the pooled QB intermediate expectile estimator are unique and given by

$$\omega_q^{\xi, \text{Var}} = \frac{(a + c + m(\gamma)b) \mathbf{V}_c^{-1} \mathbf{1} - (m(\gamma)a + b) \mathbf{V}_c^{-1} \mathbf{L}^*}{\Delta}, \quad (3.29)$$

$$\omega_\gamma^{\xi, \text{Var}} = \frac{(m(\gamma)a + b) \mathbf{V}_c^{-1} (\mathbf{1} + \mathbf{D}^2 \mathbf{1}) - (a + c + m(\gamma)b) \mathbf{V}_c^{-1} \mathbf{L}^*}{m(\gamma) \Delta}, \quad (3.30)$$

and the associated minimum asymptotic variance is

$$\sigma^\xi(\omega_\gamma^{\xi, \text{Var}}, \omega_q^{\xi, \text{Var}})^2 = \frac{(m(\gamma)^2 + 1)a + 2m(\gamma)b + c}{\Delta}. \quad (3.31)$$

Under the equal-effective-fraction condition $\mathbf{L}^* = L^* \mathbf{1}$ one has $b = L^* a$, $c = (L^*)^2 a$, $\Delta = a^2$; formulas (3.29)–(3.31) then collapse to (3.26) and (3.27), recovering [Proposition 3.4](#).

Proof. See [Appendix A.5](#). $\square$

When $m(\gamma) = 0$, the variance does not identify ω_γ ; the reduced single-vector problem is recorded in [Appendix A.5](#).

Departure from the matched-weight diagonal. Writing $\bar{L}^* = b/a$, the departure of the quantile-weight component from the Paper-3 variance weight is

$$\omega_q^{\xi, \text{Var}} - \omega^{\text{Var}} = -\frac{m(\gamma)a + b}{\Delta} \mathbf{V}_c^{-1} (\mathbf{L}^* - \bar{L}^* \mathbf{1}). \quad (3.32)$$

The correction is along the centred direction $\mathbf{V}_c^{-1}(\mathbf{L}^\star - \bar{\mathbf{L}}^\star \mathbf{1})$, so it vanishes when $\mathbf{L}^\star$ is sample-free. It is also small when $m(\gamma) + \bar{\mathbf{L}}^\star$ is small, but that condition controls only the displayed $\boldsymbol{\omega}_q$ component, not the full two-vector pair.

AMSE-optimal weights

For the closed-form stacked optimisation, assume $m(\gamma) \neq 0$.

The AMSE-optimal pair $(\boldsymbol{\omega}_\gamma^{\xi, \text{AMSE}}, \boldsymbol{\omega}_q^{\xi, \text{AMSE}})$ minimises the asymptotic mean squared error

$$\text{AMSE}^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) := (\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q))^2 + \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 \quad (3.33)$$

subject to the affine constraints $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$. Unlike the quantile AMSE problem, the bias gradient is stacked: $(m(\gamma)\mathbf{B}_c, \mathbf{B}_c^\star)^\top$, and the offset $b^{\xi/Q}$ enters the first-order conditions.

Stacked notation. To state the optimum compactly, stack the two pooling weight vectors into $\mathbf{w} := (\boldsymbol{\omega}_\gamma^\top, \boldsymbol{\omega}_q^\top)^\top \in \mathbb{R}^{2m}$ and assemble the bias gradient and variance Hessian implied by (3.22)–(3.21):

$$\mathbf{h} := \begin{pmatrix} m(\gamma) \mathbf{B}_c \\ \mathbf{B}_c^\star \end{pmatrix} \in \mathbb{R}^{2m}, \quad \mathbf{Q} := \begin{pmatrix} m(\gamma)^2 \mathbf{V}_c & m(\gamma) \mathbf{D} \mathbf{V}_c \\ m(\gamma) \mathbf{D} \mathbf{V}_c & \mathbf{V}_c + \mathbf{D}^2 \mathbf{V}_c \end{pmatrix} \in \mathbb{R}^{2m \times 2m}, \quad (3.34)$$

so that $\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = \mathbf{h}^\top \mathbf{w} + b^{\xi/Q}$ and $\sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 = \mathbf{w}^\top \mathbf{Q} \mathbf{w}$. Write $\mathbf{A} \in \mathbb{R}^{2m \times 2}$ for the matrix with columns $\mathbf{1}_g := (\mathbf{1}^\top, \mathbf{0}^\top)^\top$ and $\mathbf{1}_q := (\mathbf{0}^\top, \mathbf{1}^\top)^\top$, so that the sum-to-one constraints read $\mathbf{A}^\top \mathbf{w} = \mathbf{1}_2$.

Proposition 3.6 (AMSE-optimal weights). *Under the assumptions of Theorem 3.3 and $m(\gamma) \neq 0$, the matrix $\mathbf{M} := \mathbf{Q} + \mathbf{h}\mathbf{h}^\top$ is positive definite, and the AMSE objective (3.33) is strictly convex on the affine constraint set $\{\mathbf{w} \in \mathbb{R}^{2m} : \mathbf{A}^\top \mathbf{w} = \mathbf{1}_2\}$. The unique minimiser admits the closed form*

$$\mathbf{w}^{\xi, \text{AMSE}} = \mathbf{M}^{-1} \left[\mathbf{A} (\mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A})^{-1} (\mathbf{1}_2 + b^{\xi/Q} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{h}) - b^{\xi/Q} \mathbf{h} \right], \quad (3.35)$$

and is equivalently characterised by the first-order conditions

$$\begin{aligned} m(\gamma)^2 \mathbf{V}_c \boldsymbol{\omega}_\gamma + m(\gamma) \mathbf{D} \mathbf{V}_c \boldsymbol{\omega}_q &= \tfrac{1}{2} \lambda_\gamma \mathbf{1} - m(\gamma) \mu^\xi \mathbf{B}_c, \\ m(\gamma) \mathbf{D} \mathbf{V}_c \boldsymbol{\omega}_\gamma + (\mathbf{V}_c + \mathbf{D}^2 \mathbf{V}_c) \boldsymbol{\omega}_q &= \tfrac{1}{2} \lambda_q \mathbf{1} - \mu^\xi \mathbf{B}_c^\star, \end{aligned} \quad (3.36)$$

together with the sum-to-one constraints $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$ and the self-referential identity $\mu^\xi = m(\gamma) \boldsymbol{\omega}_\gamma^\top \mathbf{B}_c + \boldsymbol{\omega}_q^\top \mathbf{B}_c^\star + b^{\xi/Q}$. The Lagrange multipliers $\lambda_\gamma, \lambda_q \in \mathbb{R}$ are uniquely determined by these constraints.

Proof. See Appendix A.6. □

Equal-fraction AMSE collapse. Under $k_j/n_j = (k/n)(1 + o(1))$, $\mathbf{L}^* = L^*\mathbf{1}$. The compatibility identity following (3.18) gives

$$\sqrt{c_j S} \lambda_j = \Lambda^\bullet (e^{L^*})^{-\rho}, \quad S = \sum_{i=1}^m c_i^{-1},$$

which is sample-free, and therefore

$$\mathbf{B}_c = \beta_H \mathbf{1}, \quad \mathbf{B}_c^* = \beta_W \mathbf{1}$$

for constants β_H, β_W independent of j . Hence, for all admissible weights,

$$\mu^\xi(\omega_\gamma, \omega_q) = m(\gamma)\beta_H + \beta_W + b^{\xi/Q},$$

which is constant on the constraint set. Thus AMSE equals variance plus a constant, and if $m(\gamma) \neq 0$ the AMSE-optimal pair is the same matched pair as in Proposition 3.4,

$$\omega_\gamma^{\xi, \text{AMSE}} = \omega_q^{\xi, \text{AMSE}} = \omega^{\text{Var}},$$

with the same $m(\gamma) = 0$ indeterminacy as in Proposition 3.4.

Comparison with the Daouia–Padoan–Stupfler AMSE optimum away from equal fractions. Outside equal fractions, the expectile AMSE optimum generally differs from Proposition 2.17(ii): the bias gradient is $(m(\gamma)\mathbf{B}_c, \mathbf{B}_c^*)^\top$, the offset $b^{\xi/Q}$ enters the self-consistent mean, and the optimum is a pair rather than a single matched vector. Exact coincidence can occur only through compatibility of the full FOC system (3.36). The robust point is that equal fractions preserve the Paper-3 weights for both variance and AMSE, while unequal effective fractions generically make AMSE target-dependent.

3.6 Estimated weights and confidence intervals

In practice the optimal weights are replaced by consistent plug-ins, as in Daouia, Padoan, et al. (2024, §2.2). This section records the needed plug-in inputs and the resulting variance-only confidence interval under undersmoothing.

Feasible plug-in weights

The three optima have increasing plug-in demands:

- (a) *Equal-fraction variance-optimal weights* (Proposition 3.4). Under $k_j/n_j = (k/n)(1 + o(1))$, the optimum is the effective-sample-size weight

$$\hat{\omega}_j^{\text{Var}} = \frac{k_j}{k}, \quad j = 1, \dots, m. \quad (3.37)$$

The interval below uses (3.37) by default.

- (b) *General variance-optimal weights* (Proposition 3.5). Away from equal fractions, the closed forms also require $L_{j,n} \rightarrow L_j^*$ and $\hat{m} \xrightarrow{\mathbb{P}} m(\gamma)$. We seed $\hat{m}$ from the effective-sample-size Hill pool,

$$\hat{m}^{(0)} := m\left(\sum_{j=1}^m (k_j/k) \hat{\gamma}_j(k_j)\right), \quad \hat{m}^{(0)} \xrightarrow{\mathbb{P}} m(\gamma), \quad (3.38)$$

which is consistent under Theorem 2.6 and avoids self-reference. The formulas requiring $m(\gamma) \neq 0$ are used away from the root of $m(\cdot)$; near the root one uses the reduced $m(\gamma) = 0$ problem.

- (c) *AMSE-optimal weights* (Proposition 3.6). The AMSE formula additionally requires consistent estimators of the second-order bias quantities

$$\widehat{B}_c \xrightarrow{\mathbb{P}} B_c, \quad \widehat{B}_c^* \xrightarrow{\mathbb{P}} B_c^*, \quad \hat{b}^{\xi/Q} \xrightarrow{\mathbb{P}} b^{\xi/Q}. \quad (3.39)$$

These convergences require an additional second-order working model, such as the Hall-type parametrisation used by Daouia, Padoan, et al. (2024, §2.2); they do not follow from $\mathcal{C}_2(\gamma, \rho, A)$ alone. Without such a model, AMSE weights are best read as oracle or sensitivity-analysis weights.

By continuity of the closed forms on their nondegenerate regions, the plug-in convergences above give consistent weights and a consistent plug-in variance through (3.21). The formulas are normalised to satisfy the affine constraints exactly, which is the property used in Corollary 3.7.

Practical recommendation. Because second-order estimators are often unstable, the variance-optimal plug-ins are the conservative default unless the application supplies a reliable calibration of ρ and A .

Confidence interval under undersmoothing

The *undersmoothing calibration*

$$\sqrt{k_j} A(n_j/k_j) \rightarrow 0 \quad \text{for every } j = 1, \dots, m, \quad (3.40)$$

forces $B_c = B_c^* = \mathbf{0}$, $\Lambda^\bullet = 0$ and $b^{\xi/Q} = 0$, hence $\mu^\xi = 0$ for every admissible weight pair. This is the calibration under which the variance-only interval below is valid.

Corollary 3.7 (Asymptotic confidence interval for ξ_{τ_n}). *Assume the setting of Theorem 3.3 together with the undersmoothing condition (3.40), and let $(\hat{\omega}_\gamma, \hat{\omega}_q)$ be estimated weight pairs satisfying the affine sum-to-one constraints $\hat{\omega}_\gamma^\top \mathbf{1} = \hat{\omega}_q^\top \mathbf{1} = 1$ exactly and the consistency conditions*

$$\hat{\omega}_\gamma \xrightarrow{\mathbb{P}} \omega_\gamma, \quad \hat{\omega}_q \xrightarrow{\mathbb{P}} \omega_q, \quad \widehat{V}_c \xrightarrow{\mathbb{P}} V_c, \quad \hat{m} \xrightarrow{\mathbb{P}} m(\gamma), \quad (3.41)$$

for some deterministic limit pair $(\omega_\gamma, \omega_q)$ on the affine constraint set. By the right-heavy-tailed setup and the expectile-quantile equivalence, $\xi_{\tau_n} > 0$ eventually; the local Weissman factors, and hence the pooled estimator, are positive with probability tending one. Thus the logarithms below are well defined on an event whose probability tends to one. Then, for every $\alpha \in (0, 1)$, the

random log-scale interval

$$I_n^{\log} := \log \hat{\xi}_{\tau_n}^{\text{pool}}(\hat{\omega}_\gamma, \hat{\omega}_q) \pm z_{1-\alpha/2} \frac{\hat{\sigma}^\xi(\hat{\omega}_\gamma, \hat{\omega}_q)}{\sqrt{k}} \quad (3.42)$$

and the associated original-scale interval

$$I_n := \exp(I_n^{\log}) = [\exp(a_n), \exp(b_n)] \quad \text{when } I_n^{\log} = [a_n, b_n],$$

are asymptotic $(1 - \alpha)$ confidence intervals for $\log \xi_{\tau_n}$ and ξ_{τ_n} respectively:

$$\mathbb{P}(\log \xi_{\tau_n} \in I_n^{\log}) \rightarrow 1 - \alpha, \quad \mathbb{P}(\xi_{\tau_n} \in I_n) \rightarrow 1 - \alpha,$$

as $n \rightarrow \infty$, where $z_{1-\alpha/2}$ is the standard-normal quantile of order $1 - \alpha/2$ and $\hat{\sigma}^\xi$ is the plug-in version of (3.21).

Proof. See [Appendix A.7](#). □

Practical remarks. The log-scale interval (3.42) is symmetric in $\log \xi_{\tau_n}$; its exponential is the original-scale risk measure interval. Under equal fractions, the variance-optimal weights are matched. With $\hat{\omega}^k = (k_1/k, \dots, k_m/k)^\top$ of (3.37), the finite-sample plug-in variance is the direct substitution

$$\hat{\sigma}^\xi(\hat{\omega}^k, \hat{\omega}^k)^2 = \sum_{j=1}^m (\hat{\omega}_j^k)^2 \hat{V}_{jj} [(\hat{m} + L_{j,n})^2 + 1], \quad L_{j,n} = \log(k_j/(n_j(1 - \tau_n))),$$

where $\hat{V}_{jj}$ is the j th diagonal entry of $\widehat{\mathbf{V}}_{\mathbf{c}}$.

CI validity at $m(\gamma) = 0$. The CI does not require $m(\gamma) \neq 0$; it requires only consistent affine weights and a consistent positive variance estimate. At $m(\gamma) = 0$, the equal-fraction weights (3.37) remain valid and the variance reduces to

$$\hat{\sigma}^{\xi,2} = \sum_{j=1}^m (\hat{\omega}_j^k)^2 \hat{V}_{jj} (L_{j,n}^2 + 1),$$

with the usual common- L_n simplification when the finite- n log rates coincide.

Remark 3.8 (Bias-corrected confidence interval). Without undersmoothing, the limit-bias in (3.22) need not vanish and (3.42) is not nominally centred. The log-scale bias-corrected interval is

$$I_{n,\text{bc}}^{\log} := \log \hat{\xi}_{\tau_n}^{\text{pool}}(\hat{\omega}_\gamma, \hat{\omega}_q) - \frac{\hat{\mu}^\xi(\hat{\omega}_\gamma, \hat{\omega}_q)}{\sqrt{k}} \pm z_{1-\alpha/2} \frac{\hat{\sigma}^\xi(\hat{\omega}_\gamma, \hat{\omega}_q)}{\sqrt{k}},$$

where $\hat{\mu}^\xi$ substitutes $\widehat{\mathbf{B}}_{\mathbf{c}}$, $\widehat{\mathbf{B}}_{\mathbf{c}}^*$, $\hat{b}^{\xi/Q}$ and $\hat{m}$ into (3.22). It has the same coverage argument as [Corollary 3.7](#) provided

$$\hat{\mu}^\xi(\hat{\omega}_\gamma, \hat{\omega}_q) \xrightarrow{\mathbb{P}} \mu^\xi(\omega_\gamma, \omega_q), \quad \hat{\sigma}^\xi(\hat{\omega}_\gamma, \hat{\omega}_q) \xrightarrow{\mathbb{P}} \sigma^\xi(\omega_\gamma, \omega_q) > 0,$$

with consistent affine weights. For AMSE weights, this additionally requires the second-order bias-estimator consistency in (3.39).

Thus the variance-only interval under (3.40) is the conservative default; the bias-corrected version is appropriate only when the second-order calibration is credible.

3.7 A root- k diagnostic for high-expectile homogeneity

The statistic below is a root- k diagnostic for equality of the high-expectile targets being pooled. It can reveal target-level incompatibilities, but non-rejection does not validate the full common-marginal model, just as Theorem 2.20 diagnoses tail-index homogeneity rather than equality of distributions.

Let

$$\hat{\xi}_{\tau_n, j} := \psi(\hat{\gamma}_j(k_j)) \hat{q}_j^*(\tau_n | k_j), \quad j = 1, \dots, m, \quad (3.43)$$

denote the per-sample QB intermediate-expectile estimators, and collect their logarithms into

$$\hat{\Xi}_n := (\log \hat{\xi}_{\tau_n, 1}, \dots, \log \hat{\xi}_{\tau_n, m})^\top.$$

As before, logarithms are understood on the high-probability domain.

Joint limit of the per-sample log-expectiles. Applying the row $(m(\gamma), 1)$ to (3.10) and aggregating the independent samples on the $\sqrt{k}$ scale gives

$$\sqrt{k} (\hat{\Xi}_n - \log \xi_{\tau_n} \mathbf{1}) \xrightarrow{d} \mathcal{N}(\beta, \Sigma_\Xi), \quad (3.44)$$

with diagonal covariance matrix

$$\Sigma_\Xi = \text{diag}\left([(m(\gamma) + L_j^*)^2 + 1] (\mathbf{V}_c)_{j,j}\right)_{j=1}^m, \quad (\Sigma_\Xi)_{j,j} = [(m(\gamma) + L_j^*)^2 + 1] (\mathbf{V}_c)_{j,j}, \quad (3.45)$$

and asymptotic mean vector

$$\beta = m(\gamma) \mathbf{B}_c + \mathbf{B}_c^* + b^{\xi/Q} \mathbf{1}, \quad (3.46)$$

where the quantities are those of Proposition 3.2 and Theorem 3.3. Diagonality follows from independence, and (3.46) is the per-sample analogue of (3.22); the offset $b^{\xi/Q}$ enters every coordinate equally.

The GLS deviance statistic. Under the common-marginal assumption all $\hat{\xi}_{\tau_n, j}$ estimate the same population ξ_{τ_n} , and their log-scale dispersion about the generalised-least-squares (GLS) weighted mean is captured by the deviance statistic

$$\Lambda_n^\xi := k (\hat{\Xi}_n - \hat{\nu}_n \mathbf{1})^\top \hat{\Sigma}_{\Xi, n}^{-1} (\hat{\Xi}_n - \hat{\nu}_n \mathbf{1}), \quad \hat{\nu}_n = \frac{\mathbf{1}^\top \hat{\Sigma}_{\Xi, n}^{-1} \hat{\Xi}_n}{\mathbf{1}^\top \hat{\Sigma}_{\Xi, n}^{-1} \mathbf{1}}, \quad (3.47)$$

in which $\hat{\nu}_n$ is the GLS estimator of the common log-expectile under (3.44) and $\hat{\Sigma}_{\Xi,n}$ is the plug-in image of (3.45) assembled from the consistent estimators of Section 3.6,

$$\hat{\Sigma}_{\Xi,n} = \text{diag}((\hat{m} + L_{j,n})^2 + 1)_{j=1}^m \widehat{\mathbf{V}}_c, \quad L_{j,n} = \log(k_j / (n_j(1 - \tau_n))), \quad (3.48)$$

with $\hat{m} = \hat{m}^{(0)}$ from (3.38). The GLS residual removes the direction $\mathbf{1}$, so common centring and bias components collinear with $\mathbf{1}$ do not affect the statistic.

Because (3.47) is scaled by k , the formal null behind the chi-square calibration is root- k log-expectile homogeneity. If

$$\Xi_n^* := (\log \xi_{\tau_n,1}, \dots, \log \xi_{\tau_n,m})^\top$$

denotes the vector of population per-sample log-expectiles in a diagnostic reading beyond the common-marginal model, this condition is

$$H_{0,k}^\xi: \quad \sqrt{k} \min_{c \in \mathbb{R}} \|\Xi_n^* - c \mathbf{1}\| \rightarrow 0.$$

This is automatic under the common-marginal assumption. First-order expectile equivalence is useful for interpretation, but not sufficient for a root- k chi-square calibration.

The theorem gives the general noncentral limit. The central calibration requires the additional condition $\beta \in \text{span}\{\mathbf{1}\}$, extracted in the corollary.

Theorem 3.9 (Noncentral high-expectile homogeneity diagnostic). *Adopt the assumptions of Theorem 3.3, assume $m \geq 2$, and work on the high-probability logarithmic domain described above. Suppose $\hat{\Sigma}_{\Xi,n}$ is symmetric positive definite with probability tending one and $\hat{\Sigma}_{\Xi,n} \xrightarrow{\mathbb{P}} \Sigma_\Xi$, with Σ_Ξ positive definite. Under the chapter's common-marginal working model, which implies the common-target centring $\Xi_n^* = \log \xi_{\tau_n} \mathbf{1}$, the statistic (3.47) converges to a noncentral chi-square,*

$$\Lambda_n^\xi \xrightarrow{d} \chi_{m-1}^2(\delta), \quad \delta = \beta^\top \Sigma_\Xi^{-1} \beta - \frac{(\mathbf{1}^\top \Sigma_\Xi^{-1} \beta)^2}{\mathbf{1}^\top \Sigma_\Xi^{-1} \mathbf{1}} \geq 0, \quad (3.49)$$

with β, Σ_Ξ as in (3.45)–(3.46). The noncentrality δ is the squared Σ_Ξ^{-1} -norm of β projected off the direction $\mathbf{1}$, and vanishes — so that the limit is the central χ_{m-1}^2 — exactly when $\beta \in \text{span}\{\mathbf{1}\}$. Moreover, if the statistic is used diagnostically outside the common-marginal model in a setting where $\hat{\Xi}_n - \Xi_n^* = O_p(k^{-1/2})$ and the plug-in precision matrices $\hat{\Sigma}_{\Xi,n}^{-1}$ remain bounded and uniformly positive definite in probability, then under any separated alternative for which the population log-expectiles are separated faster than $k^{-1/2}$ — formally, $\sqrt{k} \min_{c \in \mathbb{R}} \|(\log \xi_{\tau_n,j})_{j=1}^m - c \mathbf{1}\| \rightarrow \infty$ — the statistic diverges, $\Lambda_n^\xi \xrightarrow{\mathbb{P}} \infty$. This alternative-side statement is only a divergence result. It does not supply a central null calibration for heterogeneous margins satisfying root- k target equality; such a calibration would require the corresponding heterogeneous covariance and bias structure.

Proof. See Appendix A.8. □

Corollary 3.10 (Central chi-square calibration). *Under the assumptions of Theorem 3.9, the*

central chi-square rejection rule

$$\Lambda_n^\xi > \chi_{m-1, 1-\alpha}^2,$$

with $\chi_{m-1, 1-\alpha}^2$ the $(1 - \alpha)$ -quantile of the central χ_{m-1}^2 law, has asymptotic size α whenever $\beta \in \text{span}\{\mathbf{1}\}$. In particular this holds in two standing regimes of the chapter:

- (i) the undersmoothing calibration (3.40), under which $\mathbf{B}_c = \mathbf{B}_c^* = \mathbf{0}$ and $b^{\xi/Q} = 0$, hence $\beta = \mathbf{0}$;
- (ii) the equal-fraction distributed specialisation $k_j/n_j = (k/n)(1 + o(1))$ described in Section 3.1, under which every coordinate of β collapses to one common value.

If $\delta > 0$ under the common-marginal working model, the same central critical value has limiting rejection probability $\mathbb{P}\{\chi_{m-1}^2(\delta) > \chi_{m-1, 1-\alpha}^2\}$, which is larger than α .

Proof. See Appendix A.8. □

Discussion. Under the common-marginal assumption the root- k null holds automatically, so the statistic is a specification check on target-level pooling rather than a test of equality of full marginal laws. The central calibration is the same variance-dominated regime used for Corollary 3.7; outside undersmoothing or equal fractions, retained bias gives the noncentral limit in (3.49). Finally, the diagnostic is complementary to Theorem 2.20. The non-injectivity of ψ does not hide fixed unequal-tail-index alternatives because the quantile factor also changes; its distinct strength is scale sensitivity when $(\mathcal{H})$ fails and tail-index equality alone leaves the high-expectile targets unequal.

Chapter 4

Extrapolation to very-extreme expectile levels

4.1 Motivation

The intermediate-level theory of [Chapter 3](#) estimates ξ_{τ_n} when the expected number of upper-tail observations diverges. This chapter moves to the very-extreme regime

$$\tau'_n \uparrow 1, \quad n(1 - \tau'_n) \rightarrow c \in [0, \infty),$$

where intermediate-level asymptotics no longer apply directly. As in the iid theory recalled in [Section 2.4.3](#), the stable anchor remains an intermediate expectile estimator, and the target $\xi_{\tau'_n}$ is reached through a Weissman-type power factor.

The pooling protocol is unchanged: samples communicate only the Paper-3 Hill and Weissman primitives, with covariance information for weights and standard errors. Hence the available intermediate anchor is the canonical pooled QB estimator

$$\hat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)) \hat{q}_n^*(\tau_n \mid \boldsymbol{\omega}_q)$$

from [Equation \(3.8\)](#). This is a consequence of the communication protocol, not a claim that QB extrapolation dominates procedures using expectile-specific primitives.

The role of [Proposition 2.14](#), the thesis-level plug-in lemma adapted from Davison et al. (2023, Supplement, Proposition C.6), is to make this transfer modular. Once an intermediate estimator and a tail-index estimator satisfy a joint root-rate limit, the extrapolated expectile has a very-extreme limit governed only by the tail-index component. Chapter 3 supplies the required joint limit for $(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma), \hat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q))$. The rest of the chapter specifies the regime, defines the pooled lift, and reads the limiting distribution and optimal weights from the pooled Hill component.

4.2 Extrapolation regime and input limit

Throughout this chapter we keep the independent common-margin setup of [Section 3.1](#): fixed m , $n = \sum_j n_j$, $k = \sum_j k_j$, $n_j \rightarrow \infty$, $k_j \rightarrow \infty$, $k_j/n_j \rightarrow 0$, and the proportionality conditions (2.12). The common margin satisfies $\mathcal{C}_2(\gamma, \rho, A)$ with $\gamma \in (0, 1)$ and

$$\mathbb{E} X^- < \infty.$$

The bias-rate limits $\sqrt{k_j} A(n_j/k_j) \rightarrow \lambda_j \in \mathbb{R}$ hold, the intermediate anchor level satisfies

$$L_j(\tau_n) = \log \frac{k_j}{n_j(1 - \tau_n)} \longrightarrow L_j^* \in \mathbb{R}, \quad j = 1, \dots, m,$$

and the moment-negligibility condition is

$$\frac{\sqrt{k}}{Q_X(\tau_n)} \longrightarrow 0. \quad (4.1)$$

All logarithms are understood on the high-probability positivity event of [Section 3.2](#). The only strengthening relative to the intermediate chapter is

$$\rho < 0. \quad (4.2)$$

This is the strict second-order condition used in [Proposition 2.14](#) and in the pooled Weissman theorem [Theorem 2.19](#). All results below remain in the independent common-margin distributed specialisation.

The new target level is a very-extreme sequence

$$\tau'_n \uparrow 1, \quad n(1 - \tau'_n) \rightarrow c \in [0, \infty), \quad \frac{1 - \tau'_n}{1 - \tau_n} \rightarrow 0. \quad (4.3)$$

Write the associated Weissman log-rate as

$$\ell_n := \log \left(\frac{1 - \tau_n}{1 - \tau'_n} \right). \quad (4.4)$$

Then $\ell_n \rightarrow \infty$, and the extrapolation is not allowed to be so far that the Weissman log-rate overtakes the available root- k information:

$$\frac{\sqrt{k}}{\ell_n} \longrightarrow \infty. \quad (4.5)$$

The finite-log condition already implies $n(1 - \tau_n) \rightarrow \infty$, so τ_n is intermediate and τ'_n is very-extreme in the sense required by [Proposition 2.14](#). If $\alpha_j = \lim n_j/n \in (0, 1)$, then

$$\alpha_j = \frac{b_j^{-1}}{\sum_{i=1}^m b_i^{-1}}, \quad n_j(1 - \tau'_n) \rightarrow \alpha_j c.$$

Thus the global very-extreme level has the expected local interpretation.

The logarithmic rate used here is also aligned with the pooled Weissman rate in [Theorem 2.19](#). The finite-log condition and proportionality assumptions imply

$$\log \frac{k}{n(1 - \tau_n)} = O(1),$$

and hence

$$\log \frac{k}{n(1 - \tau'_n)} = \ell_n + \log \frac{k}{n(1 - \tau_n)} = \ell_n + O(1). \quad (4.6)$$

Thus the pooled-Weissman logarithmic normaliser and ℓ_n are asymptotically equivalent.

For the application of [Proposition 2.14](#) we take

$$a_n = \sqrt{k}, \quad \hat{\gamma}_n = \hat{\gamma}_n(\omega_\gamma), \quad \hat{\xi}_{\tau_n}^I = \hat{\xi}_{\tau_n}^{\text{pool}}(\omega_\gamma, \omega_q).$$

The rate assumptions above verify the remaining rate hypotheses of that proposition: $a_n \rightarrow \infty$, τ_n is intermediate, τ'_n is very-extreme, (4.5) gives $a_n/\ell_n \rightarrow \infty$, [Equation \(3.18\)](#) gives $a_n A((1 - \tau_n)^{-1}) \rightarrow \Lambda^\bullet$, and (4.1) gives $a_n/Q_X(\tau_n) \rightarrow 0$. The positivity condition holds on the log-domain event already used in [Chapter 3](#). It remains only to record the deterministic ratio control and the joint Gaussian input.

Lemma 4.1 (Deterministic very-extreme ratio control). *Under the tail and rate assumptions of this section,*

$$\frac{\sqrt{k}}{\ell_n} \{ \log \xi_{\tau'_n} - \log \xi_{\tau_n} - \gamma \ell_n \} \rightarrow 0.$$

This lemma records, in the present notation, the deterministic population step behind the finite-mean plug-in extrapolation result of Davison et al. (2023, Supplement, Proposition C.6). The proof of that proposition reduces the argument to the same log-decomposition as Daouia, Girard, et al. (2020, Theorem 5); the control is used here as a source-paper ingredient, not as a new contribution of the thesis.

Proof. See [Appendix A.9](#). □

It remains to record the exact intermediate Gaussian input that will be fed into the plug-in extrapolation lemma. Put

$$\mathbf{D}^\star := \text{diag}(\mathbf{L}^\star), \quad \mathbf{L}^\star = (L_1^\star, \dots, L_m^\star)^\top.$$

Proposition 4.2 (Joint input for the extrapolation step). *Under the assumptions stated above, for any deterministic affine weight vectors $\omega_\gamma, \omega_q \in \mathbb{R}^m$ satisfying $\omega_\gamma^\top \mathbf{1} = \omega_q^\top \mathbf{1} = 1$,*

$$\sqrt{k} \begin{pmatrix} \hat{\gamma}_n(\omega_\gamma) - \gamma \\ \hat{\xi}_{\tau_n}^{\text{pool}}(\omega_\gamma, \omega_q)/\xi_{\tau_n} - 1 \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(\begin{pmatrix} \omega_\gamma^\top \mathbf{B}_c \\ \mu^\xi(\omega_\gamma, \omega_q) \end{pmatrix}, \begin{pmatrix} \omega_\gamma^\top \mathbf{V}_c \omega_\gamma & s_{\gamma, \xi}(\omega_\gamma, \omega_q) \\ s_{\gamma, \xi}(\omega_\gamma, \omega_q) & \sigma^\xi(\omega_\gamma, \omega_q)^2 \end{pmatrix} \right), \quad (4.7)$$

where μ^ξ and σ^ξ are given in (3.22) and (3.21), and

$$s_{\gamma,\xi}(\omega_\gamma, \omega_q) = m(\gamma) \omega_\gamma^\top V_c \omega_\gamma + \omega_\gamma^\top D^* V_c \omega_q. \quad (4.8)$$

The same convergence holds with random weights $(\hat{\omega}_\gamma, \hat{\omega}_q)$ satisfying

$$\hat{\omega}_\gamma^\top \mathbf{1} = \hat{\omega}_q^\top \mathbf{1} = 1 \quad \text{exactly,} \quad \hat{\omega}_\gamma \xrightarrow{\mathbb{P}} \omega_\gamma, \quad \hat{\omega}_q \xrightarrow{\mathbb{P}} \omega_q.$$

In this estimated-weight version, the mean and covariance on the right-hand side of (4.7) are evaluated at the deterministic probability limits $(\omega_\gamma, \omega_q)$.

Proof. See [Appendix A.10](#). □

[Proposition 4.2](#) is more information than the final very-extreme theorem visibly retains. In the notation of [Proposition 2.14](#), its first component is

$$\Gamma \sim \mathcal{N}(\omega_\gamma^\top B_c, \omega_\gamma^\top V_c \omega_\gamma),$$

while its second component is the intermediate pooled-QB limit Δ . The division by the diverging ℓ_n removes Δ from the first-order extrapolated law, so the only surviving bias is $\omega_\gamma^\top B_c$.

4.3 The pooled extrapolated expectile estimator

The extrapolated estimator is obtained by applying the Weissman-type lift of [Equation \(2.10\)](#) to the pooled intermediate anchor of [Equation \(3.8\)](#). For affine weights $\omega_\gamma, \omega_q \in \mathbb{R}^m$ satisfying $\omega_\gamma^\top \mathbf{1} = \omega_q^\top \mathbf{1} = 1$, define

$$\hat{\xi}_{\tau'_n}^{\text{pool},*}(\omega_\gamma, \omega_q) := \left(\frac{1 - \tau'_n}{1 - \tau_n} \right)^{-\hat{\gamma}_n(\omega_\gamma)} \hat{\xi}_{\tau_n}^{\text{pool}}(\omega_\gamma, \omega_q). \quad (4.9)$$

Although ω_q affects the finite-sample anchor and the intermediate-level component of the joint input limit, it disappears from the first-order very-extreme CLT after normalisation by $\sqrt{k}/\ell_n$. It remains part of the definition because different choices of ω_q give different finite-sample estimators and higher-order terms. Equivalently, using [Equation \(3.8\)](#),

$$\hat{\xi}_{\tau'_n}^{\text{pool},*}(\omega_\gamma, \omega_q) = \exp\{\ell_n \hat{\gamma}_n(\omega_\gamma)\} \psi(\hat{\gamma}_n(\omega_\gamma)) \hat{q}_n^*(\tau_n \mid \omega_q),$$

where ℓ_n is the log-rate in (4.4). Thus the two primitive weights remain visible: ω_γ controls the QB factor and extrapolation power, while ω_q controls the intermediate quantile anchor. As in [Section 3.2](#), all log-scale and real-power expressions are read on the high-probability positivity event; arbitrary definitions off that event do not affect weak limits.

The construction targets the same expectile level τ'_n that appears in $\xi_{\tau'_n}$; no adjusted quantile level is introduced. The bridge $\xi_\tau/Q_X(\tau) \rightarrow \psi(\gamma)$ is same-level, and [Proposition 2.14](#) extrapolates the intermediate expectile from τ_n to τ'_n .

At matched weights, (4.9) is exactly the QB plug-in applied to the pooled Weissman quantile at the very-extreme level. Indeed, by the local Weissman identity,

$$\hat{q}_j^*(\tau'_n | k_j) = \left(\frac{1 - \tau'_n}{1 - \tau_n} \right)^{-\hat{\gamma}_j(k_j)} \hat{q}_j^*(\tau_n | k_j),$$

and therefore, for every affine ω ,

$$\hat{q}_n^*(\tau'_n | \omega) = \left(\frac{1 - \tau'_n}{1 - \tau_n} \right)^{-\hat{\gamma}_n(\omega)} \hat{q}_n^*(\tau_n | \omega). \quad (4.10)$$

Combining (4.10) with (4.9) gives the exact matched-weight identity

$$\hat{\xi}_{\tau'_n}^{\text{pool},*}(\omega, \omega) = \psi(\hat{\gamma}_n(\omega)) \hat{q}_n^*(\tau'_n | \omega). \quad (4.11)$$

Thus, on the matched diagonal, the Chapter 4 estimator is not a new third primitive: it is the same pooled-Hill / pooled-Weissman QB pipeline as in Chapter 3, evaluated after the Weissman quantile input has been extrapolated to τ'_n .

Remark 4.3 (Matched local-QB equivalence). There is also a local-QB analogue of the construction, parallel to Equation (3.7). Define the marginal extrapolated QB expectile on sample j by

$$\hat{\xi}_{\tau'_n, j}^* := \left(\frac{1 - \tau'_n}{1 - \tau_n} \right)^{-\hat{\gamma}_j(k_j)} \psi(\hat{\gamma}_j(k_j)) \hat{q}_j^*(\tau_n | k_j) = \psi(\hat{\gamma}_j(k_j)) \hat{q}_j^*(\tau'_n | k_j),$$

and geometrically pool these m scalars:

$$\hat{\xi}_{\tau'_n}^{\text{loc},*}(\omega) := \prod_{j=1}^m [\hat{\xi}_{\tau'_n, j}^*]^{\omega_j}. \quad (4.12)$$

The comparison with (4.11) reduces to the same ψ -placement difference as in Proposition 3.1:

$$\log \frac{\hat{\xi}_{\tau'_n}^{\text{pool},*}(\omega, \omega)}{\hat{\xi}_{\tau'_n}^{\text{loc},*}(\omega)} = \log \psi(\hat{\gamma}_n(\omega)) - \sum_{j=1}^m \omega_j \log \psi(\hat{\gamma}_j(k_j)),$$

so the unscaled log-ratio is $O_p(k^{-1})$, hence negligible on both the intermediate $\sqrt{k}$ scale and the extreme $\sqrt{k}/\ell_n$ scale. Thus (4.9) agrees first order with the local-QB interpretation on the matched diagonal.

The matched-weight qualification is essential. If $\omega_\gamma \neq \omega_q$, replacing (4.9) by

$$\psi(\hat{\gamma}_n(\omega_\gamma)) \hat{q}_n^*(\tau'_n | \omega_q)$$

changes the extrapolation exponent from $\hat{\gamma}_n(\omega_\gamma)$ to $\hat{\gamma}_n(\omega_q)$. The log-ratio contains

$$\ell_n \{ \hat{\gamma}_n(\omega_q) - \hat{\gamma}_n(\omega_\gamma) \}.$$

After the extreme normalisation $\sqrt{k}/\ell_n$, this contributes $\sqrt{k} \{ \hat{\gamma}_n(\omega_q) - \hat{\gamma}_n(\omega_\gamma) \}$, which is generally non-negligible.

4.4 Main extrapolation theorem

We can now state the chapter's main limit theorem. It is a direct application of the plug-in extrapolation result [Proposition 2.14](#) to the pooled intermediate input of [Proposition 4.2](#). The theorem is intentionally short: all expectile-specific work has already been done at the intermediate level, and the very-extreme scale keeps only the pooled-Hill component.

Theorem 4.4 (Pooled Weissman extrapolation for extreme expectiles). *Assume the setup and rate conditions of [Section 4.2](#); in particular, $\rho < 0$, $\sqrt{k}/Q_X(\tau_n) \rightarrow 0$, $\ell_n \rightarrow \infty$ and $\sqrt{k}/\ell_n \rightarrow \infty$. Use the positivity and log-domain convention of [Section 4.3](#), and the deterministic ratio control verified in [Lemma 4.1](#). Let $\omega_\gamma, \omega_q \in \mathbb{R}^m$ be deterministic affine weight vectors satisfying $\omega_\gamma^\top \mathbf{1} = \omega_q^\top \mathbf{1} = 1$, and define $\hat{\xi}_{\tau'_n}^{\text{pool},*}(\omega_\gamma, \omega_q)$ by [\(4.9\)](#). Then*

$$\frac{\sqrt{k}}{\ell_n} \left(\frac{\hat{\xi}_{\tau'_n}^{\text{pool},*}(\omega_\gamma, \omega_q)}{\xi_{\tau'_n}} - 1 \right) \xrightarrow{d} \mathcal{N}(\omega_\gamma^\top \mathbf{B}_c, \omega_\gamma^\top \mathbf{V}_c \omega_\gamma), \quad (4.13)$$

where $\ell_n = \log\{(1 - \tau_n)/(1 - \tau'_n)\}$.

The same conclusion holds with estimated weights $(\hat{\omega}_\gamma, \hat{\omega}_q)$, provided

$$\hat{\omega}_\gamma^\top \mathbf{1} = \hat{\omega}_q^\top \mathbf{1} = 1 \quad \text{exactly,} \quad \hat{\omega}_\gamma \xrightarrow{\mathbb{P}} \omega_\gamma, \quad \hat{\omega}_q \xrightarrow{\mathbb{P}} \omega_q.$$

Proof. See [Appendix A.11](#). □

[Theorem 4.4](#) is the expectile analogue of the pooled Weissman limit in the independent common-margin setting. The diverging logarithmic distance magnifies tail-index error, while the intermediate anchor is lower order. Thus $m(\gamma)$, $\mathbf{B}_c^*$ and $b^{\xi/Q}$ need not vanish; they simply do not survive the $\sqrt{k}/\ell_n$ normalisation.

4.5 First-order optimal Hill weights for extrapolation

After normalisation by $\sqrt{k}/\ell_n$, the intermediate anchor is first-order invisible. The limiting mean and variance are

$$\mu^{\xi,*}(\omega_\gamma) = \omega_\gamma^\top \mathbf{B}_c, \quad \sigma^{\xi,*}(\omega_\gamma)^2 = \omega_\gamma^\top \mathbf{V}_c \omega_\gamma. \quad (4.14)$$

This is the same quadratic structure as in [Proposition 2.17](#). The first-order criterion therefore identifies only the extrapolation exponent, namely ω_γ ; it does not identify the anchor weight ω_q .

As in [Proposition 2.17](#), weights are real affine weights; no non-negativity constraint is imposed. If convex weights are required, the solution is the corresponding constrained quadratic programme and need not equal the closed forms below.

Proposition 4.5 (First-order optimal Hill weights for extrapolated expectiles). *Assume the hypotheses of [Theorem 4.4](#), and suppose that $\mathbf{V}_c$ is positive definite. Among affine Hill weights*

$\omega_\gamma^\top \mathbf{1} = 1$, the variance-optimal extrapolation weight is

$$\omega_\gamma^{\xi, \text{Var}, \star} = \omega^{\text{Var}} = \frac{\mathbf{V}_c^{-1} \mathbf{1}}{\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}}, \quad (4.15)$$

and the minimum first-order variance is

$$\sigma^{\xi, \star}(\omega_\gamma^{\xi, \text{Var}, \star})^2 = \frac{1}{\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}}. \quad (4.16)$$

If the limiting bias is retained, the first-order AMSE criterion is

$$\text{AMSE}^{\xi, \star}(\omega_\gamma) = (\omega_\gamma^\top \mathbf{B}_c)^2 + \omega_\gamma^\top \mathbf{V}_c \omega_\gamma, \quad \omega_\gamma^\top \mathbf{1} = 1, \quad (4.17)$$

whose unique minimiser is

$$\omega_\gamma^{\xi, \text{AMSE}, \star} = \omega^{\text{AMSE}} = \frac{(1 + \mathbf{B}_c^\top \mathbf{V}_c^{-1} \mathbf{B}_c) \mathbf{V}_c^{-1} \mathbf{1} - (\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{B}_c) \mathbf{V}_c^{-1} \mathbf{B}_c}{(1 + \mathbf{B}_c^\top \mathbf{V}_c^{-1} \mathbf{B}_c)(\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}) - (\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{B}_c)^2}. \quad (4.18)$$

The corresponding minimum AMSE is

$$\text{AMSE}^{\xi, \star}(\omega_\gamma^{\xi, \text{AMSE}, \star}) = \frac{1 + \mathbf{B}_c^\top \mathbf{V}_c^{-1} \mathbf{B}_c}{(1 + \mathbf{B}_c^\top \mathbf{V}_c^{-1} \mathbf{B}_c)(\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}) - (\mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{B}_c)^2}. \quad (4.19)$$

The denominator in (4.18)–(4.19) is strictly positive. The quantile-anchor weight ω_q is not identified by either first-order criterion: for the variance objective, every pair $(\omega^{\text{Var}}, \nu)$ with $\nu^\top \mathbf{1} = 1$ has the same first-order variance, and for the AMSE objective the same is true of every pair $(\omega^{\text{AMSE}}, \nu)$.

Proof. See [Appendix A.12](#). □

In the independent common-distribution distributed setting used throughout the contribution chapters, $R_{j,\ell} \equiv 0$ for $j \neq \ell$ and [Equation \(2.14\)](#) gives a diagonal covariance matrix. Consequently (4.15) reduces to

$$\omega_{\gamma,j}^{\xi, \text{Var}, \star} = \left(\sum_{i=1}^m c_i^{-1} \right)^{-1} c_j^{-1} \sim \frac{k_j}{k}, \quad j = 1, \dots, m. \quad (4.20)$$

Since c_j^{-1} is proportional to the limiting effective sample size, the variance-optimal distributed rule is again effective-sample-size weighting. If $\mathbf{B}_c$ is collinear with $\mathbf{1}$ — for instance under undersmoothing, or under equal effective fractions in the common-margin regime — then $\omega_\gamma^\top \mathbf{B}_c$ is constant on the affine constraint set, so $\omega_\gamma^{\xi, \text{AMSE}, \star} = \omega_\gamma^{\xi, \text{Var}, \star}$.

This is where Chapter 4 differs from [Section 3.5](#). The intermediate AMSE contains $m(\gamma)$, $\mathbf{L}^\star$, $\mathbf{B}_c^\star$ and $b^{\xi/Q}$, whereas the very-extreme first-order AMSE contains only $\mathbf{B}_c$ and $\mathbf{V}_c$. Outside collapse regimes, Chapter 4 AMSE weights therefore need not be the Chapter 3 expectile AMSE weights.

The non-identification of ω_q is only first-order. The anchor weight still changes the finite-sample point estimator and may affect higher-order accuracy. A natural canonical representative is the

matched choice

$$\omega_q = \omega_\gamma,$$

with ω_γ chosen as in (4.15) or (4.18). This is an implementation convention, not a first-order optimality consequence. It keeps the estimator in the matched Weissman form $\psi(\hat{\gamma}_n(\omega_\gamma)) \hat{q}_n^*(\tau'_n | \omega_\gamma)$ and gives a single default for Section 4.7.

4.6 Bias-reduction

On the $\sqrt{k}/\ell_n$ scale, Theorem 4.4 retains only the pooled-Hill bias

$$\mu^{\xi,*}(\omega_\gamma) = \omega_\gamma^\top B_c.$$

The intermediate QB bias terms enter only the second coordinate of Proposition 4.2 and are divided out by the diverging ℓ_n . Thus very-extreme bias reduction is inherited directly from the tail-index correction of Daouia, Padoan, et al. (2024, §2.2).

Assume that a consistent estimator of the limiting Hill-bias vector is available,

$$\widehat{B}_c \xrightarrow{\mathbb{P}} B_c. \quad (4.21)$$

This is an additional second-order calibration; it does not follow from $\mathcal{C}_2(\gamma, \rho, A)$ alone. For any affine Hill-weight vector $\omega_\gamma^\top \mathbf{1} = 1$, define the bias-reduced pooled tail-index estimator

$$\bar{\gamma}_n(\omega_\gamma) := \hat{\gamma}_n(\omega_\gamma) - \frac{\omega_\gamma^\top \widehat{B}_c}{\sqrt{k}}. \quad (4.22)$$

The associated bias-reduced extrapolated expectile keeps the same intermediate anchor as (4.9), but replaces the extrapolation exponent by (4.22):

$$\hat{\xi}_{\tau'_n}^{\text{pool},*,\text{bc}}(\omega_\gamma, \omega_q) := \left(\frac{1 - \tau'_n}{1 - \tau_n} \right)^{-\bar{\gamma}_n(\omega_\gamma)} \hat{\xi}_{\tau_n}^{\text{pool}}(\omega_\gamma, \omega_q). \quad (4.23)$$

For estimated weights, the same formula is used after substituting $(\hat{\omega}_\gamma, \hat{\omega}_q)$ for $(\omega_\gamma, \omega_q)$. Only the exponent is corrected because only the exponent contributes at the $\sqrt{k}/\ell_n$ scale. Replacing the occurrence of $\hat{\gamma}_n(\omega_\gamma)$ inside the intermediate QB factor $\psi(\hat{\gamma}_n(\omega_\gamma))$ by $\bar{\gamma}_n(\omega_\gamma)$, on the same high-probability compact event used for Taylor expansion of $\log \psi$, would change (4.23) by a log-ratio of order $O_p(k^{-1/2})$, hence by $o_p(1)$ after multiplication by $\sqrt{k}/\ell_n$. The minimal correction above is therefore enough for the first-order theorem and introduces no new log-domain restriction.

Corollary 4.6 (Bias-reduced extrapolated expectile CLT). *Assume the hypotheses of Theorem 4.4, and assume (4.21); in particular $\ell_n \rightarrow \infty$ and $\sqrt{k}/\ell_n \rightarrow \infty$, and the positivity convention of Section 4.3 is in force. Let $\omega_\gamma, \omega_q \in \mathbb{R}^m$ be deterministic affine weight vectors satisfying*

$\omega_\gamma^\top \mathbf{1} = \omega_q^\top \mathbf{1} = 1$. Then the estimator (4.23) satisfies

$$\frac{\sqrt{k}}{\ell_n} \left(\frac{\hat{\xi}_{\tau'_n}^{\text{pool},*,\text{bc}}(\omega_\gamma, \omega_q)}{\xi_{\tau'_n}} - 1 \right) \xrightarrow{d} \mathcal{N}\left(0, \omega_\gamma^\top \mathbf{V}_c \omega_\gamma\right). \quad (4.24)$$

The same conclusion holds with estimated weights $(\hat{\omega}_\gamma, \hat{\omega}_q)$ satisfying the exact affine constraints $\hat{\omega}_\gamma^\top \mathbf{1} = \hat{\omega}_q^\top \mathbf{1} = 1$ and $\hat{\omega}_\gamma \xrightarrow{\mathbb{P}} \omega_\gamma$, $\hat{\omega}_q \xrightarrow{\mathbb{P}} \omega_q$.

Proof. See [Appendix A.13](#). □

Corollary 4.6 is deliberately first-order: $\mathbf{B}_c^*$, $b^{\xi/Q}$ and ω_q may still matter for finite-sample or higher-order behaviour.

The practical cost is the same as in Daouia, Padoan, et al. (2024): estimating $\mathbf{B}_c$ usually requires a second-order working model, such as a Hall-type form $A(t) = \gamma\beta t^\rho$, and inherits the finite-sample instability of second-order bias estimation (Daouia, Padoan, et al. 2024, Remark 4); see also the iid expectile discussion in Daouia, Girard, et al. (2018, §3.1.1).

After bias correction, the first-order objective is variance rather than AMSE, so the natural companion weight is ω^{Var} of (4.15). The AMSE weight remains valid but is not generally first-order variance-optimal. No extra first-order variance term appears from $\widehat{\mathbf{B}}_c$, since $\omega_\gamma^\top (\widehat{\mathbf{B}}_c - \mathbf{B}_c) = o_p(1)$.

4.7 Asymptotic confidence intervals

The limiting theorem is used on the log scale, as in the pooled Weissman-quantile and extreme-expectile intervals of Daouia, Padoan, et al. (2024) and Davison et al. (2023). Because the intermediate anchor is first-order invisible, the standard error is the pooled-Hill standard error multiplied by ℓ_n .

For estimated Hill weights define

$$\hat{\sigma}_{\gamma,n}^{*2} := \hat{\omega}_\gamma^\top \widehat{\mathbf{V}}_c \hat{\omega}_\gamma, \quad \ell_n = \log\left(\frac{1 - \tau_n}{1 - \tau'_n}\right). \quad (4.25)$$

The anchor weight $\hat{\omega}_q$ enters the point estimator but not the first-order standard error.

Corollary 4.7 (Variance-only confidence interval for $\xi_{\tau'_n}$). Assume the setting of [Theorem 4.4](#). Let $(\hat{\omega}_\gamma, \hat{\omega}_q)$ be estimated weights satisfying the affine constraints

$$\hat{\omega}_\gamma^\top \mathbf{1} = \hat{\omega}_q^\top \mathbf{1} = 1 \quad \text{exactly,} \quad \hat{\omega}_\gamma \xrightarrow{\mathbb{P}} \omega_\gamma, \quad \hat{\omega}_q \xrightarrow{\mathbb{P}} \omega_q,$$

for deterministic affine limits $(\omega_\gamma, \omega_q)$. Assume also that

$$\widehat{\mathbf{V}}_c \xrightarrow{\mathbb{P}} \mathbf{V}_c, \quad \omega_\gamma^\top \mathbf{V}_c \omega_\gamma > 0, \quad \hat{\sigma}_{\gamma,n}^{*2} > 0 \text{ with probability tending to one.}$$

Let $\widehat{\xi}_{\tau'_n}$ denote either

$$\widehat{\xi}_{\tau'_n}^{\text{pool},\star}(\widehat{\omega}_\gamma, \widehat{\omega}_q)$$

under the centring condition $\omega_\gamma^\top \mathbf{B}_c = 0$ — in particular under the undersmoothing calibration (3.40) — or

$$\widehat{\xi}_{\tau'_n}^{\text{pool},\star,\text{bc}}(\widehat{\omega}_\gamma, \widehat{\omega}_q)$$

under the bias-reduction assumptions of [Corollary 4.6](#). Under the positivity convention of [Section 4.3](#) and the expectile-quantile equivalence, $\widehat{\xi}_{\tau'_n}$ is positive with probability tending to one and $\xi_{\tau'_n}$ is positive eventually. Then, for every $\alpha \in (0, 1)$, with

$$r_{n,\star} := z_{1-\alpha/2} \ell_n \sqrt{\frac{\widehat{\sigma}_{\gamma,n}^{\star 2}}{k}}, \quad I_{n,\star}^{\log} := [\log \widehat{\xi}_{\tau'_n} - r_{n,\star}, \log \widehat{\xi}_{\tau'_n} + r_{n,\star}], \quad (4.26)$$

the log-scale interval $I_{n,\star}^{\log}$ and its exponential image

$$I_{n,\star} := \exp(I_{n,\star}^{\log}) = [\exp(L_{n,-}), \exp(L_{n,+})] \quad \text{when } I_{n,\star}^{\log} = [L_{n,-}, L_{n,+}],$$

are asymptotic $(1 - \alpha)$ confidence intervals for $\log \xi_{\tau'_n}$ and $\xi_{\tau'_n}$ respectively:

$$\mathbb{P}(\log \xi_{\tau'_n} \in I_{n,\star}^{\log}) \rightarrow 1 - \alpha, \quad \mathbb{P}(\xi_{\tau'_n} \in I_{n,\star}) \rightarrow 1 - \alpha.$$

Proof. See [Appendix A.14](#). □

Remark 4.8 (Bias-centred interval). If the asymptotic bias in [Theorem 4.4](#) is retained rather than removed, the variance-only interval (4.26) is not centred correctly unless $\omega_\gamma^\top \mathbf{B}_c = 0$. Suppose instead that

$$\widehat{\mathbf{B}}_c \xrightarrow{\mathbb{P}} \mathbf{B}_c, \quad \widehat{\mu}_{\gamma,n}^\star := \widehat{\omega}_\gamma^\top \widehat{\mathbf{B}}_c \xrightarrow{\mathbb{P}} \omega_\gamma^\top \mathbf{B}_c.$$

On the very-extreme scale this is the only surviving first-order bias. With the same weights and $\widehat{\mathbf{B}}_c$, centering is algebraically identical to using (4.23), since

$$\log \widehat{\xi}_{\tau'_n}^{\text{pool},\star,\text{bc}}(\widehat{\omega}_\gamma, \widehat{\omega}_q) = \log \widehat{\xi}_{\tau'_n}^{\text{pool},\star}(\widehat{\omega}_\gamma, \widehat{\omega}_q) - \widehat{\mu}_{\gamma,n}^\star \frac{\ell_n}{\sqrt{k}}.$$

Then, for every $\alpha \in (0, 1)$, using the radius $r_{n,\star}$ from (4.26), the log-scale bias-centred interval

$$C_{n,\star,\text{bcen}} := \log \widehat{\xi}_{\tau'_n}^{\text{pool},\star}(\widehat{\omega}_\gamma, \widehat{\omega}_q) - \widehat{\mu}_{\gamma,n}^\star \frac{\ell_n}{\sqrt{k}}, \quad I_{n,\star,\text{bcen}}^{\log} := [C_{n,\star,\text{bcen}} - r_{n,\star}, C_{n,\star,\text{bcen}} + r_{n,\star}] \quad (4.27)$$

has the same asymptotic coverage as (4.26), and so does its exponential image. Feasibility again depends on consistent Hill-bias estimation, usually via a second-order working model.

The width of (4.26) is

$$2z_{1-\alpha/2} \ell_n \sqrt{\widehat{\omega}_\gamma^\top \widehat{\mathbf{V}}_c \widehat{\omega}_\gamma / k}.$$

Thus the width grows linearly with the extrapolation distance and shrinks with the pooled Hill root- k information. The standard error does not involve $m(\gamma)$, $\mathbf{L}^\star$, $\mathbf{B}_c^\star$ or $b^{\xi/Q}$, and there is no

special restriction at $m(\gamma) = 0$.

In the distributed common-margin case, the variance-dominated default is effective-sample-size Hill weighting (4.20) with the matched convention $\hat{\omega}_q = \hat{\omega}_\gamma$. Under undersmoothing, or whenever $\omega_\gamma^\top \mathbf{B}_c = 0$, this gives the variance-only interval without estimating $\mathbf{B}_c$. Otherwise one needs either (4.23) or the bias-centred interval (4.27), both of which require consistent Hill-bias estimation.

Chapter 5

Finite-sample performance

5.1 Purpose and scope

The asymptotic results of [Chapters 3 and 4](#) are first-order statements. They identify the limiting distribution of the pooled QB expectile estimator, the weight vectors that minimise the leading variance or AMSE, and the log-scale confidence intervals that follow from the corresponding Gaussian approximations. A finite-sample study is useful for a different question: whether those first-order rules already behave sensibly at sample sizes comparable to the distributed extreme-value examples of Daouia, Padoan, et al. ([2024](#)).

The study in this chapter is simulation-only. It deliberately remains inside the thesis setting: observations are iid within each machine and independent across machines, all machines have the same marginal distribution, and only the communicated primitives

$$(\hat{\gamma}_j(k_j), \hat{q}_j^*(\cdot \mid k_j)), \quad j = 1, \dots, m,$$

are used by the pooled procedures. No local LAWS expectile, raw observation, time-series feature, or multivariate observation is introduced. This keeps the numerical experiment aligned with the distributed-inference specialisation of [Section 2.5.6](#), rather than the more general tail-dependent pooling framework of Daouia, Padoan, et al. ([2024](#), §2).

5.2 Simulation design

The simulation has two targets. The intermediate target is ξ_{τ_n} with $\tau_n = 0.99$. The very-extreme target is $\xi_{\tau'_n}$ with $\tau'_n = 1 - 1/n$, estimated by the Weissman-style extrapolation of [Chapter 4](#) from the intermediate anchor. The true expectiles are computed by deterministic root solving of

$$\tau \mathbb{E}[(X - x)_+] = (1 - \tau) \mathbb{E}[(x - X)_+],$$

using numerical integration of the distribution function and survival function of the data-generating model.

Six heavy-tailed marginal distributions are used. The first four lie in the finite-variance range $\gamma < 1/2$, where comparison with direct LAWS theory is meaningful. The final two are finite-mean but infinite-variance stress cases with $\gamma = 0.6$; they are included to illustrate the QB estimator in the wider range allowed by the theory of [Theorem 2.12](#).

Model	Parameters	Tail index
Pareto	survival $x^{-1/\gamma}$, $x \geq 1$	$\gamma = 0.25$
Fréchet	$F(x) = \exp(-x^{-1/\gamma})$	$\gamma = 0.25$
Burr	$F(x) = 1 - (1 + x^{-\rho/\gamma})^{1/\rho}$, $\rho = -1$	$\gamma = 0.25$
Absolute Student	$X = T_3 $	$\gamma = 1/3$
ParetoHeavy	survival $x^{-1/\gamma}$, $x \geq 1$	$\gamma = 0.6$
BurrHeavy	$F(x) = 1 - (1 + x^{-\rho/\gamma})^{1/\rho}$, $\rho = -1$	$\gamma = 0.6$

All six distributions are nonnegative, so the lower-tail integrability condition is automatic. The two heavier cases deliberately fall outside the direct-LAWS finite-variance regime while remaining inside the finite-mean QB regime $0 < \gamma < 1$.

The machine dimension is $m \in \{5, 10, 20\}$. For each m , the final grid includes balanced, moderately unbalanced and strongly unbalanced sample allocations. The main effective-sample rule is the equal-fraction choice $k_j/n_j = k/n$ with $k/n = 0.05$, the regime in which the variance-optimal expectile weights collapse to the Daouia–Padoan–Stupfler distributed weights. A sensitivity block varies k/n over $\{0.02, 0.05, 0.10, 0.15\}$ in the strongly unbalanced regime, and a separate stress block keeps k_j equal across machines under strongly unbalanced n_j to make the unequal-fraction AMSE effects visible. These fixed threshold fractions are finite-sample tuning choices. They are not intended as a numerical verification of the asymptotic rate conditions $k \rightarrow \infty$ and $k/n \rightarrow 0$, but as a controlled comparison of the proposed pooling rules under thresholds commonly used in simulation studies.

5.3 Estimators and metrics

For each generated dataset, the local Hill and Weissman primitives are first computed on every machine. The feasible pooled estimators are:

- (i) the naive pooled QB estimator with equal weights;
- (ii) the variance-weighted estimator with $\omega_j^{\text{Var}} = k_j/k$;
- (iii) the plug-in AMSE estimator obtained by inserting estimated second-order quantities into the Chapter 3 stacked AMSE formula.

The plug-in AMSE row is exploratory. It follows the implementation strategy of Daouia, Padoan, et al. (2024, §2.2 and §3.2): when the R package `evt0` is available, the simulation scripts attempt to obtain second-order estimates from its MOP routines and insert them into the bias vector. This feasibility does not follow from the abstract $\mathcal{C}_2(\gamma, \rho, A)$ condition alone; it relies on a working second-order calibration. Because such estimation is often unstable in finite

samples, two diagnostic comparators are also reported: an oracle AMSE estimator, using the known second-order constants of the data-generating model, and an infeasible full-sample QB benchmark computed from the combined raw sample.

Performance is evaluated on the log-relative-error scale

$$\log(\widehat{\xi}/\xi),$$

which matches the log-scale form of the confidence intervals in [Sections 3.6](#) and [4.7](#). For each estimator and target the scripts report bias, variance, MSE, empirical coverage of nominal 95% confidence intervals, average confidence-interval length, the frequency with which AMSE weights are negative, and the average maximum absolute AMSE weight. For the heavier DGPs, a finite-sample Hill estimate may occasionally fall outside the QB domain $(0, 1)$. The scripts do not truncate such estimates. They record the corresponding estimator as invalid, compute log-error and coverage summaries over valid replications only, and report the invalid-domain rate separately.

5.4 Reproducibility

The simulation code lives in the repository under `simulation/`. The command

```
Rscript simulation/run_simulation.R -pilot
```

runs a 300-replication debugging grid. The command

```
Rscript simulation/run_simulation.R -final
```

runs the 5,000-replication publication grid. Results are stored as RDS files under `simulation/results/`; rendered tables and figures are generated by

```
Rscript simulation/render_results.R -final.
```

The random seed is fixed at 20260602. If `evt0` is not installed, the feasible plug-in AMSE rows are skipped and the scripts still report the naive, variance-optimal, oracle-AMSE and full-sample benchmarks. The package `ExtremeRisks` is used only as an external validation reference for standard expectile routines when it is available; the pooled expectile estimators themselves are implemented directly from the formulas of [Chapters 3](#) and [4](#).

5.5 Generated summaries

Additional diagnostics are reported in [Appendix B](#). The domain-validity table [Table B.1](#) checks how often finite-sample Hill estimates leave the QB domain, while [Table B.2](#) records AMSE-weight instability away from the equal-fraction collapse.

Target	DGP	Estimator	Bias	MSE	Coverage
intermediate	AbsStudent3	full_sample	-0.046	0.004	0.816
intermediate	AbsStudent3	variance	-0.048	0.004	0.813
intermediate	Burr	full_sample	-0.124	0.016	0.005
intermediate	Burr	variance	-0.127	0.017	0.004
intermediate	BurrHeavy	full_sample	-0.002	0.018	0.949
intermediate	BurrHeavy	variance	-0.006	0.018	0.947
intermediate	Frechet	full_sample	-0.141	0.020	0.000
intermediate	Frechet	variance	-0.143	0.021	0.000
intermediate	Pareto	full_sample	-0.156	0.025	0.000
intermediate	Pareto	variance	-0.158	0.025	0.000
intermediate	ParetoHeavy	full_sample	-0.056	0.019	0.895
intermediate	ParetoHeavy	variance	-0.062	0.020	0.888
very_extreme	AbsStudent3	full_sample	0.189	0.050	0.376
very_extreme	AbsStudent3	variance	0.190	0.050	0.381
very_extreme	Burr	full_sample	-0.008	0.006	0.829
very_extreme	Burr	variance	-0.009	0.006	0.823
very_extreme	BurrHeavy	full_sample	0.112	0.080	0.643
very_extreme	BurrHeavy	variance	0.108	0.079	0.651
very_extreme	Frechet	full_sample	-0.028	0.006	0.799
very_extreme	Frechet	variance	-0.030	0.006	0.800
very_extreme	Pareto	full_sample	-0.045	0.007	0.741
very_extreme	Pareto	variance	-0.047	0.008	0.730
very_extreme	ParetoHeavy	full_sample	0.004	0.064	0.665
very_extreme	ParetoHeavy	variance	-0.002	0.064	0.664

Table 5.1: Core Monte Carlo summary for the variance-weighted distributed estimator and the infeasible full-sample QB benchmark. Rows use $m = 10$, the strongly unbalanced allocation, $k/n = 0.05$, and the 5,000-replication final grid.

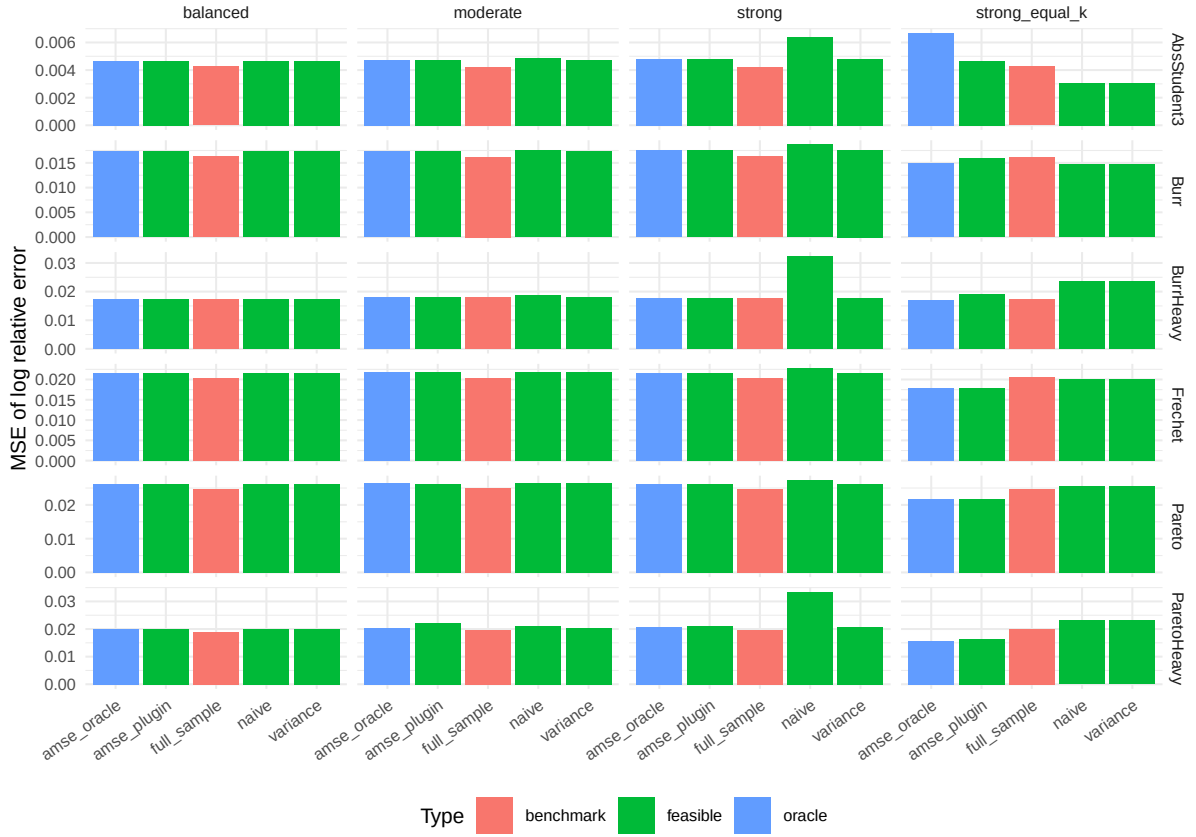
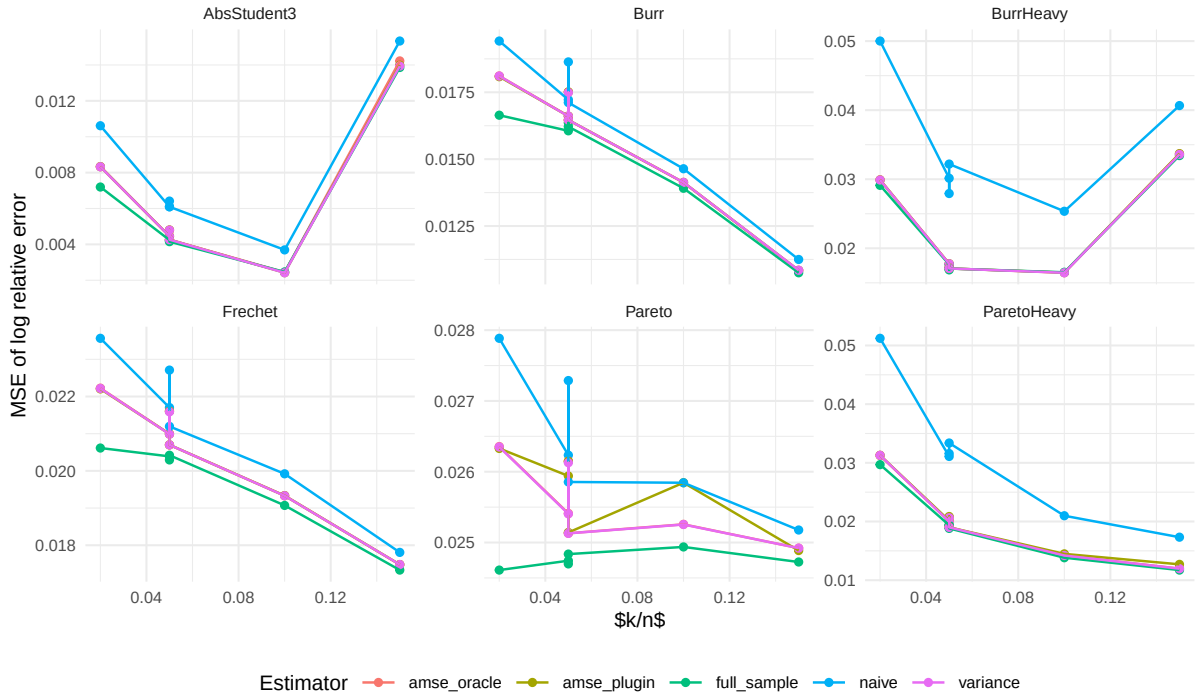


Figure 5.1: MSE of log-relative error across estimators, DGPs and sample allocation regimes.

Figure 5.2: Sensitivity of the intermediate-expectile MSE to the effective sample fraction k/n in the strongly unbalanced regime.

Interpretation. The results give a compact finite-sample check of the theory. In the equal-fraction blocks, the variance rule is almost indistinguishable from the oracle and plug-in AMSE rules; this is the numerical analogue of [Proposition 3.4](#). It also stays close to the infeasible full-sample benchmark in [Table 5.1](#), so the main practical loss is not the choice of distributed weights. The naive rule is mainly a foil: [Figure 5.1](#) shows the expected MSE penalty when unbalanced machines are averaged as if they carried equal tail information.

Coverage should be read as a first-order diagnostic, not as evidence for exact finite-sample calibration. The severe undercoverage in some intermediate finite-variance rows is driven by visible second-order bias under the fixed threshold fraction $k/n = 0.05$; the intervals used here are variance-only and not bias-centred. Finally, [Table B.2](#) shows why the feasible AMSE rule is treated as exploratory: once the equal-fraction collapse is broken, AMSE weights can become negative or large, especially in the strong-equal- k stress block.

Chapter 6

Discussion and future work

6.1 Summary of contributions

[**TODO:** Restate the contributions of Chapters 3 and 4 in the language of Section 1.5. Emphasise the modular architecture: the pooled inputs $(\hat{\gamma}_n, \hat{q}_n^*)$ of Daouia, Padoan, et al. (2024) compose cleanly with the expectile–quantile asymptotic equivalence via the plug-in CLT of Davison et al. (2023, Prop. C.6), so the only expectile-specific computation is the delta-method calculus around $\psi(\gamma) = (\gamma^{-1} - 1)^{-\gamma}$.]

6.2 Connections to Expected Shortfall

[**TODO:** For a heavy-tailed loss with index $\gamma \in (0, 1)$ there is an asymptotic relation $\xi_\tau \sim \text{ES}_{\tau'}$ for a specific matching level $\tau' = \tau'(\tau, \gamma)$ that can be expressed in closed form (cf. Daouia, Girard, et al. (2018)). Thus pooled expectile inference yields, as an immediate corollary, pooled inference on the Expected Shortfall, sidestepping the non-elicitability of ES. Sketch the level-matching identity and the corresponding plug-in estimator.]

6.3 Limitations

[**TODO:** (i) Tail-homogeneity across samples is assumed when pooling extreme quantiles geometrically (cf. assumption $(\mathcal{H})$ in Daouia, Padoan, et al. (2024, §2.3)) and the corresponding expectile-tail target should also be checked before pooling expectiles. The diagnostic of Section 3.7 is only a target-level expectile-tail-homogeneity check, not a validation of equality of the full marginal distributions.

(ii) The choice of effective sample sizes k_j is data-driven and introduces variability that the asymptotic theory does not account for.

(iii) The quantile-based route remains theoretically available for finite-mean tails $\gamma \in [1/2, 1)$, but standard direct-LAWS finite-variance CLTs no longer apply there; simulation evidence in that range should therefore be interpreted as QB stress testing rather than LAWS comparison.

(iv) The framework is iid within and across samples. Time-series dependence within a sample

and dependence across samples are excluded.]

6.4 Future work

[TODO: (i) *Time-series dependence within each sample.* Extending the framework to β -mixing observations within a sample, in the spirit of Davison et al. (2023), would require inflating the within-sample asymptotic variance by a serial-dependence factor. The pooled inference architecture itself is preserved.

(ii) *Multivariate observations.* Each sample could be d -dimensional, with the goal of pooling marginal-component-wise expectiles. The joint expectile machinery of Padoan and Stupfler (2022) provides the within-sample tools; the cross-sample step is identical to the univariate case.

(iii) *Tail-heteroscedasticity.* A simple parametric model for $q_j(\tau)/q_\ell(\tau) \rightarrow \kappa_{j,\ell}$ would allow pooling even when assumption $(\mathcal{H})$ fails.

(iv) *Empirical applications.* Distributed-inference settings in operational risk (cross-bank loss data), insurance (multi-line claims), and climate (multi-station extremes) are natural use cases.

(v) *Bias-robust pooled estimators.* Use bias-reduced tail-index estimators (e.g. corrected Hill) in place of the plain Hill estimator in the pooled construction.]

Appendix A

Proofs

This appendix collects the detailed proofs of the named results of [Chapters 2 to 4](#). Each section is self-contained and re-uses only the notation and named results of the body of the thesis.

A.1 Proof of [Proposition 2.14](#)

We prove the thesis-level plug-in CLT used in [Chapter 4](#). The argument is the log-scale core of Davison et al. ([2023](#), Proposition C.6), with the normalising sequence written as a generic a_n and the deterministic population extrapolation error imposed explicitly.

Let

$$\ell_n = \log\left(\frac{1 - \tau_n}{1 - \tau'_n}\right),$$

so $\ell_n \rightarrow \infty$ by $(1 - \tau'_n)/(1 - \tau_n) \rightarrow 0$, and $\ell_n/a_n \rightarrow 0$ by the assumption $a_n/\ell_n \rightarrow \infty$. Work on the event on which $\hat{\xi}_{\tau_n}^I > 0$; this event has probability tending one by assumption, and arbitrary definitions on its complement do not affect the weak limit. On this event,

$$\hat{\xi}_{\tau'_n}^* = \exp(\ell_n \hat{\gamma}_n) \hat{\xi}_{\tau_n}^I.$$

Define the log-relative extrapolation error

$$Z_n := \log \frac{\hat{\xi}_{\tau'_n}^*}{\hat{\xi}_{\tau_n}^I}.$$

A direct decomposition gives

$$Z_n = \ell_n(\hat{\gamma}_n - \gamma) + \log \frac{\hat{\xi}_{\tau_n}^I}{\xi_{\tau_n}} - \{\log \xi_{\tau'_n} - \log \xi_{\tau_n} - \gamma \ell_n\}. \quad (\text{A.1})$$

Multiplying [\(A.1\)](#) by a_n/ℓ_n yields

$$\frac{a_n}{\ell_n} Z_n = a_n(\hat{\gamma}_n - \gamma) + \frac{a_n}{\ell_n} \log \frac{\hat{\xi}_{\tau_n}^I}{\xi_{\tau_n}} - \frac{a_n}{\ell_n} \{\log \xi_{\tau'_n} - \log \xi_{\tau_n} - \gamma \ell_n\}.$$

The first term converges in distribution to Γ by the joint input limit (2.11). The third term converges to zero by the deterministic extrapolation-error assumption. For the middle term, the same input limit gives

$$\frac{\widehat{\xi}_{\tau_n}^I}{\xi_{\tau_n}} - 1 = O_p(a_n^{-1}) = o_p(1).$$

Therefore

$$\log \frac{\widehat{\xi}_{\tau_n}^I}{\xi_{\tau_n}} = O_p(a_n^{-1}),$$

by the Taylor expansion $\log(1+x) = x + O(x^2)$ for $x = o_p(1)$, and so

$$\frac{a_n}{\ell_n} \log \frac{\widehat{\xi}_{\tau_n}^I}{\xi_{\tau_n}} = O_p(\ell_n^{-1}) = o_p(1).$$

Slutsky's theorem gives the log-scale limit

$$\frac{a_n}{\ell_n} \log \frac{\widehat{\xi}_{\tau'_n}^{\star}}{\xi_{\tau'_n}} \xrightarrow{d} \Gamma. \quad (\text{A.2})$$

It remains only to pass from the log-relative error to the relative error in the statement of Proposition 2.14. From (A.2), $Z_n = O_p(\ell_n/a_n) = o_p(1)$. Hence

$$\exp(Z_n) - 1 - Z_n = O_p(Z_n^2), \quad \frac{a_n}{\ell_n} \{\exp(Z_n) - 1 - Z_n\} = O_p\left(\frac{\ell_n}{a_n}\right) = o_p(1).$$

Since $\exp(Z_n) - 1 = \widehat{\xi}_{\tau'_n}^{\star}/\xi_{\tau'_n} - 1$, combining this display with (A.2) proves the stated relative-error convergence. $\square$

A.2 Proof of Proposition 3.1

We prove the $\sqrt{k}$ -asymptotic equivalence of the two pooled QB intermediate-expectile estimators $\widehat{\xi}_{\tau_n}^{(A)}$ and $\widehat{\xi}_{\tau_n}^{(B)}$ introduced in Section 3.2, applied to a common, *deterministic* weight vector $\boldsymbol{\omega} \in \mathbb{R}^m$ with $\boldsymbol{\omega}^\top \mathbf{1} = 1$, the number of samples m being fixed as $n \rightarrow \infty$. The estimated-weight extension stated in the proposition is proved after the deterministic case.

Reduction to a ψ -difference. From the definitions (3.6) and (3.7),

$$\begin{aligned} \widehat{\xi}_{\tau_n}^{(A)}(\boldsymbol{\omega}, \boldsymbol{\omega}) &= \psi(\hat{\gamma}_n(\boldsymbol{\omega})) \widehat{q}_n^{\star}(\tau_n \mid \boldsymbol{\omega}), \\ \widehat{\xi}_{\tau_n}^{(B)}(\boldsymbol{\omega}) &= \left(\prod_{j=1}^m \psi(\hat{\gamma}_j(k_j))^{\omega_j} \right) \widehat{q}_n^{\star}(\tau_n \mid \boldsymbol{\omega}). \end{aligned}$$

The Weissman factor is identical in the two estimators, so the ratio reduces to a function of the Hill estimators only:

$$\log \frac{\widehat{\xi}_{\tau_n}^{(A)}(\boldsymbol{\omega}, \boldsymbol{\omega})}{\widehat{\xi}_{\tau_n}^{(B)}(\boldsymbol{\omega})} = \log \psi(\hat{\gamma}_n(\boldsymbol{\omega})) - \sum_{j=1}^m \omega_j \log \psi(\hat{\gamma}_j(k_j)). \quad (\text{A.3})$$

The proposition therefore reduces to

$$\sqrt{k} \left[\log \psi(\hat{\gamma}_n(\boldsymbol{\omega})) - \sum_j \omega_j \log \psi(\hat{\gamma}_j(k_j)) \right] \xrightarrow{\mathbb{P}} 0. \quad (\text{A.4})$$

The log-geometric Weissman factors are defined on the high-probability positivity event described after (3.5). They cancel exactly in (A.3), so no Weissman-specific stochastic term enters the remainder of this proof.

Good event supporting the Taylor expansion. The map $g \mapsto \log \psi(g) = -g \log(g^{-1} - 1)$ is defined on the open interval $(0, 1)$ and singular at its endpoints, so the Taylor expansion below is carried out on a compact subinterval of $(0, 1)$. Choose $\eta \in (0, \min\{\gamma, 1 - \gamma\})$, so that $\gamma \in (\eta, 1 - \eta)$. By Theorem 2.6, $\hat{\gamma}_j(k_j) \xrightarrow{\mathbb{P}} \gamma$ for every j . The pooled estimator also satisfies $\hat{\gamma}_n(\boldsymbol{\omega}) = \sum_j \omega_j \hat{\gamma}_j(k_j) \xrightarrow{\mathbb{P}} \gamma$, because it is a finite weighted sum of consistent terms with deterministic weights summing to one. Hence the compact good event

$$E_{n,\eta} := \{ \hat{\gamma}_1(k_1), \dots, \hat{\gamma}_m(k_m), \hat{\gamma}_n(\boldsymbol{\omega}) \in [\eta, 1 - \eta] \}$$

satisfies $\mathbb{P}(E_{n,\eta}) \rightarrow 1$. It suffices to establish the claimed $O_p(k^{-1})$ bound on $E_{n,\eta}$: modifying a sequence on an event of probability tending to zero does not affect convergence in probability or $O_p(\cdot)$ bounds. Throughout the deterministic part of the proof we therefore work on $E_{n,\eta}$ without further comment.

Taylor expansion with Lagrange remainder. The map $g \mapsto \log \psi(g) = -g \log(g^{-1} - 1)$ is C^∞ on the open interval $(0, 1)$, with first derivative $m(g) = (1 - g)^{-1} - \log(g^{-1} - 1)$ (the function introduced in (3.15)) and second derivative

$$m'(g) = \frac{1}{(1 - g)^2} + \frac{1}{1 - g} + \frac{1}{g} > 0, \quad g \in (0, 1).$$

By Taylor's theorem with Lagrange remainder, for each $j \in \{1, \dots, m\}$ there exists a random $\tilde{\gamma}_j$ lying between $\hat{\gamma}_j(k_j)$ and γ such that

$$\log \psi(\hat{\gamma}_j(k_j)) - \log \psi(\gamma) = m(\gamma) (\hat{\gamma}_j(k_j) - \gamma) + \frac{1}{2} m'(\tilde{\gamma}_j) (\hat{\gamma}_j(k_j) - \gamma)^2, \quad (\text{A.5})$$

and similarly there exists $\tilde{\gamma}_\star$ between $\hat{\gamma}_n(\boldsymbol{\omega})$ and γ with

$$\log \psi(\hat{\gamma}_n(\boldsymbol{\omega})) - \log \psi(\gamma) = m(\gamma) (\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma) + \frac{1}{2} m'(\tilde{\gamma}_\star) (\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma)^2. \quad (\text{A.6})$$

Subtract the $\boldsymbol{\omega}$ -weighted sum of (A.5) across j from (A.6). The constant terms $\log \psi(\gamma)$ cancel because $\sum_j \omega_j = 1$, and the linear terms cancel because the pooled Hill estimator is the weighted linear combination $\hat{\gamma}_n(\boldsymbol{\omega}) = \sum_j \omega_j \hat{\gamma}_j(k_j)$, which implies $\sum_j \omega_j (\hat{\gamma}_j - \gamma) = \hat{\gamma}_n(\boldsymbol{\omega}) - \gamma$. We obtain

$$\log \psi(\hat{\gamma}_n(\boldsymbol{\omega})) - \sum_j \omega_j \log \psi(\hat{\gamma}_j(k_j)) = \frac{1}{2} m'(\tilde{\gamma}_\star) (\hat{\gamma}_n(\boldsymbol{\omega}) - \gamma)^2 - \frac{1}{2} \sum_j \omega_j m'(\tilde{\gamma}_j) (\hat{\gamma}_j(k_j) - \gamma)^2. \quad (\text{A.7})$$

Order of the remainder terms. By Theorem 2.6, $\sqrt{k_j}(\hat{\gamma}_j(k_j) - \gamma) \xrightarrow{d} \mathcal{N}(\lambda_j/(1 - \rho), \gamma^2)$ for every j , hence $\hat{\gamma}_j(k_j) - \gamma = O_p(k_j^{-1/2})$. The proportionality condition (2.12), $k_1/k_j \rightarrow c_j \in (0, \infty)$, gives $k_j \asymp k$ uniformly in j , so

$$\hat{\gamma}_j(k_j) - \gamma = O_p(k^{-1/2}), \quad (\hat{\gamma}_j(k_j) - \gamma)^2 = O_p(k^{-1}), \quad j = 1, \dots, m. \quad (\text{A.8})$$

The pooled Hill estimator inherits the rate as a finite weighted combination of $\sqrt{k}$ -consistent terms:

$$\hat{\gamma}_n(\omega) - \gamma = O_p(k^{-1/2}), \quad (\hat{\gamma}_n(\omega) - \gamma)^2 = O_p(k^{-1}). \quad (\text{A.9})$$

For the Lagrange-remainder pre-factors, all intermediate points $\tilde{\gamma}_j$ and $\tilde{\gamma}_\star$ lie in $[\eta, 1 - \eta]$ on $E_{n,\eta}$. Since m' is continuous on this compact interval,

$$C_\eta := \sup_{g \in [\eta, 1 - \eta]} |m'(g)| < \infty.$$

Thus the Lagrange factors are deterministically bounded on the good event.

Inserting these orders into (A.7): each summand on the right-hand side is the product of an $O_p(1)$ factor ($m'(\tilde{\gamma}_\cdot)$) and an $O_p(k^{-1})$ factor (a squared deviation), so each is $O_p(k^{-1})$. Since the sum on the right has $m + 1$ terms and m is fixed, the right-hand side of (A.7) is itself $O_p(k^{-1})$.

Conclusion for deterministic weights. Combining the bound in (A.7) with the reduction (A.3) gives

$$\log \frac{\tilde{\xi}_{\tau_n}^{(A)}(\omega, \omega)}{\tilde{\xi}_{\tau_n}^{(B)}(\omega)} = O_p(k^{-1}),$$

hence

$$\sqrt{k} \log \frac{\tilde{\xi}_{\tau_n}^{(A)}(\omega, \omega)}{\tilde{\xi}_{\tau_n}^{(B)}(\omega)} = O_p(k^{-1/2}) = o_p(1),$$

which proves the convergence in probability claimed by Proposition 3.1. Since $e^x - 1 \sim x$ as $x \rightarrow 0$ and our log-ratio is $O_p(k^{-1})$, the continuous mapping theorem applied to exp further yields

$$\frac{\tilde{\xi}_{\tau_n}^{(A)}(\omega, \omega)}{\tilde{\xi}_{\tau_n}^{(B)}(\omega)} - 1 = O_p(k^{-1}).$$

The k^{-1} rate is one full order below the $k^{-1/2}$ scale of Theorem 3.3: the two constructions agree asymptotically not only at the CLT scale but at a strictly finer scale, so the two estimators are interchangeable to leading order in every quantity downstream of the joint Gaussian limit.

Estimated-weight extension. Let now $\hat{\omega}$ be a random weight vector satisfying $\hat{\omega} \xrightarrow{\mathbb{P}} \omega$ for a deterministic ω with $\omega^\top \mathbf{1} = 1$. Then $\|\hat{\omega}\| = O_p(1)$. If $\hat{\omega}^\top \mathbf{1} = 1$ exactly, the previous proof applies with ω replaced by $\hat{\omega}$ on the joint high-probability event where $\|\hat{\omega}\|$ is bounded and all local and pooled Hill estimators lie in $[\eta, 1 - \eta]$. The weighted Hill deviation remains $O_p(k^{-1/2})$, because

$$\left| \sum_{j=1}^m \hat{\omega}_j (\hat{\gamma}_j(k_j) - \gamma) \right| \leq \|\hat{\omega}\| \|\hat{\gamma}_n - \gamma \mathbf{1}\| = O_p(k^{-1/2}),$$

and the weighted sum of squared local deviations remains $O_p(k^{-1})$, since m is fixed, $\|\hat{\omega}\| = O_p(1)$, and each $(\hat{\gamma}_j(k_j) - \gamma)^2 = O_p(k^{-1})$. Hence the unscaled log-ratio is still $O_p(k^{-1})$.

If exact normalisation is relaxed, write $s_n = \hat{\omega}^\top \mathbf{1}$ and $f(g) = \log \psi(g)$. Taylor expansion around γ gives

$$f\left(\sum_{j=1}^m \hat{\omega}_j \hat{\gamma}_j(k_j)\right) - \sum_{j=1}^m \hat{\omega}_j f(\hat{\gamma}_j(k_j)) = (s_n - 1)\{\gamma f'(\gamma) - f(\gamma)\} + O_p(k^{-1}) + o_p(k^{-1/2}),$$

where the remainders use $\|\hat{\omega}\| = O_p(1)$, $\hat{\gamma}_j(k_j) - \gamma = O_p(k^{-1/2})$ uniformly over fixed j , and $\sqrt{k} |s_n - 1| \xrightarrow{\mathbb{P}} 0$. Multiplication by $\sqrt{k}$ therefore still yields $o_p(1)$, which proves the stated estimated-weight $\sqrt{k}$ -equivalence. $\square$

A.3 Proof of Proposition 3.2

We establish the joint $\sqrt{k}$ -Gaussian limit (3.11) of the pair $(\hat{\gamma}_n(\omega_\gamma), \log \hat{q}_n^*(\tau_n | \omega_q))$ under the standing assumptions of Section 3.1: m independent samples sharing the common marginal F and the common tail index $\gamma \in (0, 1)$, the proportionality conditions (2.12), the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ with the per-sample bias rate $\sqrt{k_j} A(n_j/k_j) \rightarrow \lambda_j \in \mathbb{R}$, an intermediate target level $\tau_n \rightarrow 1$ with $n(1 - \tau_n) \rightarrow \infty$, and the chapter's log-rate convergence $L_j(\tau_n) \rightarrow L_j^* \in \mathbb{R}$ for every j . The number of samples m is fixed.

The proof composes four steps: (i) the within-sample joint CLT for the Hill estimator and the intermediate log-order statistic; (ii) the linear lift of (i) to the (Hill, log-Weissman) pair via the exact decomposition (3.9); (iii) cross-sample independence, which renders the $2m$ -vector of all per-sample inputs jointly Gaussian with block-diagonal covariance on the common $\sqrt{k}$ scale; (iv) the linear pooling map, which collapses the $2m$ -vector onto the 2-vector (3.11) and yields the covariance (3.12).

Step (i): within-sample joint CLT for Hill and intermediate log-order statistic.

Fix $j \in \{1, \dots, m\}$ and write $\tau_{n_j} := 1 - k_j/n_j$ for the sample- j intermediate level. The joint $\sqrt{k_j}$ -asymptotics of the Hill estimator and its companion intermediate log-order statistic is a standard tail-empirical-process result (Haan and Ferreira 2006, Theorem 2.4.8 and §5.1); in the form we use it is the input to the iid QB plug-in CLT reproduced as Theorem 2.12. Under $\mathcal{C}_2(\gamma, \rho, A)$ with $k_j \rightarrow \infty$, $k_j/n_j \rightarrow 0$, and $\sqrt{k_j} A(n_j/k_j) \rightarrow \lambda_j \in \mathbb{R}$,

$$\sqrt{k_j} \begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log X_{n_j - k_j:n_j, j} - \log Q_X(\tau_{n_j}) \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(\begin{pmatrix} b_j^H \\ 0 \end{pmatrix}, \gamma^2 \mathbf{I}_2 \right), \quad (\text{A.10})$$

where $b_j^H = \lambda_j/(1 - \rho)$ is the Hill bias and $\mathbf{I}_2$ is the 2×2 identity matrix. The *zero* second component is the zero asymptotic bias of the intermediate log-order statistic on the $\sqrt{k_j}$ scale, centred at its own population quantile $\log Q_X(\tau_{n_j}) = \log U(n_j/k_j)$. In the tail-quantile-process representation, evaluating the threshold process at $s = 1$ removes the deterministic second-order term because $\Psi_\rho(1) = 0$ for the second-order kernel (3.14); hence no $A(n_j/k_j)$ -bias remains in

the second coordinate (Haan and Ferreira 2006, Theorem 2.4.8). This is in contrast to the Hill estimator, whose bias b_j^H arises precisely from averaging the second-order tail-quantile term across the top k_j order statistics, where it does not vanish.

The load-bearing structural fact in (A.10) is the *zero off-diagonal entry*. The same Brownian representation, with W_j a standard Brownian motion arising as the tail-empirical-process limit for sample j , writes the centred threshold fluctuation as $\gamma W_j(1)$ and the centred Hill fluctuation as

$$\gamma \int_0^1 \{s^{-1} W_j(s) - W_j(1)\} ds$$

plus the deterministic Hill bias. Therefore

$$\text{Cov}\left(W_j(1), \int_0^1 \{s^{-1} W_j(s) - W_j(1)\} ds\right) = \int_0^1 \{s^{-1} \text{Cov}(W_j(1), W_j(s)) - 1\} ds = 0,$$

which gives the asymptotic independence of the two random components and their common variance γ^2 . The diagonal form $\gamma^2 \mathbf{I}_2$ is therefore essential for the simple $\mathbf{V}_c$ -and-diag($\mathbf{L}^\star$) covariance structure of (3.12) used in the rest of the chapter.

Step (ii): linear lift to (Hill, log-Weissman). The Weissman decomposition (3.9), exact for every n , reads

$$\log \hat{q}_j^\star(\tau_n | k_j) - \log Q_X(\tau_n) = L_j(\tau_n) (\hat{\gamma}_j(k_j) - \gamma) + (\log X_{n_j - k_j : n_j, j} - \log Q_X(\tau_{n_j})) + r_j(\tau_n), \quad (\text{A.11})$$

where $r_j(\tau_n) := \gamma L_j(\tau_n) + \log Q_X(\tau_{n_j}) - \log Q_X(\tau_n)$ is a deterministic remainder. Identity (A.11) follows by inserting the definition $\hat{q}_j^\star(\tau_n | k_j) = (k_j / (n_j(1 - \tau_n)))^{\hat{\gamma}_j(k_j)} X_{n_j - k_j : n_j, j}$ and adding and subtracting $\gamma L_j(\tau_n)$ and $\log Q_X(\tau_{n_j})$.

The remainder $r_j(\tau_n)$ measures the deterministic gap between the population log-quantile at the chapter level τ_n and its first-order regular-variation approximation pivoted at τ_{n_j} . By the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ applied to $U = (1/\bar{F})^\leftarrow$ at base point $t = n_j/k_j$ and ratio $y = k_j/(n_j(1 - \tau_n))$, one has $\log U(ty) - \log U(t) - \gamma \log y = A(t) \Psi_\rho(y) + o(A(t))$ uniformly on compact y -sets, with

$$\Psi_\rho(y) = \begin{cases} (y^\rho - 1)/\rho, & \rho < 0, \\ \log y, & \rho = 0, \end{cases}$$

the endpoint being read in the usual second-order convention. Therefore

$$r_j(\tau_n) = -A(n_j/k_j) \Psi_\rho(k_j/(n_j(1 - \tau_n))) + o(A(n_j/k_j)), \quad (\text{A.12})$$

and the per-sample bias rate $\sqrt{k_j} A(n_j/k_j) \rightarrow \lambda_j$ together with $L_j(\tau_n) \rightarrow L_j^\star \in \mathbb{R}$ yields $\sqrt{k_j} r_j(\tau_n) \rightarrow -\lambda_j \Psi_\rho(e^{L_j^\star}) =: b_j^R$, a finite deterministic shift. Since $r_j(\tau_n)$ is deterministic, it shifts only the limiting mean and does not enter the covariance: the asymptotic covariance below is determined entirely by the two random inputs $\hat{\gamma}_j(k_j) - \gamma$ and $\log X_{n_j - k_j : n_j, j} - \log Q_X(\tau_{n_j})$ of (A.10).

Write (A.11) in matrix form,

$$\sqrt{k_j} \begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log \hat{q}_j^*(\tau_n | k_j) - \log Q_X(\tau_n) \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ L_j(\tau_n) & 1 \end{pmatrix}}_{=: \mathbf{M}_j(\tau_n)} \sqrt{k_j} \begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log X_{n_j - k_j:n_j, j} - \log Q_X(\tau_{n_j}) \end{pmatrix} + \begin{pmatrix} 0 \\ \sqrt{k_j} r_j(\tau_n) \end{pmatrix}.$$

The deterministic matrix $\mathbf{M}_j(\tau_n) \rightarrow \mathbf{M}_j^* := (1, 0; L_j^*, 1)$ entrywise, and the deterministic shift converges to $(0, b_j^R)^\top$. The continuous mapping theorem and Slutsky combine (A.10) with these deterministic convergences to deliver

$$\sqrt{k_j} \begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log \hat{q}_j^*(\tau_n | k_j) - \log Q_X(\tau_n) \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(\boldsymbol{\mu}_j^*, \gamma^2 \begin{pmatrix} 1 & L_j^* \\ L_j^* & 1 + (L_j^*)^2 \end{pmatrix} \right), \quad (\text{A.13})$$

with mean $\boldsymbol{\mu}_j^* = \mathbf{M}_j^*(b_j^H, 0)^\top + (0, b_j^R)^\top = (b_j^H, L_j^* b_j^H + b_j^R)^\top$, the zero order-statistic asymptotic bias of (A.10) carrying through. The covariance is $\mathbf{M}_j^* (\gamma^2 \mathbf{I}_2) (\mathbf{M}_j^*)^\top = \gamma^2 (1, L_j^*; L_j^*, 1 + (L_j^*)^2)$, which is precisely (3.10).

Step (iii): cross-sample independence and rescaling to $\sqrt{k}$. The m samples are independent, so the $2m$ random variables $(\hat{\gamma}_j(k_j) - \gamma, \log \hat{q}_j^*(\tau_n | k_j) - \log Q_X(\tau_n))_{j=1}^m$ are organised as m independent 2-blocks. Stacking them into the column vector

$$\mathbf{Z}_n := (\hat{\gamma}_1 - \gamma, \dots, \hat{\gamma}_m - \gamma, \log \hat{q}_1^* - \log Q_X(\tau_n), \dots, \log \hat{q}_m^* - \log Q_X(\tau_n))^\top \in \mathbb{R}^{2m}$$

(Hill components stacked first, log-Weissman components second), the joint CLT of $\sqrt{k_j} \mathbf{Z}_n^{(j)}$ from (A.13) aggregates across j under independence to a joint $2m$ -dimensional Gaussian limit. To express the limit on the common $\sqrt{k}$ scale, multiply each per-sample $\sqrt{k_j}$ input by the deterministic factor $\sqrt{k/k_j}$, which converges to a finite limit:

$$\frac{k}{k_j} = \frac{\sum_{i=1}^m k_i}{k_j} = \sum_{i=1}^m \frac{k_1/k_j}{k_1/k_i} \longrightarrow c_j \sum_{i=1}^m c_i^{-1},$$

using (2.12). The per-sample within-block covariance (A.13) scales by the factor k/k_j , so its diagonal entry $(\sqrt{k}/\sqrt{k_j})^2 \cdot \gamma^2 \rightarrow \gamma^2 c_j \sum_i c_i^{-1} = (\mathbf{V}_c)_{j,j}$, the diagonal entry of the diagonal pooled-Hill covariance (2.14) for the independent-samples specialisation. We therefore obtain

$$\sqrt{k} \mathbf{Z}_n \xrightarrow{d} \mathcal{N}(\boldsymbol{\mu}_Z, \boldsymbol{\Sigma}_Z), \quad (\text{A.14})$$

where, with $\mathbf{D}^* := \text{diag}(\mathbf{L}^*)$, the $2m \times 2m$ covariance $\boldsymbol{\Sigma}_Z$ has the block-structured form

$$\boldsymbol{\Sigma}_Z = \begin{pmatrix} \mathbf{V}_c & \mathbf{D}^* \mathbf{V}_c \\ \mathbf{V}_c \mathbf{D}^* & \mathbf{V}_c + \mathbf{D}^* \mathbf{V}_c \mathbf{D}^* \end{pmatrix}, \quad (\text{A.15})$$

expressed in $m \times m$ blocks. Each block is diagonal: the upper-left block is $\mathbf{V}_c$ itself (with diagonal entries $(\mathbf{V}_c)_{j,j} = \gamma^2 c_j \sum_i c_i^{-1}$); the off-diagonal blocks have j th diagonal entry $L_j^*(\mathbf{V}_c)_{j,j}$; the lower-right block has j th diagonal entry $(1 + (L_j^*)^2)(\mathbf{V}_c)_{j,j}$. All non-diagonal entries within each block vanish because of cross-sample independence. The mean $\boldsymbol{\mu}_Z$ is the corresponding linear functional of the per-sample bias parameters b_j^H and b_j^R , rescaled to the $\sqrt{k}$ scale: explicitly, $(\boldsymbol{\mu}_Z)_j = \sqrt{c_j \sum_i c_i^{-1}} b_j^H = (\mathbf{B}_c)_j$ for $j = 1, \dots, m$, and $(\boldsymbol{\mu}_Z)_{m+j} = \sqrt{c_j \sum_i c_i^{-1}} (L_j^* b_j^H + b_j^R) =: (\mathbf{B}_c^*)_j$, the order-statistic bias being absent by (A.10). The first half of $\boldsymbol{\mu}_Z$ is exactly the pooled-Hill bias vector $\mathbf{B}_c$ of Theorem 2.16; the second half is the pooled-Weissman bias vector $\mathbf{B}_c^*$ of (3.14), specialised to the intermediate-level target.

Step (iv): linear pooling map. By definitions (3.4) and (3.5),

$$\begin{aligned} \hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma &= \sum_{j=1}^m \omega_{\gamma,j} (\hat{\gamma}_j(k_j) - \gamma), \\ \log \hat{q}_n^*(\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n) &= \sum_{j=1}^m \omega_{q,j} (\log \hat{q}_j^* - \log Q_X(\tau_n)), \end{aligned}$$

where the second identity uses $\boldsymbol{\omega}_q^\top \mathbf{1} = 1$ so that subtracting the common term $\log Q_X(\tau_n)$ commutes with the weighted sum. Both pooled coordinates are linear in $\mathbf{Z}_n$, and the relation is exactly

$$\begin{pmatrix} \hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma \\ \log \hat{q}_n^*(\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n) \end{pmatrix} = \mathbf{A} \mathbf{Z}_n, \quad \mathbf{A} := \begin{pmatrix} \boldsymbol{\omega}_\gamma^\top & \mathbf{0}_m^\top \\ \mathbf{0}_m^\top & \boldsymbol{\omega}_q^\top \end{pmatrix} \in \mathbb{R}^{2 \times 2m}.$$

Applying $\mathbf{A}$ to (A.14),

$$\sqrt{k} \mathbf{A} \mathbf{Z}_n \xrightarrow{d} \mathcal{N}(\mathbf{A} \boldsymbol{\mu}_Z, \mathbf{A} \boldsymbol{\Sigma}_Z \mathbf{A}^\top).$$

The mean follows from the block structure of $\boldsymbol{\mu}_Z$ identified in Step (iii):

$$\mathbf{A} \boldsymbol{\mu}_Z = (\boldsymbol{\omega}_\gamma^\top \mathbf{B}_c, \boldsymbol{\omega}_q^\top \mathbf{B}_c^*)^\top,$$

which is the chapter's $\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$.

For the covariance, partition $\mathbf{A}$ along the same Hill / log-Weissman split as $\boldsymbol{\Sigma}_Z$ and compute block-by-block, using (A.15):

$$\begin{aligned} \mathbf{A} \boldsymbol{\Sigma}_Z \mathbf{A}^\top|_{1,1} &= \boldsymbol{\omega}_\gamma^\top \mathbf{V}_c \boldsymbol{\omega}_\gamma, \\ \mathbf{A} \boldsymbol{\Sigma}_Z \mathbf{A}^\top|_{1,2} &= \boldsymbol{\omega}_\gamma^\top \mathbf{D}^* \mathbf{V}_c \boldsymbol{\omega}_q, \\ \mathbf{A} \boldsymbol{\Sigma}_Z \mathbf{A}^\top|_{2,2} &= \boldsymbol{\omega}_q^\top (\mathbf{V}_c + \mathbf{D}^* \mathbf{V}_c \mathbf{D}^*) \boldsymbol{\omega}_q, \end{aligned}$$

with the $(2, 1)$ block equal to the $(1, 2)$ block by symmetry of $\boldsymbol{\Sigma}_Z$. The diagonality of $\mathbf{V}_c$ (under independent sampling, (2.14)) ensures that $\mathbf{V}_c$ and $\mathbf{D}^*$ commute, so the off-diagonal block can equivalently be written $\boldsymbol{\omega}_\gamma^\top \mathbf{V}_c \mathbf{D}^* \boldsymbol{\omega}_q$ and is the entrywise sum $\sum_j \omega_{\gamma,j} \omega_{q,j} L_j^* (\mathbf{V}_c)_{j,j}$. This recovers the covariance (3.12) of the proposition.

Combining the two displays, the pooled 2-vector converges in distribution to a Gaussian with mean $\boldsymbol{\mu}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$ and covariance $\boldsymbol{\Sigma}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$, which is precisely (3.11). $\square$

A.4 Proof of Theorem 3.3

We prove the pooled QB intermediate-expectile CLT (3.20) under the standing assumptions of Section 3.1: m independent samples with common marginal F , common tail index $\gamma \in (0, 1)$, the proportionality conditions (2.12), the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ with per-sample bias rate $\sqrt{k_j} A(n_j/k_j) \rightarrow \lambda_j \in \mathbb{R}$, an intermediate target $\tau_n \rightarrow 1$ with $n(1 - \tau_n) \rightarrow \infty$, and the log-rate convergence $L_j(\tau_n) \rightarrow L_j^* \in \mathbb{R}$ for every j . The number of samples m is fixed. We work for arbitrary deterministic weight vectors $\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q \in \mathbb{R}^m$ satisfying $\boldsymbol{\omega}_\gamma^\top \mathbf{1} = \boldsymbol{\omega}_q^\top \mathbf{1} = 1$; the extension to estimated weights is collected in Corollary 3.7.

The proof composes four steps: (i) a deterministic identity on the log scale that decomposes the expectile log-ratio into a random Hill increment, a random log-Weissman increment, and a deterministic expectile–quantile gap; (ii) a Taylor linearisation of $\log \psi$ at γ that converts the Hill increment into a linear functional of $\hat{\gamma}_n - \gamma$ up to a $\sqrt{k}$ -negligible remainder; (iii) the second-order expectile–quantile expansion that controls the deterministic gap; (iv) the multivariate delta method applied to Proposition 3.2 to read off the asymptotic mean and covariance, followed by the continuous-mapping theorem applied to $\exp$.

Step (i): the deterministic log-ratio decomposition. By the QB factorisation (3.8),

$$\log \hat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = \log \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)) + \log \hat{q}_n^*(\tau_n \mid \boldsymbol{\omega}_q).$$

Subtract $\log \xi_{\tau_n}$ from both sides and insert $\pm \log \psi(\gamma) \pm \log Q_X(\tau_n)$:

$$\begin{aligned} \log \hat{\xi}_{\tau_n}^{\text{pool}} - \log \xi_{\tau_n} &= \underbrace{[\log \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)) - \log \psi(\gamma)]}_{\text{(H)}} \\ &\quad + \underbrace{[\log \hat{q}_n^*(\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n)]}_{\text{(W)}} \\ &\quad + \underbrace{[\log \psi(\gamma) + \log Q_X(\tau_n) - \log \xi_{\tau_n}]}_{\text{(G)}}. \end{aligned} \tag{A.16}$$

The identity (A.16) is exact for every n . Terms (H) and (W) are random fluctuations around their population centering quantities; their $\sqrt{k}$ -means are those supplied by Proposition 3.2 after the linearisation in Step (ii). Term (G) is deterministic and depends on the target level τ_n only through population quantities. The sign in (G) is chosen so that the rescaled $\sqrt{k}$ (G) converges directly to $b^{\xi/Q}$ in (3.19), eliminating the bookkeeping of an intermediate minus sign in Step (iii).

Step (ii): Taylor linearisation of $\log \psi$. The argument is identical in spirit to that of Appendix A.2. The map $g \mapsto \log \psi(g) = -g \log(g^{-1} - 1)$ is C^∞ on $(0, 1)$ with first derivative

$m(g)$ defined in (3.15) and second derivative

$$m'(g) = \frac{1}{(1-g)^2} + \frac{1}{1-g} + \frac{1}{g} > 0, \quad g \in (0, 1).$$

By Theorem 2.6 and (3.2), $\hat{\gamma}_n(\omega_\gamma) \xrightarrow{\mathbb{P}} \gamma \in (0, 1)$, so the event $E_n := \{\hat{\gamma}_n(\omega_\gamma) \in (0, 1)\}$ satisfies $\mathbb{P}(E_n) \rightarrow 1$ and we may work on it without affecting convergence in distribution. On E_n , Taylor's theorem with Lagrange remainder yields a random $\tilde{\gamma}_\star$ between $\hat{\gamma}_n(\omega_\gamma)$ and γ such that

$$\log \psi(\hat{\gamma}_n(\omega_\gamma)) - \log \psi(\gamma) = m(\gamma) (\hat{\gamma}_n(\omega_\gamma) - \gamma) + \frac{1}{2} m'(\tilde{\gamma}_\star) (\hat{\gamma}_n(\omega_\gamma) - \gamma)^2. \quad (\text{A.17})$$

Proposition 3.2 gives $\hat{\gamma}_n(\omega_\gamma) - \gamma = O_p(k^{-1/2})$, hence $(\hat{\gamma}_n - \gamma)^2 = O_p(k^{-1})$. The squeeze $\tilde{\gamma}_\star \xrightarrow{\mathbb{P}} \gamma$ together with continuity of m' on $(0, 1)$ gives $m'(\tilde{\gamma}_\star) \xrightarrow{\mathbb{P}} m'(\gamma) = O_p(1)$. The Lagrange remainder in (A.17) is therefore $O_p(k^{-1}) = o_p(k^{-1/2})$, so

$$\sqrt{k}(\text{H}) = m(\gamma) \sqrt{k} (\hat{\gamma}_n(\omega_\gamma) - \gamma) + o_p(1). \quad (\text{A.18})$$

Step (iii): the deterministic expectile–quantile gap. The first-order equivalence of Proposition 2.10,

$$\xi_\tau / Q_X(\tau) \rightarrow \psi(\gamma), \quad \tau \uparrow 1,$$

sharpens, under the second-order condition $\mathcal{C}_2(\gamma, \rho, A)$ with the finite first-moment condition in (3.2), via the second-order expansion

$$\begin{aligned} \log \xi_\tau - \log \psi(\gamma) - \log Q_X(\tau) &= c(\gamma, \rho) A((1 - \tau)^{-1}) + o(A((1 - \tau)^{-1})) \\ &\quad + O(Q_X(\tau)^{-1}), \quad \tau \uparrow 1. \end{aligned} \quad (\text{A.19})$$

the logarithmic form of (2.7) of Daouia, Girard, et al. (2020), with the explicit constant $c(\gamma, \rho)$ recorded there. The additive log form follows by a Taylor expansion of $\log(1 + x)$ around zero. The first-moment term $O(Q_X(\tau)^{-1})$ is displayed explicitly because it need not be $o(A((1 - \tau)^{-1}))$ without an additional comparison of rates. Along the sequence τ_n , however, it is $\sqrt{k}$ -negligible under (3.3). Since $\sqrt{k} A((1 - \tau_n)^{-1}) = O(1)$ by the argument below, the $o(A)$ term is also $o(k^{-1/2})$. Consequently,

$$\sqrt{k} [\log \xi_{\tau_n} - \log \psi(\gamma) - \log Q_X(\tau_n)] = c(\gamma, \rho) \sqrt{k} A((1 - \tau_n)^{-1}) + o(1). \quad (\text{A.20})$$

Only the finiteness of $c(\gamma, \rho)$ for $\gamma \in (0, 1)$ and $\rho \leq 0$ is used below.

To identify the deterministic shift in (A.20), we show that

$$\sqrt{k} A((1 - \tau_n)^{-1}) \longrightarrow \Lambda^\bullet \in \mathbb{R}, \quad (\text{A.21})$$

a finite limit determined by the chapter setup. Indeed, for any fixed j , the log-rate convergence

$L_j(\tau_n) \rightarrow L_j^*$ gives the ratio

$$\frac{(1 - \tau_n)^{-1}}{n_j/k_j} = \frac{k_j}{n_j(1 - \tau_n)} = e^{L_j(\tau_n)} \rightarrow e^{L_j^*} \in (0, \infty),$$

so by [Condition 2.3](#), whose auxiliary function A is eventually of constant sign and has signed ratio limit y^ρ ,

$$\frac{A((1 - \tau_n)^{-1})}{A(n_j/k_j)} \rightarrow (e^{L_j^*})^\rho.$$

Multiplying by $\sqrt{k_j} A(n_j/k_j) \rightarrow \lambda_j$ and then by $\sqrt{k/k_j} \rightarrow \sqrt{c_j \sum_i c_i^{-1}}$, the limit [\(A.21\)](#) exists and equals

$$\Lambda^\bullet = \sqrt{c_j \sum_{i=1}^m c_i^{-1}} \lambda_j (e^{L_j^*})^\rho \quad \text{for every } j = 1, \dots, m. \quad (\text{A.22})$$

The right-hand side of [\(A.22\)](#) appears to depend on j , but the left-hand side $\sqrt{k} A((1 - \tau_n)^{-1})$ does not: hence the per- j limits must coincide whenever they exist. This is a consistency identity implicit in the chapter setup — not an additional hypothesis — expressing the compatibility of the per-sample bias rates $\sqrt{k_j} A(n_j/k_j) \rightarrow \lambda_j$ and log-rate limits $L_j(\tau_n) \rightarrow L_j^*$ across j under a single global constant-sign auxiliary function A . Inserting [\(A.21\)](#) into the sequence-level expansion [\(A.20\)](#) and using the sign convention of (G) chosen in [\(A.16\)](#),

$$\sqrt{k}(\text{G}) = \sqrt{k} [\log \psi(\gamma) + \log Q_X(\tau_n) - \log \xi_{\tau_n}] \rightarrow -c(\gamma, \rho) \Lambda^\bullet = b^{\xi/Q}, \quad (\text{A.23})$$

which is the finite deterministic shift announced in [\(3.19\)](#).

Step (iv): the delta method and the limit. Insert [\(A.18\)](#) and the trivial identity $\sqrt{k}(\text{W}) = \sqrt{k}(\text{W})$ into the $\sqrt{k}$ -rescaled version of [\(A.16\)](#), and recall [\(A.23\)](#):

$$\sqrt{k} [\log \hat{\xi}_{\tau_n}^{\text{pool}} - \log \xi_{\tau_n}] = (m(\gamma), 1) \sqrt{k} \begin{pmatrix} \hat{\gamma}_n(\omega_\gamma) - \gamma \\ \log \hat{q}_n^*(\tau_n | \omega_q) - \log Q_X(\tau_n) \end{pmatrix} + \sqrt{k}(\text{G}) + o_p(1). \quad (\text{A.24})$$

By [Proposition 3.2](#) and the continuous-mapping theorem applied to the deterministic linear functional $(g, x) \mapsto m(\gamma)g + x$, the leading term on the right of [\(A.24\)](#) converges in distribution to a univariate Gaussian with mean $(m(\gamma), 1) \mu(\omega_\gamma, \omega_q)$ and variance $(m(\gamma), 1) \Sigma(\omega_\gamma, \omega_q) (m(\gamma), 1)^\top$. Reading off from [\(3.13\)](#) and [\(3.12\)](#),

$$(m(\gamma), 1) \mu(\omega_\gamma, \omega_q) = m(\gamma) \omega_\gamma^\top \mathbf{B}_c + \omega_q^\top \mathbf{B}_c^*,$$

and the quadratic form expands to

$$\begin{aligned} (m(\gamma), 1) \Sigma(\omega_\gamma, \omega_q) (m(\gamma), 1)^\top &= m(\gamma)^2 \omega_\gamma^\top \mathbf{V}_c \omega_\gamma + 2 m(\gamma) \omega_\gamma^\top \text{diag}(\mathbf{L}^*) \mathbf{V}_c \omega_q \\ &\quad + \omega_q^\top (\mathbf{V}_c + \text{diag}(\mathbf{L}^*)^2 \mathbf{V}_c) \omega_q = \sigma^\xi(\omega_\gamma, \omega_q)^2. \end{aligned}$$

The deterministic shift $\sqrt{k}(\text{G})$ in [\(A.24\)](#) converges to $b^{\xi/Q}$ by [\(A.23\)](#). Combining via Slutsky's theorem,

$$\sqrt{k} [\log \hat{\xi}_{\tau_n}^{\text{pool}} - \log \xi_{\tau_n}] \xrightarrow{d} \mathcal{N}(\mu^\xi(\omega_\gamma, \omega_q), \sigma^\xi(\omega_\gamma, \omega_q)^2), \quad (\text{A.25})$$

with $\mu^\xi(\omega_\gamma, \omega_q) = m(\gamma) \omega_\gamma^\top \mathbf{B}_c + \omega_q^\top \mathbf{B}_c^* + b^{\xi/Q}$ as in (3.22): the random part contributes $m(\gamma) \omega_\gamma^\top \mathbf{B}_c + \omega_q^\top \mathbf{B}_c^*$ to the mean and the deterministic shift contributes $+b^{\xi/Q}$, both with the same sign as in (A.24).

To convert the log-ratio statement (A.25) into the ratio statement (3.20), observe that (A.25) forces $\sqrt{k} \log(\hat{\xi}_{\tau_n}^{\text{pool}}/\xi_{\tau_n}) = O_p(1)$, equivalently $\log(\hat{\xi}_{\tau_n}^{\text{pool}}/\xi_{\tau_n}) = o_p(1)$. A first-order Taylor expansion of $u \mapsto e^u - 1$ around $u = 0$ then gives the asymptotic relation

$$\frac{\hat{\xi}_{\tau_n}^{\text{pool}}}{\xi_{\tau_n}} - 1 = \log \frac{\hat{\xi}_{\tau_n}^{\text{pool}}}{\xi_{\tau_n}} \cdot (1 + o_p(1)), \quad (\text{A.26})$$

since the next term in the expansion is squared and thus $O_p(1/k) = o_p(1/\sqrt{k})$. Multiplying through by $\sqrt{k}$ and applying Slutsky's theorem,

$$\sqrt{k} \left(\frac{\hat{\xi}_{\tau_n}^{\text{pool}}}{\xi_{\tau_n}} - 1 \right) = \sqrt{k} \log \frac{\hat{\xi}_{\tau_n}^{\text{pool}}}{\xi_{\tau_n}} + o_p(1) \xrightarrow{d} \mathcal{N}(\mu^\xi, \sigma^{\xi,2}),$$

which is (3.20). □

A.5 Proof of Propositions 3.4 and 3.5

We prove Proposition 3.5 (the general case). The equal-fraction Proposition 3.4 is recovered as the specialisation $\mathbf{L}^* = L^* \mathbf{1}$ at the end of the section.

We minimise the asymptotic variance $\sigma^\xi(\omega_\gamma, \omega_q)^2$ of (3.21) over deterministic weight vectors $\omega_\gamma, \omega_q \in \mathbb{R}^m$ satisfying $\omega_\gamma^\top \mathbf{1} = \omega_q^\top \mathbf{1} = 1$, under the standing assumptions of Section 3.1. Recall $\mathbf{D} := \text{diag}(\mathbf{L}^*)$ and the scalars a, b, c, Δ defined in (3.28).

Step (i): the change of variables. The factorisation (3.25),

$$\sigma^\xi(\omega_\gamma, \omega_q)^2 = \mathbf{u}^\top \mathbf{V}_c \mathbf{u} + \omega_q^\top \mathbf{V}_c \omega_q, \quad \mathbf{u} := m(\gamma) \omega_\gamma + \mathbf{D} \omega_q,$$

relies only on the diagonality of $\mathbf{V}_c$ and is verified by direct expansion. For $m(\gamma) \neq 0$, the linear map $(\omega_\gamma, \omega_q) \mapsto (\mathbf{u}, \omega_q)$ is a bijection of $\mathbb{R}^m \times \mathbb{R}^m$ onto itself, with inverse $\omega_\gamma = m(\gamma)^{-1}(\mathbf{u} - \mathbf{D} \omega_q)$. The sum-to-one constraints translate accordingly: from $\omega_\gamma^\top \mathbf{1} = 1$ and $\mathbf{D} \mathbf{1} = \mathbf{L}^*$,

$$\mathbf{u}^\top \mathbf{1} - \omega_q^\top \mathbf{L}^* = m(\gamma), \quad \omega_q^\top \mathbf{1} = 1. \quad (\text{A.27})$$

The optimisation (3.25) reduces to a strictly convex quadratic programme on $\mathbb{R}^{2m}$, with the block-diagonal positive-definite Hessian $\text{diag}(\mathbf{V}_c, \mathbf{V}_c)$ and the two linear constraints (A.27).

Step (ii): the dual variables. Introduce Lagrange multipliers $\mu, \nu \in \mathbb{R}$ and form

$$\mathcal{L}(\mathbf{u}, \omega_q; \mu, \nu) = \mathbf{u}^\top \mathbf{V}_c \mathbf{u} + \omega_q^\top \mathbf{V}_c \omega_q - \mu (\mathbf{u}^\top \mathbf{1} - \omega_q^\top \mathbf{L}^* - m(\gamma)) - \nu (\omega_q^\top \mathbf{1} - 1).$$

The first-order conditions read

$$2\mathbf{V}_c \mathbf{u} = \mu \mathbf{1}, \quad 2\mathbf{V}_c \boldsymbol{\omega}_q = \nu \mathbf{1} - \mu \mathbf{L}^*. \quad (\text{A.28})$$

Both equations admit the unique solutions

$$\mathbf{u} = (\mu/2) \mathbf{V}_c^{-1} \mathbf{1}, \quad \boldsymbol{\omega}_q = (\nu/2) \mathbf{V}_c^{-1} \mathbf{1} - (\mu/2) \mathbf{V}_c^{-1} \mathbf{L}^*, \quad (\text{A.29})$$

displaying $\mathbf{u}$ in $\text{span}\{\mathbf{V}_c^{-1} \mathbf{1}\}$ and $\boldsymbol{\omega}_q$ in $\text{span}\{\mathbf{V}_c^{-1} \mathbf{1}, \mathbf{V}_c^{-1} \mathbf{L}^*\}$ at the optimum.

Step (iii): solving for the multipliers. Substitute (A.29) into the constraints (A.27). Recall $a = \mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{1}$, $b = \mathbf{1}^\top \mathbf{V}_c^{-1} \mathbf{L}^* = (\mathbf{L}^*)^\top \mathbf{V}_c^{-1} \mathbf{1}$ (using symmetry of $\mathbf{V}_c^{-1}$), and $c = (\mathbf{L}^*)^\top \mathbf{V}_c^{-1} \mathbf{L}^*$:

$$\mathbf{u}^\top \mathbf{1} = (\mu/2) a, \quad \boldsymbol{\omega}_q^\top \mathbf{L}^* = (\nu/2) b - (\mu/2) c, \quad \boldsymbol{\omega}_q^\top \mathbf{1} = (\nu/2) a - (\mu/2) b.$$

The constraints (A.27) become the linear system

$$\begin{pmatrix} a+c & -b \\ -b & a \end{pmatrix} \begin{pmatrix} \mu/2 \\ \nu/2 \end{pmatrix} = \begin{pmatrix} m(\gamma) \\ 1 \end{pmatrix}. \quad (\text{A.30})$$

The coefficient matrix has determinant $a(a+c) - b^2 = \Delta$. Cauchy–Schwarz in the $\mathbf{V}_c^{-1}$ -inner product gives $b^2 \leq ac$, with equality iff $\mathbf{L}^* \in \text{span}\{\mathbf{1}\}$, that is, iff $\mathbf{L}^* = L^* \mathbf{1}$ for some scalar L^* . Therefore $ac - b^2 \geq 0$ and, since $a > 0$,

$$\Delta = a^2 + (ac - b^2) > 0, \quad (\text{A.31})$$

strictly, with $\Delta = a^2$ exactly under the equal-effective-fraction specialisation $\mathbf{L}^* = L^* \mathbf{1}$. Cramer's rule applied to (A.30) yields

$$\mu/2 = \frac{m(\gamma) a + b}{\Delta}, \quad \nu/2 = \frac{a + c + m(\gamma) b}{\Delta}. \quad (\text{A.32})$$

Step (iv): the primal solution. Inserting (A.32) into (A.29) gives the optimal $\boldsymbol{\omega}_q$ of (3.29). For $\boldsymbol{\omega}_\gamma$, invert the change of variables: $\boldsymbol{\omega}_\gamma^* = m(\gamma)^{-1}(\mathbf{u}^* - \mathbf{D} \boldsymbol{\omega}_q^*)$. Using the diagonality identities $\mathbf{D} \mathbf{V}_c^{-1} \mathbf{1} = \mathbf{V}_c^{-1} \mathbf{L}^*$ and $\mathbf{D} \mathbf{V}_c^{-1} \mathbf{L}^* = \mathbf{V}_c^{-1} \mathbf{D}^2 \mathbf{1}$ (both consequences of $\mathbf{V}_c, \mathbf{D}$ being diagonal and hence commuting), one obtains

$$\mathbf{u}^* - \mathbf{D} \boldsymbol{\omega}_q^* = \frac{m(\gamma) a + b}{\Delta} \mathbf{V}_c^{-1} \mathbf{1} - \frac{a + c + m(\gamma) b}{\Delta} \mathbf{V}_c^{-1} \mathbf{L}^* + \frac{m(\gamma) a + b}{\Delta} \mathbf{V}_c^{-1} \mathbf{D}^2 \mathbf{1},$$

which is (3.30) after the factor $m(\gamma)^{-1}$. The minimum variance follows from Lagrangian duality: at the optimum of a strictly convex quadratic minimised under linear constraints, the optimal value equals $\frac{1}{2}(\mu^* d_1 + \nu^* d_2)$ where $d_1 = m(\gamma)$, $d_2 = 1$ are the constraint right-hand sides. Substituting (A.32):

$$\sigma^\xi(\boldsymbol{\omega}_\gamma^{\xi, \text{Var}}, \boldsymbol{\omega}_q^{\xi, \text{Var}})^2 = \frac{m(\gamma) [m(\gamma) a + b] + [a + c + m(\gamma) b]}{\Delta} = \frac{(m(\gamma)^2 + 1) a + 2 m(\gamma) b + c}{\Delta},$$

which is (3.31).

Step (v): the departure formula. A direct computation gives

$$\omega_q^{\xi, \text{Var}} - \omega^{\text{Var}} = \frac{(a + c + m(\gamma)b) \mathbf{V}_c^{-1} \mathbf{1} - (m(\gamma)a + b) \mathbf{V}_c^{-1} \mathbf{L}^*}{\Delta} - \frac{\mathbf{V}_c^{-1} \mathbf{1}}{a}.$$

Bring to a common denominator $a\Delta$ and collect the $\mathbf{V}_c^{-1} \mathbf{1}$ coefficient: $a(a + c + m(\gamma)b) - \Delta = a^2 + ac + m(\gamma)ab - a^2 - ac + b^2 = b(m(\gamma)a + b)$. Hence

$$\omega_q^{\xi, \text{Var}} - \omega^{\text{Var}} = \frac{m(\gamma)a + b}{a\Delta} [b \mathbf{V}_c^{-1} \mathbf{1} - a \mathbf{V}_c^{-1} \mathbf{L}^*] = -\frac{m(\gamma)a + b}{\Delta} \mathbf{V}_c^{-1} (\mathbf{L}^* - \bar{L}^* \mathbf{1}),$$

with $\bar{L}^* = b/a$, which is (3.32).

Specialisation to equal effective fractions. Substituting $\mathbf{L}^* = L^* \mathbf{1}$ gives $b = L^*a$, $c = (L^*)^2a$, $\Delta = a^2$. From (A.32):

$$\begin{aligned} \mu/2 &= \frac{(m(\gamma) + L^*)a}{a^2} = \frac{m(\gamma) + L^*}{a}, \\ \nu/2 &= \frac{(1 + L^{*2} + m(\gamma)L^*)a}{a^2} = \frac{1 + L^{*2} + m(\gamma)L^*}{a}, \end{aligned}$$

and $\omega_q^* = (\nu/2 - (\mu/2)L^*) \mathbf{V}_c^{-1} \mathbf{1} = ((1 + L^{*2} + m(\gamma)L^*) - L^*(m(\gamma) + L^*)) \mathbf{V}_c^{-1} \mathbf{1}/a = \mathbf{V}_c^{-1} \mathbf{1}/a = \omega^{\text{Var}}$. For the Hill-weight component, $\mathbf{u}^* = (m(\gamma) + L^*) \mathbf{V}_c^{-1} \mathbf{1}/a$ and $\mathbf{D}\omega_q^* = L^* \mathbf{V}_c^{-1} \mathbf{1}/a$, so

$$\omega_\gamma^* = m(\gamma)^{-1}(\mathbf{u}^* - \mathbf{D}\omega_q^*) = \mathbf{V}_c^{-1} \mathbf{1}/a = \omega^{\text{Var}}.$$

The minimum-variance value reduces to $[(m(\gamma) + L^*)^2 + 1]/a$, recovering Proposition 3.4.

The case $m(\gamma) = 0$. When $m(\gamma) = 0$, the change-of-variables map is no longer a bijection: $\mathbf{u} = \mathbf{D}\omega_q$ depends only on ω_q and ω_γ disappears from the variance. The factorisation reduces to $\sigma^\xi(\omega_\gamma, \omega_q)^2 = \omega_q^\top \mathbf{V}_c (\mathbf{I} + \mathbf{D}^2) \omega_q$, a strictly convex quadratic in ω_q alone (since $\mathbf{V}_c (\mathbf{I} + \mathbf{D}^2)$ is the product of two positive-definite diagonal matrices, hence positive definite). Minimisation under $\omega_q^\top \mathbf{1} = 1$ gives, by the standard Lagrangian calculation for a single linear constraint,

$$\omega_q^{\xi, \text{Var}}|_{m(\gamma)=0} = \frac{(\mathbf{I} + \mathbf{D}^2)^{-1} \mathbf{V}_c^{-1} \mathbf{1}}{\mathbf{1}^\top (\mathbf{I} + \mathbf{D}^2)^{-1} \mathbf{V}_c^{-1} \mathbf{1}}, \quad (\text{A.33})$$

with ω_γ indeterminate. In the equal-effective-fraction case $\mathbf{D} = L^* \mathbf{I}$, $(\mathbf{I} + \mathbf{D}^2)^{-1}$ is a scalar multiple of $\mathbf{I}$ and (A.33) reduces to ω^{Var} ; in general it differs from ω^{Var} . $\square$

A.6 Proof of Proposition 3.6

We prove existence, uniqueness, and the closed form (3.35) of the AMSE minimiser under the standing assumptions of Section 3.1 and $m(\gamma) \neq 0$. Recall the stacked notation $\mathbf{w} = (\omega_\gamma^\top, \omega_q^\top)^\top$, the bias gradient and variance Hessian $\mathbf{h}, \mathbf{Q}$ of (3.34), the constraint matrix $\mathbf{A} \in \mathbb{R}^{2m \times 2}$ with

columns $\mathbf{1}_g = (\mathbf{1}^\top, \mathbf{0}^\top)^\top$ and $\mathbf{1}_q = (\mathbf{0}^\top, \mathbf{1}^\top)^\top$, and the matrix $\mathbf{M} := \mathbf{Q} + \mathbf{h}\mathbf{h}^\top$.

Step (i): rewriting the AMSE as a quadratic form. By construction $\mu^\xi(\omega_\gamma, \omega_q) = \mathbf{h}^\top \mathbf{w} + b^{\xi/Q}$ and $\sigma^\xi(\omega_\gamma, \omega_q)^2 = \mathbf{w}^\top \mathbf{Q} \mathbf{w}$, so

$$\begin{aligned} \text{AMSE}^\xi(\mathbf{w}) &= (\mathbf{h}^\top \mathbf{w} + b^{\xi/Q})^2 + \mathbf{w}^\top \mathbf{Q} \mathbf{w} \\ &= \mathbf{w}^\top \mathbf{M} \mathbf{w} + 2b^{\xi/Q} \mathbf{h}^\top \mathbf{w} + (b^{\xi/Q})^2, \end{aligned}$$

recognising $(\mathbf{h}^\top \mathbf{w})^2 = \mathbf{w}^\top (\mathbf{h}\mathbf{h}^\top) \mathbf{w}$ and $\mathbf{M} = \mathbf{Q} + \mathbf{h}\mathbf{h}^\top$. The additive constant $(b^{\xi/Q})^2$ does not influence the argmin and is dropped below.

Step (ii): positive-definiteness of $\mathbf{M}$. $\mathbf{Q}$ is positive definite. Indeed, the variance factorisation $\mathbf{w}^\top \mathbf{Q} \mathbf{w} = \mathbf{u}^\top \mathbf{V}_c \mathbf{u} + \omega_q^\top \mathbf{V}_c \omega_q$ with $\mathbf{u} = m(\gamma)\omega_\gamma + \mathbf{D}\omega_q$ exhibits $\mathbf{Q}$ as the pull-back of the block-diagonal positive-definite matrix $\text{diag}(\mathbf{V}_c, \mathbf{V}_c)$ under the linear bijection $\mathbf{w} \mapsto (\mathbf{u}, \omega_q)$ (valid for $m(\gamma) \neq 0$, see [Appendix A.5](#)). Hence $\mathbf{w}^\top \mathbf{Q} \mathbf{w} \geq 0$ with equality iff $\mathbf{u} = 0$ and $\omega_q = 0$, which forces $\mathbf{w} = 0$. The rank-one positive semi-definite update $\mathbf{h}\mathbf{h}^\top$ preserves positive definiteness, so $\mathbf{M} = \mathbf{Q} + \mathbf{h}\mathbf{h}^\top$ is positive definite. Strict convexity of $\text{AMSE}^\xi(\mathbf{w}) - (b^{\xi/Q})^2$ on $\mathbb{R}^{2m}$, and a fortiori on the closed affine constraint set $\{\mathbf{A}^\top \mathbf{w} = \mathbf{1}_2\}$, follows. Positive definiteness also makes the quadratic objective coercive, so a minimiser exists on this non-empty closed affine set.

Step (iii): existence and uniqueness via the Lagrangian. Form

$$\mathcal{L}(\mathbf{w}; \boldsymbol{\nu}) = \mathbf{w}^\top \mathbf{M} \mathbf{w} + 2b^{\xi/Q} \mathbf{h}^\top \mathbf{w} - \boldsymbol{\nu}^\top (\mathbf{A}^\top \mathbf{w} - \mathbf{1}_2),$$

with $\boldsymbol{\nu} \in \mathbb{R}^2$. The first-order conditions read

$$2\mathbf{M} \mathbf{w} + 2b^{\xi/Q} \mathbf{h} - \mathbf{A} \boldsymbol{\nu} = \mathbf{0}, \tag{A.34}$$

together with $\mathbf{A}^\top \mathbf{w} = \mathbf{1}_2$. From (A.34),

$$\mathbf{w} = \mathbf{M}^{-1} \left[\frac{1}{2} \mathbf{A} \boldsymbol{\nu} - b^{\xi/Q} \mathbf{h} \right]. \tag{A.35}$$

Imposing the constraint $\mathbf{A}^\top \mathbf{w} = \mathbf{1}_2$:

$$\frac{1}{2} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A} \boldsymbol{\nu} - b^{\xi/Q} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{h} = \mathbf{1}_2.$$

The matrix $\mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A} \in \mathbb{R}^{2 \times 2}$ is positive definite: $\mathbf{M}^{-1}$ is PD, and $\mathbf{A}$ has full column rank since its two columns $\mathbf{1}_g, \mathbf{1}_q$ have disjoint nonzero supports. Hence

$$\boldsymbol{\nu} = 2(\mathbf{A}^\top \mathbf{M}^{-1} \mathbf{A})^{-1} (\mathbf{1}_2 + b^{\xi/Q} \mathbf{A}^\top \mathbf{M}^{-1} \mathbf{h}). \tag{A.36}$$

Substituting (A.36) back into (A.35) delivers the closed form (3.35). Strict convexity of the objective on the affine constraint set together with the linear FOC (A.34) forces this $\mathbf{w}$ to be the unique constrained minimiser.

Step (iv): equivalence with the block FOC system. Expand (A.34) block-by-block. The product $M\mathbf{w} = Q\mathbf{w} + (\mathbf{h}^\top \mathbf{w}) \mathbf{h} = Q\mathbf{w} + (\mu^\xi - b^{\xi/Q})\mathbf{h}$, using $\mu^\xi = \mathbf{h}^\top \mathbf{w} + b^{\xi/Q}$. Substituting, (A.34) becomes

$$2Q\mathbf{w} + 2(\mu^\xi - b^{\xi/Q})\mathbf{h} + 2b^{\xi/Q}\mathbf{h} = A\boldsymbol{\nu} \iff Q\mathbf{w} + \mu^\xi \mathbf{h} = \frac{1}{2}A\boldsymbol{\nu}.$$

Writing $\boldsymbol{\nu} = (\lambda_\gamma, \lambda_q)^\top$ and reading off the upper and lower blocks of size m (these λ 's are Lagrange multipliers, not the second-order bias-rate constants λ_j):

$$\begin{aligned} [Q\mathbf{w}]_{1,\dots,m} &= m(\gamma)^2 V_c \boldsymbol{\omega}_\gamma + m(\gamma) D V_c \boldsymbol{\omega}_q, \\ [Q\mathbf{w}]_{m+1,\dots,2m} &= m(\gamma) D V_c \boldsymbol{\omega}_\gamma + (V_c + D^2 V_c) \boldsymbol{\omega}_q, \end{aligned}$$

$[\mathbf{h}]_{1,\dots,m} = m(\gamma) \mathbf{B}_c$, $[\mathbf{h}]_{m+1,\dots,2m} = \mathbf{B}_c^*$, $[A\boldsymbol{\nu}]_{1,\dots,m} = \lambda_\gamma \mathbf{1}$, $[A\boldsymbol{\nu}]_{m+1,\dots,2m} = \lambda_q \mathbf{1}$. The block decomposition reproduces (3.36). Uniqueness from strict convexity, combined with the consistency identity $\mu^\xi = m(\gamma) \boldsymbol{\omega}_\gamma^\top \mathbf{B}_c + \boldsymbol{\omega}_q^\top \mathbf{B}_c^* + b^{\xi/Q}$ at the minimiser, completes the equivalence. $\square$

A.7 Proof of Corollary 3.7

We prove the asymptotic-coverage statement of Corollary 3.7 under the standing assumptions of Section 3.1, the undersmoothing condition (3.40), and the consistency assumptions (3.41) on the estimated weight pair $(\hat{\boldsymbol{\omega}}_\gamma, \hat{\boldsymbol{\omega}}_q)$, the plug-in covariance $\widehat{V}_c$, and the plug-in QB Jacobian $\hat{m}$. We retain the affine sum-to-one constraints $\hat{\boldsymbol{\omega}}_\gamma^\top \mathbf{1} = \hat{\boldsymbol{\omega}}_q^\top \mathbf{1} = 1$ *exactly*, which holds for the variance- and AMSE-optimal plug-ins of Propositions 2.17, 3.5 and 3.6 by construction. All logarithms and square roots below are understood on high-probability events on which they are well defined; the right-tail setup, the expectile–quantile equivalence, and the consistency of the variance plug-in make the complements of these events negligible.

The proof composes four steps: (i) recast the conclusion of Theorem 3.3 on the log scale; (ii) lift the deterministic-weight CLT to the estimated weights via the exact sum-to-one constraint and Slutsky; (iii) substitute the plug-in variance to obtain the asymptotic pivot under undersmoothing; (iv) transport coverage to the original scale by strict monotonicity of exp.

Step (i): log-scale recasting. By the QB factorisation (3.8),

$$\log \widehat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) = \log \psi(\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)) + \log \hat{q}_n^*(\tau_n | \boldsymbol{\omega}_q). \quad (\text{A.37})$$

The population expectile is positive eventually by the right-heavy-tailed setup and Proposition 2.10. The local Weissman factors, and hence their weighted-geometric pool, are positive with probability tending one, so (A.37) is legitimate on an event whose probability tends to one. Theorem 3.3 delivers $\sqrt{k}(\widehat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)/\xi_{\tau_n} - 1) \xrightarrow{d} \mathcal{N}(\mu^\xi, \sigma^{\xi,2})$. The displayed log-CLT (A.25) in the proof of Theorem 3.3 (Appendix A.4) is the same statement on the log scale,

$$\sqrt{k}(\log \widehat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) - \log \xi_{\tau_n}) \xrightarrow{d} \mathcal{N}(\mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q), \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2). \quad (\text{A.38})$$

Under undersmoothing (3.40), every per-sample bias rate $\lambda_j = \lim \sqrt{k_j} A(n_j/k_j)$ is zero. The formulas (3.14) and Theorem 2.16 then give $\mathbf{B}_c = \mathbf{0}$ and $\mathbf{B}_c^* = \mathbf{0}$. Moreover, the compatibility identity

$$\Lambda^\bullet = \sqrt{c_j S} \lambda_j e^{\rho L_j^*}, \quad j = 1, \dots, m,$$

forces $\Lambda^\bullet = 0$, and therefore $b^{\xi/Q} = -c(\gamma, \rho) \Lambda^\bullet = 0$ by (3.19). Hence $\mu^\xi(\omega_\gamma, \omega_q) = 0$ and the limit in (A.38) reduces to $\mathcal{N}(0, \sigma^\xi(\omega_\gamma, \omega_q)^2)$.

Step (ii): estimated-weight lift. We show that replacing $(\omega_\gamma, \omega_q)$ by $(\hat{\omega}_\gamma, \hat{\omega}_q)$ in (A.37) changes the left-hand side of (A.38) by $o_p(1)$. Decompose this change into a Hill and a Weissman contribution,

$$\begin{aligned} \Delta_n^\gamma &:= \log \psi(\hat{\gamma}_n(\hat{\omega}_\gamma)) - \log \psi(\hat{\gamma}_n(\omega_\gamma)), \\ \Delta_n^q &:= \log \hat{q}_n^*(\tau_n \mid \hat{\omega}_q) - \log \hat{q}_n^*(\tau_n \mid \omega_q). \end{aligned}$$

Hill contribution. Linearity of the pooled Hill in its weights ((3.4)) gives

$$\hat{\gamma}_n(\hat{\omega}_\gamma) - \hat{\gamma}_n(\omega_\gamma) = (\hat{\omega}_\gamma - \omega_\gamma)^\top \hat{\gamma}_n = (\hat{\omega}_\gamma - \omega_\gamma)^\top (\hat{\gamma}_n - \gamma \mathbf{1}),$$

the deterministic offset $\gamma \mathbf{1}$ vanishing by the exact sum-to-one identity $(\hat{\omega}_\gamma - \omega_\gamma)^\top \mathbf{1} = 1 - 1 = 0$ on both the plug-in and the limit weight vectors. The per-sample Hill CLT (within-sample component of (3.10)) gives $\sqrt{k_j}(\hat{\gamma}_j - \gamma) = O_p(1)$, hence $\sqrt{k}(\hat{\gamma}_n - \gamma \mathbf{1}) = \sqrt{k/k_j} O_p(1) = O_p(1)$ because $k/k_j \rightarrow c_j \sum_i c_i^{-1} < \infty$ by (2.12). Therefore $\sqrt{k}(\hat{\gamma}_n - \gamma \mathbf{1}) = O_p(1)$, and the Cauchy–Schwarz inequality combined with $\hat{\omega}_\gamma - \omega_\gamma = o_p(1)$ yields

$$\sqrt{k}(\hat{\gamma}_n(\hat{\omega}_\gamma) - \hat{\gamma}_n(\omega_\gamma)) = (\hat{\omega}_\gamma - \omega_\gamma)^\top \sqrt{k}(\hat{\gamma}_n - \gamma \mathbf{1}) = o_p(1) \cdot O_p(1) = o_p(1). \quad (\text{A.39})$$

The map $\log \psi$ is continuously differentiable on $(0, 1)$ with derivative $m(\cdot)$ defined in (3.15). Let

$$E_n^\gamma = \{\hat{\gamma}_n(\omega_\gamma) \in (0, 1), \hat{\gamma}_n(\hat{\omega}_\gamma) \in (0, 1)\}.$$

By consistency, $\hat{\gamma}_n(\hat{\omega}_\gamma), \hat{\gamma}_n(\omega_\gamma) \xrightarrow{\mathbb{P}} \gamma$, so $\mathbb{P}(E_n^\gamma) \rightarrow 1$. On this event, the mean value theorem yields a random $\tilde{\gamma}_n$ between the two pooled values such that

$$\Delta_n^\gamma = m(\tilde{\gamma}_n) [\hat{\gamma}_n(\hat{\omega}_\gamma) - \hat{\gamma}_n(\omega_\gamma)].$$

Squeeze gives $\tilde{\gamma}_n \xrightarrow{\mathbb{P}} \gamma$ and, by continuity of $m(\cdot)$, $m(\tilde{\gamma}_n) \xrightarrow{\mathbb{P}} m(\gamma)$. Combining with (A.39), $\sqrt{k} \Delta_n^\gamma = O_p(1) \cdot o_p(1) = o_p(1)$.

Weissman contribution. Weighted-geometric linearity of $\log \hat{q}_n^*$ in ω_q ((3.5)) gives

$$\Delta_n^q = (\hat{\omega}_q - \omega_q)^\top (\log \hat{q}_n^* - \log Q_X(\tau_n) \mathbf{1}),$$

where the deterministic offset $\log Q_X(\tau_n) \mathbf{1}$ cancels via the sum-to-one identity $(\hat{\omega}_q - \omega_q)^\top \mathbf{1} = 0$. The within-sample Weissman component of (3.10) delivers $\sqrt{k_j}(\log \hat{q}_j^* - \log Q_X(\tau_n)) = O_p(1)$

under $L_j(\tau_n) \rightarrow L_j^* \in \mathbb{R}$, and the same $\sqrt{k/k_j} = O(1)$ rescaling promotes this to $\sqrt{k}(\log \hat{\mathbf{q}}_n^* - \log Q_X(\tau_n)\mathbf{1}) = O_p(1)$. Hence

$$\sqrt{k} \Delta_n^q = (\hat{\omega}_q - \omega_q)^\top \sqrt{k}(\log \hat{\mathbf{q}}_n^* - \log Q_X(\tau_n)\mathbf{1}) = o_p(1) \cdot O_p(1) = o_p(1). \quad (\text{A.40})$$

Assembly. Combining (A.38) (under undersmoothing, limit-mean zero) with $\sqrt{k} \Delta_n^\gamma + \sqrt{k} \Delta_n^q = o_p(1)$ and Slutsky's theorem,

$$\sqrt{k}(\log \hat{\xi}_{\tau_n}^{\text{pool}}(\hat{\omega}_\gamma, \hat{\omega}_q) - \log \xi_{\tau_n}) \xrightarrow{d} \mathcal{N}(0, \sigma^\xi(\omega_\gamma, \omega_q)^2). \quad (\text{A.41})$$

Step (iii): self-normalisation. The variance (3.21) is a polynomial in $(\mathbf{V}_c, \mathbf{L}^*, m, \omega_\gamma, \omega_q)$, and $\sigma^\xi(\omega_\gamma, \omega_q)^2 > 0$ because $\mathbf{V}_c$ is positive definite, $\omega_q^\top \mathbf{1} = 1$ forces $\omega_q \neq \mathbf{0}$, and the structurally non-negative contribution $\omega_q^\top \mathbf{V}_c \omega_q$ to (3.21) is strictly positive. The plug-in substitutes $\widehat{\mathbf{V}}_c, L_{j,n} := L_j(\tau_n), \hat{m}, \hat{\omega}_\gamma, \hat{\omega}_q$ for their population analogues. The log-rate plug-in $L_{j,n} = L_j(\tau_n) = \log(k_j/(n_j(1 - \tau_n)))$ is deterministic and converges to L_j^* by the chapter setup of Section 3.1; the remaining four convergences are the assumptions (3.41). The continuous-mapping theorem applied first to the polynomial (3.21) and then to $\sqrt{\cdot}$ on $(0, \infty)$ yields

$$\hat{\sigma}^\xi(\hat{\omega}_\gamma, \hat{\omega}_q) \xrightarrow{\mathbb{P}} \sigma^\xi(\omega_\gamma, \omega_q) > 0.$$

In particular, $\mathbb{P}\{\hat{\sigma}^\xi(\hat{\omega}_\gamma, \hat{\omega}_q) > 0\} \rightarrow 1$, so the division below is well defined with probability tending one. Slutsky's theorem applied to the ratio of (A.41) and this consistent variance estimator produces the asymptotic pivot

$$Z_n := \frac{\sqrt{k}(\log \hat{\xi}_{\tau_n}^{\text{pool}}(\hat{\omega}_\gamma, \hat{\omega}_q) - \log \xi_{\tau_n})}{\hat{\sigma}^\xi(\hat{\omega}_\gamma, \hat{\omega}_q)} \xrightarrow{d} \mathcal{N}(0, 1). \quad (\text{A.42})$$

For every $\alpha \in (0, 1)$, $\mathbb{P}(|Z_n| \leq z_{1-\alpha/2}) \rightarrow 1 - \alpha$. Inverting the inequality $|Z_n| \leq z_{1-\alpha/2}$ for $\log \xi_{\tau_n}$ recovers exactly the event $\log \xi_{\tau_n} \in I_n^{\log}$ with $I_n^{\log}$ defined in (3.42), so $\mathbb{P}(\log \xi_{\tau_n} \in I_n^{\log}) \rightarrow 1 - \alpha$.

Step (iv): transport to the original scale. The map $x \mapsto e^x$ is strictly increasing and continuous on $\mathbb{R}$, so for the random closed interval $I_n^{\log} = [a_n, b_n]$,

$$\log \xi_{\tau_n} \in [a_n, b_n] \iff \xi_{\tau_n} \in [e^{a_n}, e^{b_n}] = \exp(I_n^{\log}) = I_n.$$

The two events coincide pointwise on the underlying probability space, hence $\mathbb{P}(\xi_{\tau_n} \in I_n) = \mathbb{P}(\log \xi_{\tau_n} \in I_n^{\log}) \rightarrow 1 - \alpha$. $\square$

A.8 Proof of Theorem 3.9 and Corollary 3.10

We prove the high-expectile homogeneity diagnostic of Theorem 3.9 and the central calibration of Corollary 3.10 under the standing assumptions of Section 3.1, with $\hat{\Sigma}_{\Xi, n} \xrightarrow{\mathbb{P}} \Sigma_\Xi$ and Σ_Ξ positive definite. The argument composes six steps: (i) the per-sample QB log-expectile CLT, obtained by the single-sample specialisation of Appendix A.4; (ii) aggregation across the independent

samples and rescaling to the common $\sqrt{k}$ scale, which yields the joint limit (3.44) with covariance (3.45) and mean (3.46); (iii) the GLS algebra reducing the statistic to a quadratic form in a single asymptotically Gaussian vector; (iv) the resulting noncentral $\chi^2_{m-1}(\delta)$ limit; (v) the two regimes in which $\delta = 0$; (vi) consistency under the alternative. Throughout, $S := \sum_{i=1}^m c_i^{-1}$, so that $(\mathbf{V}_c)_{j,j} = \gamma^2 c_j S$ by (2.14) and $k/k_j \rightarrow c_j S$ by (2.12), as established in Appendix A.3.

Step (i): per-sample QB log-expectile CLT. Fix $j \in \{1, \dots, m\}$. The per-sample QB log-expectile (3.43) decomposes exactly, as in (A.16) with $m = 1$ and unit weight, into

$$\log \hat{\xi}_{\tau_n, j} - \log \xi_{\tau_n} = \underbrace{[\log \psi(\hat{\gamma}_j(k_j)) - \log \psi(\gamma)]}_{(H)_j} + \underbrace{[\log \hat{q}_j^*(\tau_n | k_j) - \log Q_X(\tau_n)]}_{(W)_j} + (G),$$

with $(G) = \log \psi(\gamma) + \log Q_X(\tau_n) - \log \xi_{\tau_n}$ the deterministic expectile–quantile gap, which is common to all samples. The Taylor linearisation of Step (ii) of Appendix A.4 gives $(H)_j = m(\gamma)(\hat{\gamma}_j(k_j) - \gamma) + O_p(k_j^{-1})$, so on the $\sqrt{k_j}$ scale

$$\sqrt{k_j} (\log \hat{\xi}_{\tau_n, j} - \log \xi_{\tau_n}) = (m(\gamma), 1) \sqrt{k_j} \begin{pmatrix} \hat{\gamma}_j(k_j) - \gamma \\ \log \hat{q}_j^*(\tau_n | k_j) - \log Q_X(\tau_n) \end{pmatrix} + \sqrt{k_j} (G) + o_p(1).$$

The within-sample 2-vector obeys the joint CLT (3.10) (proved as (A.13)), with mean $\boldsymbol{\mu}_j^* = (b_j^H, L_j^* b_j^H + b_j^R)^\top$, $b_j^R = -\lambda_j \Psi_\rho(e^{L_j^*})$ (the order-statistic bias being zero by (A.10)), and covariance $\gamma^2(1, L_j^*; L_j^*, 1 + (L_j^*)^2)$. The deterministic gap is controlled by the second-order expansion (A.19) and the regular variation of A used in (A.21). The moment remainder is $o(k_j^{-1/2})$ because $k_j \asymp k$ and (3.3) is in force. Consequently,

$$\sqrt{k_j} (G) = -c(\gamma, \rho) \sqrt{k_j} A((1 - \tau_n)^{-1}) + o(1) \rightarrow -c(\gamma, \rho) \lambda_j (e^{L_j^*})^\rho =: g_j.$$

Reading the row $(m(\gamma), 1)$ through the within-sample covariance gives the scalar variance $\gamma^2[(m(\gamma) + L_j^*)^2 + 1]$ (the computation of (3.21) at $m = 1$, unit weight), so Slutsky's theorem yields

$$\sqrt{k_j} (\log \hat{\xi}_{\tau_n, j} - \log \xi_{\tau_n}) \xrightarrow{d} \mathcal{N}(\beta_j^\circ, \gamma^2[(m(\gamma) + L_j^*)^2 + 1]), \quad \beta_j^\circ = (m(\gamma) + L_j^*) b_j^H + b_j^R + g_j. \quad (\text{A.43})$$

Step (ii): aggregation and rescaling to $\sqrt{k}$. The m samples are independent (3.1), so the scalars in (A.43) are independent across j . Joint convergence follows by the Cramér–Wold device: for any fixed vector $\mathbf{a}$, the corresponding linear combination is a sum of independent per-sample terms whose characteristic functions factor, and the marginal limits in (A.43) therefore combine into the stated Gaussian law. Thus the vector $\sqrt{k}(\hat{\boldsymbol{\Xi}}_n - \log \xi_{\tau_n} \mathbf{1})$ has independent limiting coordinates obtained by multiplying (A.43) by $\sqrt{k/k_j} \rightarrow \sqrt{c_j S}$. The j th limiting variance is

$$c_j S \gamma^2[(m(\gamma) + L_j^*)^2 + 1] = [(m(\gamma) + L_j^*)^2 + 1] (\mathbf{V}_c)_{j,j},$$

which is the j th diagonal entry of Σ_{Ξ} in (3.45); the off-diagonal entries vanish by independence, so the limiting covariance is exactly Σ_{Ξ} . The j th limiting mean is $\sqrt{c_j S} \beta_j^{\circ}$. Splitting β_j° as in (A.43) and using the $\sqrt{k}$ -rescaled bias identities of Appendix A.3,

$$\sqrt{c_j S} b_j^H = (\mathbf{B}_c)_j, \quad \sqrt{c_j S} (L_j^* b_j^H + b_j^R) = (\mathbf{B}_c^*)_j,$$

the first from Theorem 2.16 and the second from (3.14), we obtain

$$\sqrt{c_j S} \beta_j^{\circ} = \sqrt{c_j S} [m(\gamma) b_j^H + (L_j^* b_j^H + b_j^R)] + \sqrt{c_j S} g_j = m(\gamma) (\mathbf{B}_c)_j + (\mathbf{B}_c^*)_j + \sqrt{c_j S} g_j.$$

For the last term, the per- j identity (A.22) reads $\sqrt{c_j S} \lambda_j (e^{L_j^*})^{\rho} = \Lambda^{\bullet}$, so

$$\sqrt{c_j S} g_j = -c(\gamma, \rho) \sqrt{c_j S} \lambda_j (e^{L_j^*})^{\rho} = -c(\gamma, \rho) \Lambda^{\bullet} = b^{\xi/Q}$$

by (3.19), *independently of j* . Hence $\sqrt{c_j S} \beta_j^{\circ} = m(\gamma) (\mathbf{B}_c)_j + (\mathbf{B}_c^*)_j + b^{\xi/Q}$, i.e. the limiting mean vector is $\beta = m(\gamma) \mathbf{B}_c + \mathbf{B}_c^* + b^{\xi/Q} \mathbf{1}$ of (3.46). This proves (3.44).

Step (iii): GLS reduction to a quadratic form. Write $\mathbf{Y}_n := \sqrt{k}(\hat{\Xi}_n - \log \xi_{\tau_n} \mathbf{1})$, so $\mathbf{Y}_n \xrightarrow{d} \mathbf{Y} \sim \mathcal{N}(\beta, \Sigma_{\Xi})$ by Step (ii). Introduce the random oblique projection

$$\mathbf{P}_n := \mathbf{I} - \frac{\mathbf{1} \mathbf{1}^{\top} \hat{\Sigma}_{\Xi, n}^{-1}}{\mathbf{1}^{\top} \hat{\Sigma}_{\Xi, n}^{-1} \mathbf{1}},$$

which satisfies $\mathbf{P}_n \mathbf{1} = \mathbf{0}$ and, from the definition of $\hat{\nu}_n$ in (3.47), $\hat{\Xi}_n - \hat{\nu}_n \mathbf{1} = \mathbf{P}_n \hat{\Xi}_n = \mathbf{P}_n (\hat{\Xi}_n - \log \xi_{\tau_n} \mathbf{1}) = k^{-1/2} \mathbf{P}_n \mathbf{Y}_n$, the second equality because $\mathbf{P}_n \mathbf{1} = \mathbf{0}$ removes the common centring. Substituting into (3.47),

$$\Lambda_n^{\xi} = k (k^{-1/2} \mathbf{P}_n \mathbf{Y}_n)^{\top} \hat{\Sigma}_{\Xi, n}^{-1} (k^{-1/2} \mathbf{P}_n \mathbf{Y}_n) = \mathbf{Y}_n^{\top} \mathbf{P}_n^{\top} \hat{\Sigma}_{\Xi, n}^{-1} \mathbf{P}_n \mathbf{Y}_n. \quad (\text{A.44})$$

By hypothesis $\hat{\Sigma}_{\Xi, n} \xrightarrow{\mathbb{P}} \Sigma_{\Xi}$, so $\mathbf{P}_n \xrightarrow{\mathbb{P}} \mathbf{P} := \mathbf{I} - \mathbf{1} \mathbf{1}^{\top} \Sigma_{\Xi}^{-1} / (\mathbf{1}^{\top} \Sigma_{\Xi}^{-1} \mathbf{1})$ and $\mathbf{P}_n^{\top} \hat{\Sigma}_{\Xi, n}^{-1} \mathbf{P}_n \xrightarrow{\mathbb{P}} \mathbf{M} := \mathbf{P}^{\top} \Sigma_{\Xi}^{-1} \mathbf{P}$, both by the continuous-mapping theorem. A direct computation, writing $\mathbf{w} := \Sigma_{\Xi}^{-1} \mathbf{1}$ and $s := \mathbf{1}^{\top} \Sigma_{\Xi}^{-1} \mathbf{1} = \mathbf{1}^{\top} \mathbf{w}$, gives the standard GLS identity

$$\mathbf{M} = \mathbf{P}^{\top} \Sigma_{\Xi}^{-1} \mathbf{P} = \Sigma_{\Xi}^{-1} - \frac{\Sigma_{\Xi}^{-1} \mathbf{1} \mathbf{1}^{\top} \Sigma_{\Xi}^{-1}}{\mathbf{1}^{\top} \Sigma_{\Xi}^{-1} \mathbf{1}}, \quad (\text{A.45})$$

because the two first-order cross terms in the expansion of $(\mathbf{I} - \mathbf{w} \mathbf{1}^{\top} / s) \Sigma_{\Xi}^{-1} (\mathbf{I} - \mathbf{1} \mathbf{w}^{\top} / s)$ each contribute $-\mathbf{w} \mathbf{w}^{\top} / s$, while the second-order term contributes $+\mathbf{w} \mathbf{w}^{\top} / s$ (using $\mathbf{1}^{\top} \mathbf{w} = s$ and symmetry of Σ_{Ξ}^{-1}), leaving $\Sigma_{\Xi}^{-1} - \mathbf{w} \mathbf{w}^{\top} / s$. The matrix $\mathbf{M}$ is symmetric positive semi-definite and satisfies $\mathbf{M} \mathbf{1} = \mathbf{w} - \mathbf{w} (s/s) = \mathbf{0}$. By (A.44), Slutsky's theorem, and the continuous-mapping theorem applied to the (jointly continuous) quadratic-form map $(\mathbf{A}, \mathbf{y}) \mapsto \mathbf{y}^{\top} \mathbf{A} \mathbf{y}$,

$$\Lambda_n^{\xi} \xrightarrow{d} \mathbf{Y}^{\top} \mathbf{M} \mathbf{Y}, \quad \mathbf{Y} \sim \mathcal{N}(\beta, \Sigma_{\Xi}). \quad (\text{A.46})$$

Step (iv): the noncentral chi-square limit. Standardise $\mathbf{Z} := \Sigma_{\Xi}^{-1/2} \mathbf{Y} \sim \mathcal{N}(\Sigma_{\Xi}^{-1/2} \boldsymbol{\beta}, \mathbf{I}_m)$, where $\Sigma_{\Xi}^{-1/2}$ is the symmetric positive-definite square root of Σ_{Ξ}^{-1} . Then $\mathbf{Y}^{\top} \mathbf{M} \mathbf{Y} = \mathbf{Z}^{\top} \widetilde{\mathbf{M}} \mathbf{Z}$ with

$$\widetilde{\mathbf{M}} := \Sigma_{\Xi}^{1/2} \mathbf{M} \Sigma_{\Xi}^{1/2} = \mathbf{I}_m - \frac{\mathbf{v} \mathbf{v}^{\top}}{\mathbf{v}^{\top} \mathbf{v}}, \quad \mathbf{v} := \Sigma_{\Xi}^{-1/2} \mathbf{1},$$

using (A.45) and $\Sigma_{\Xi}^{1/2} \mathbf{w} = \Sigma_{\Xi}^{-1/2} \mathbf{1} = \mathbf{v}$, $s = \mathbf{v}^{\top} \mathbf{v}$. The matrix $\widetilde{\mathbf{M}}$ is the orthogonal projection onto $\mathbf{v}^{\perp}$: it is symmetric, idempotent, and of rank $m - 1$. A quadratic form $\mathbf{Z}^{\top} \widetilde{\mathbf{M}} \mathbf{Z}$ in a unit-covariance Gaussian $\mathbf{Z} \sim \mathcal{N}(\boldsymbol{\mu}_Z, \mathbf{I}_m)$ with $\widetilde{\mathbf{M}}$ a rank- r orthogonal projection is noncentral χ_r^2 with noncentrality $\boldsymbol{\mu}_Z^{\top} \widetilde{\mathbf{M}} \boldsymbol{\mu}_Z$. Here $r = m - 1$ and $\boldsymbol{\mu}_Z = \Sigma_{\Xi}^{-1/2} \boldsymbol{\beta}$, so

$$\Lambda_n^{\xi} \xrightarrow{d} \chi_{m-1}^2(\delta), \quad \delta = \boldsymbol{\beta}^{\top} \Sigma_{\Xi}^{-1/2} \widetilde{\mathbf{M}} \Sigma_{\Xi}^{-1/2} \boldsymbol{\beta} = \boldsymbol{\beta}^{\top} \mathbf{M} \boldsymbol{\beta} = \boldsymbol{\beta}^{\top} \Sigma_{\Xi}^{-1} \boldsymbol{\beta} - \frac{(\mathbf{1}^{\top} \Sigma_{\Xi}^{-1} \boldsymbol{\beta})^2}{\mathbf{1}^{\top} \Sigma_{\Xi}^{-1} \mathbf{1}},$$

the last equality by (A.45). This is (3.49). Since $\widetilde{\mathbf{M}}$ is positive semi-definite, $\delta \geq 0$, with $\delta = 0$ iff $\widetilde{\mathbf{M}} \Sigma_{\Xi}^{-1/2} \boldsymbol{\beta} = \mathbf{0}$, i.e. $\Sigma_{\Xi}^{-1/2} \boldsymbol{\beta} \in \text{span}\{\mathbf{v}\} = \text{span}\{\Sigma_{\Xi}^{-1/2} \mathbf{1}\}$, equivalently $\boldsymbol{\beta} \in \text{span}\{\mathbf{1}\}$.

Step (v): the two regimes with $\delta = 0$. (i) *Undersmoothing.* Under (3.40), $\lambda_j = 0$ for every j , so the per-sample $\sqrt{k_j}$ -biases $b_j^H = \lambda_j/(1 - \rho)$ and $b_j^R = -\lambda_j \Psi_{\rho}(e^{L_j^*})$ both vanish (the order-statistic bias being identically zero), and $\Lambda^{\bullet} = 0$, whence $\mathbf{B}_c = \mathbf{B}_c^{\star} = \mathbf{0}$ and $b^{\xi/Q} = 0$. Therefore $\boldsymbol{\beta} = \mathbf{0} \in \text{span}\{\mathbf{1}\}$ and $\delta = 0$.

(ii) *Equal-fraction distributed inference.* Here $L_j^{\star} = L^{\star}$ is sample-free. The two per-sample $\sqrt{k_j}$ -biases are proportional to λ_j with coefficients depending only on the common parameters $(\gamma, \rho, L^{\star})$: $b_j^H = \lambda_j/(1 - \rho)$ and $b_j^R = -\lambda_j \Psi_{\rho}(e^{L^{\star}})$, the order-statistic bias being identically zero. The $\sqrt{k}$ -rescaling factor satisfies, by (A.22) at $L_j^{\star} = L^{\star}$,

$$\sqrt{c_j S} \lambda_j = \Lambda^{\bullet} (e^{L^{\star}})^{-\rho}, \quad \text{sample-free.}$$

Consequently $(\mathbf{B}_c)_j = \sqrt{c_j S} b_j^H = \Lambda^{\bullet} (e^{L^{\star}})^{-\rho}/(1 - \rho)$ and $(\mathbf{B}_c^{\star})_j = \sqrt{c_j S} (L^{\star} b_j^H + b_j^R) = \Lambda^{\bullet} (e^{L^{\star}})^{-\rho} [L^{\star}/(1 - \rho) - \Psi_{\rho}(e^{L^{\star}})]$ are both sample-free, and $b^{\xi/Q} \mathbf{1}$ is collinear with $\mathbf{1}$ by construction. Hence every coordinate of $\boldsymbol{\beta} = m(\gamma) \mathbf{B}_c + \mathbf{B}_c^{\star} + b^{\xi/Q} \mathbf{1}$ equals the common constant

$$\beta_0 = \Lambda^{\bullet} (e^{L^{\star}})^{-\rho} \left[\frac{m(\gamma) + L^{\star}}{1 - \rho} - \Psi_{\rho}(e^{L^{\star}}) \right] + b^{\xi/Q},$$

so $\boldsymbol{\beta} = \beta_0 \mathbf{1} \in \text{span}\{\mathbf{1}\}$ and $\delta = 0$. This is the expectile-level counterpart of the bias-collinearity that renders the tail-index test Theorem 2.20 asymptotically central under the same equal-fraction distributed regime.

Step (vi): consistency under diagnostic alternatives. This final step proves the consistency assertion when (3.47) is read as a diagnostic outside the chapter's common-marginal working model. It therefore uses the additional alternative-side hypotheses stated in Theorem 3.9, rather than the common-target CLT of Steps (i)–(v). Write $\boldsymbol{\Xi}_n^{\star} := (\log \xi_{\tau_n, 1}, \dots, \log \xi_{\tau_n, m})^{\top}$ for the vector of *population* per-sample log-expectiles, and abbreviate $\mathbf{D}_n := \hat{\boldsymbol{\Sigma}}_{\Xi, n}^{-1}$, with induced norm $\|\mathbf{x}\|_{\mathbf{D}_n}^2 := \mathbf{x}^{\top} \mathbf{D}_n \mathbf{x}$. Assume $\hat{\boldsymbol{\Xi}}_n - \boldsymbol{\Xi}_n^{\star} = O_p(k^{-1/2})$, $\|\mathbf{D}_n\| = O_p(1)$, and $\mathbb{P}\{\lambda_{\min}(\mathbf{D}_n) > \eta\} \rightarrow 1$

for some $\eta > 0$. These conditions say, respectively, that the local log-expectile estimators remain root- k consistent for their own targets and that the plug-in precision matrix does not degenerate. Because the GLS residual $\hat{\Xi}_n - \hat{\nu}_n \mathbf{1} = \mathbf{P}_n \hat{\Xi}_n$ is, by definition of $\hat{\nu}_n$, the minimiser of $c \mapsto \|\hat{\Xi}_n - c\mathbf{1}\|_{\mathbf{D}_n}^2$, the statistic is the squared GLS distance of $\hat{\Xi}_n$ from $\text{span}\{\mathbf{1}\}$:

$$\Lambda_n^\xi = k \min_{c \in \mathbb{R}} \|\hat{\Xi}_n - c\mathbf{1}\|_{\mathbf{D}_n}^2.$$

On the event $\lambda_{\min}(\mathbf{D}_n) > \eta$, the norm $\|\cdot\|_{\mathbf{D}_n}$ is well defined and the triangle inequality gives, with $S_1 := \text{span}\{\mathbf{1}\}$ and $d_{\mathbf{D}_n}(\mathbf{x}, S_1) := \inf_{s \in S_1} \|\mathbf{x} - s\|_{\mathbf{D}_n}$ (while d denotes Euclidean distance),

$$\sqrt{\Lambda_n^\xi} = \sqrt{k} d_{\mathbf{D}_n}(\hat{\Xi}_n, S_1) \geq \sqrt{k} d_{\mathbf{D}_n}(\Xi_n^*, S_1) - \sqrt{k} \|\hat{\Xi}_n - \Xi_n^*\|_{\mathbf{D}_n}.$$

Moreover $d_{\mathbf{D}_n}(\Xi_n^*, S_1) \geq \sqrt{\lambda_{\min}(\mathbf{D}_n)} d(\Xi_n^*, S_1)$, since the Rayleigh quotient bound applies to every vector $\Xi_n^* - c\mathbf{1}$. Therefore, on the same event,

$$\sqrt{\Lambda_n^\xi} \geq \lambda_{\min}(\mathbf{D}_n)^{1/2} \sqrt{k} \min_c \|\Xi_n^* - c\mathbf{1}\| - \sqrt{k} \|\hat{\Xi}_n - \Xi_n^*\|_{\mathbf{D}_n}.$$

The subtracted term is $O_p(1)$ since $\sqrt{k}(\hat{\Xi}_n - \Xi_n^*) = O_p(1)$ and $\|\mathbf{D}_n\| = O_p(1)$. Under the stated alternative $\sqrt{k} \min_c \|\Xi_n^* - c\mathbf{1}\| \rightarrow \infty$, while $\lambda_{\min}(\mathbf{D}_n)$ is bounded away from zero in probability. The first term on the right therefore diverges in probability and dominates the $O_p(1)$ correction, so $\Lambda_n^\xi \xrightarrow{\mathbb{P}} \infty$. $\square$

A.9 Proof of Lemma 4.1

Put $t_n = (1 - \tau_n)^{-1}$ and $y_n = (1 - \tau_n)/(1 - \tau'_n)$, so $y_n \rightarrow \infty$ and $\ell_n = \log y_n$. At the quantile level, the source proofs use, under $\rho < 0$, the sequential Weissman form of the second-order log-quantile control (a consequence of Haan and Ferreira (2006, Theorem B.2.18); see also the use in Daouia, Padoan, et al. (2024, Supplement, Eq. (B.18))),

$$\log U(t_n y_n) - \log U(t_n) - \gamma \log y_n = O(|A(t_n)|). \quad (\text{A.47})$$

This is the point at which the strict condition $\rho < 0$ enters the deterministic extrapolation step: the fixed- y statement of Condition 2.3 alone is not the form being used along the diverging sequence $y_n \rightarrow \infty$.

Decompose

$$\begin{aligned} & \log \xi_{\tau'_n} - \log \xi_{\tau_n} - \gamma \ell_n \\ &= [\log Q_X(\tau'_n) - \log Q_X(\tau_n) - \gamma \ell_n] \\ & \quad + [\log\{\xi_{\tau'_n}/Q_X(\tau'_n)\} - \log\{\xi_{\tau_n}/Q_X(\tau_n)\}]. \end{aligned}$$

The first bracket is exactly (A.47). The second bracket is controlled by the logged expectile-quantile expansion (3.16), obtained from (2.7). Since $A \in RV_\rho$ with $\rho < 0$, Potter bounds give $|A(t_n y_n)| = O(|A(t_n)|)$ along $y_n \geq 1$, and the monotonicity of the upper quantile gives

$Q_X(\tau'_n) \geq Q_X(\tau_n)$ eventually. Hence

$$\begin{aligned} & \log \xi_{\tau'_n} - \log \xi_{\tau_n} - \gamma \ell_n \\ &= O\left(\left|A((1 - \tau_n)^{-1})\right|\right) + O(Q_X(\tau_n)^{-1}) + o\left(\left|A((1 - \tau_n)^{-1})\right|\right) + o(Q_X(\tau_n)^{-1}). \end{aligned}$$

Multiplying by $\sqrt{k}/\ell_n$ gives the result, because $\sqrt{k}A((1 - \tau_n)^{-1})$ converges to the finite limit $\Lambda^\bullet$, $\sqrt{k}/Q_X(\tau_n) \rightarrow 0$, and $\ell_n \rightarrow \infty$. $\square$

A.10 Proof of Proposition 4.2

Proposition 3.2 gives the joint Gaussian limit of $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$ and $\log \hat{q}_n^*(\tau_n \mid \boldsymbol{\omega}_q)$. The first-order Taylor expansion of $\log \psi$ around γ , combined with (3.19), gives

$$\begin{aligned} & \sqrt{k} \log \frac{\hat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau_n}} \\ &= m(\gamma) \sqrt{k} \{\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma\} + \sqrt{k} \{\log \hat{q}_n^*(\tau_n \mid \boldsymbol{\omega}_q) - \log Q_X(\tau_n)\} + b^{\xi/Q} + o_p(1). \end{aligned}$$

Thus the second coordinate is the image of the joint limit in Proposition 3.2 under the row $(m(\gamma), 1)$, plus the deterministic offset $b^{\xi/Q}$. This yields the mean and variance (3.22)–(3.21). Its covariance with the first coordinate is the covariance of the pooled-Hill component with $m(\gamma)$ times itself plus the pooled log-Weissman component, namely

$$m(\gamma) \boldsymbol{\omega}_\gamma^\top \mathbf{V}_c \boldsymbol{\omega}_\gamma + \boldsymbol{\omega}_\gamma^\top \mathbf{D}^* \mathbf{V}_c \boldsymbol{\omega}_q,$$

by (3.12). Passing from the centred log-ratio to the relative error, and from its covariance with the Hill component, does not change the limit. Indeed, with

$$R_n = \log \frac{\hat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau_n}}, \quad R_n = O_p(k^{-1/2}),$$

we have

$$\sqrt{k} \left[\left\{ \frac{\hat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau_n}} - 1 \right\} - R_n \right] = \sqrt{k} O_p(R_n^2) = o_p(1).$$

For estimated weights, exact affine normalisation removes the deterministic centring. For example,

$$\sqrt{k} \{\hat{\gamma}_n(\hat{\boldsymbol{\omega}}_\gamma) - \gamma\} = \hat{\boldsymbol{\omega}}_\gamma^\top \sqrt{k}(\hat{\boldsymbol{\gamma}}_n - \gamma \mathbf{1}).$$

Since the vector on the right is tight and $\hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma$, this differs by $o_p(1)$ from the same display with $\boldsymbol{\omega}_\gamma$. The weighted-geometric Weissman input obeys the identical argument on the log scale, using $\hat{\boldsymbol{\omega}}_q^\top \mathbf{1} = 1$ and $\hat{\boldsymbol{\omega}}_q \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_q$. Substituting these two Slutsky steps into the Taylor expansion above gives the stated estimated-weight extension, with limiting mean and covariance computed at the deterministic probability limits. $\square$

A.11 Proof of Theorem 4.4

Section 4.2 verifies the non-Gaussian hypotheses of Proposition 2.14 with $a_n = \sqrt{k}$, $\hat{\gamma}_n = \hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$ and $\hat{\xi}_{\tau_n}^I = \hat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)$. The required joint input is precisely Proposition 4.2; its first coordinate has limiting law

$$\Gamma \sim \mathcal{N}(\boldsymbol{\omega}_\gamma^\top \mathbf{B}_c, \boldsymbol{\omega}_\gamma^\top \mathbf{V}_c \boldsymbol{\omega}_\gamma).$$

Applying Proposition 2.14 therefore gives (4.13). We spell out the log-scale mechanism, because it is the structural reason why $\boldsymbol{\omega}_q$ vanishes from the first-order limit.

On the high-probability positivity event fixed in Section 4.3,

$$\begin{aligned} \log \frac{\hat{\xi}_{\tau'_n}^{\text{pool},*}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau'_n}} &= \ell_n \{\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma\} + \log \frac{\hat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau_n}} \\ &\quad - \{\log \xi_{\tau'_n} - \log \xi_{\tau_n} - \gamma \ell_n\}. \end{aligned} \tag{A.48}$$

Multiplication by $\sqrt{k}/\ell_n$ sends the first term to

$$\sqrt{k} \{\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma\} \xrightarrow{d} \mathcal{N}(\boldsymbol{\omega}_\gamma^\top \mathbf{B}_c, \boldsymbol{\omega}_\gamma^\top \mathbf{V}_c \boldsymbol{\omega}_\gamma),$$

the pooled-Hill Gaussian limit of Theorem 2.16. For the second term, Proposition 4.2 gives

$$\sqrt{k} \left\{ \frac{\hat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau_n}} - 1 \right\} = O_p(1),$$

and the corresponding log-ratio is also $O_p(k^{-1/2})$ because the relative error is $o_p(1)$. Since $\ell_n \rightarrow \infty$, this term is $o_p(1)$ after multiplication by $\sqrt{k}/\ell_n$. The deterministic log-ratio condition in Proposition 2.14 holds by Lemma 4.1; therefore the third term in (A.48) is $o(\ell_n/\sqrt{k})$. Hence

$$\frac{\sqrt{k}}{\ell_n} \log \frac{\hat{\xi}_{\tau'_n}^{\text{pool},*}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau'_n}} \xrightarrow{d} \Gamma.$$

Let

$$Z_n = \log \frac{\hat{\xi}_{\tau'_n}^{\text{pool},*}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)}{\xi_{\tau'_n}}.$$

The preceding display gives $(\sqrt{k}/\ell_n)Z_n \xrightarrow{d} \Gamma$, hence $Z_n = O_p(\ell_n/\sqrt{k}) = o_p(1)$ because $\ell_n/\sqrt{k} \rightarrow 0$. Therefore

$$\exp(Z_n) - 1 - Z_n = O_p(Z_n^2), \quad \frac{\sqrt{k}}{\ell_n} \{\exp(Z_n) - 1 - Z_n\} = O_p\left(\frac{\ell_n}{\sqrt{k}}\right) = o_p(1).$$

Thus the relative-error normalisation has the same limit as the log-ratio normalisation, proving (4.13).

For estimated weights, use the estimated-weight part of Proposition 4.2. Exact affine normalisation removes any deterministic centring error, and $\hat{\boldsymbol{\omega}}_\gamma \xrightarrow{\mathbb{P}} \boldsymbol{\omega}_\gamma$ carries the pooled-Hill mean and variance to the deterministic limits in (4.13) by Slutsky's theorem. The convergence of $\hat{\boldsymbol{\omega}}_q$

affects only the intermediate-anchor term, which has already been shown to vanish on the $\sqrt{k}/\ell_n$ scale. $\square$

A.12 Proof of Proposition 4.5

Proposition 2.14, applied in Theorem 4.4, gives the first-order mean and variance (4.14). The variance problem is therefore the same constrained quadratic programme as Proposition 2.17(i), giving (4.15)–(4.16).

For AMSE, set $\mathbf{M} = \mathbf{V}_c + \mathbf{B}_c \mathbf{B}_c^\top$. Since $\mathbf{V}_c$ is positive definite, so is $\mathbf{M}$. The constrained quadratic programme

$$\min_{\boldsymbol{\omega}_\gamma^\top \mathbf{1} = 1} \boldsymbol{\omega}_\gamma^\top \mathbf{M} \boldsymbol{\omega}_\gamma$$

has solution

$$\boldsymbol{\omega}_\gamma = \frac{\mathbf{M}^{-1} \mathbf{1}}{\mathbf{1}^\top \mathbf{M}^{-1} \mathbf{1}}.$$

Applying the Sherman–Morrison identity,

$$\mathbf{M}^{-1} = \mathbf{V}_c^{-1} - \frac{\mathbf{V}_c^{-1} \mathbf{B}_c \mathbf{B}_c^\top \mathbf{V}_c^{-1}}{1 + \mathbf{B}_c^\top \mathbf{V}_c^{-1} \mathbf{B}_c},$$

yields (4.18). The minimum value is $(\mathbf{1}^\top \mathbf{M}^{-1} \mathbf{1})^{-1}$, which gives (4.19). Positivity follows from positive definiteness of $\mathbf{M}$. Since neither (4.14) nor (4.17) contains $\boldsymbol{\omega}_q$, the whole affine hyperplane $\{\boldsymbol{\nu} \in \mathbb{R}^m : \boldsymbol{\nu}^\top \mathbf{1} = 1\}$ is first-order equivalent for the anchor weight once $\boldsymbol{\omega}_\gamma$ is fixed at the relevant optimum. $\square$

A.13 Proof of Corollary 4.6

By Theorem 2.16 and (4.21),

$$\sqrt{k}\{\bar{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma\} = \sqrt{k}\{\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma\} - \boldsymbol{\omega}_\gamma^\top \widehat{\mathbf{B}}_c \xrightarrow{d} \mathcal{N}(0, \boldsymbol{\omega}_\gamma^\top \mathbf{V}_c \boldsymbol{\omega}_\gamma).$$

Moreover,

$$\sqrt{k}\{\bar{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma\} = \sqrt{k}\{\hat{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma\} - \boldsymbol{\omega}_\gamma^\top \mathbf{B}_c + o_p(1).$$

Combining this recentering with Proposition 4.2 gives the joint input

$$\sqrt{k} \begin{pmatrix} \bar{\gamma}_n(\boldsymbol{\omega}_\gamma) - \gamma \\ \widehat{\xi}_{\tau_n}^{\text{pool}}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) / \xi_{\tau_n} - 1 \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(\begin{pmatrix} 0 \\ \mu^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \end{pmatrix}, \begin{pmatrix} \boldsymbol{\omega}_\gamma^\top \mathbf{V}_c \boldsymbol{\omega}_\gamma & s_{\gamma, \xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) \\ s_{\gamma, \xi}(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q) & \sigma^\xi(\boldsymbol{\omega}_\gamma, \boldsymbol{\omega}_q)^2 \end{pmatrix} \right).$$

Subtracting the consistent estimate of the deterministic bias shifts only the first-coordinate mean and leaves the covariance matrix unchanged. Now repeat the log-decomposition (A.48), with $\bar{\gamma}_n(\boldsymbol{\omega}_\gamma)$ in place of $\hat{\gamma}_n(\boldsymbol{\omega}_\gamma)$. The intermediate-anchor term is still $o_p(1)$ on the $\sqrt{k}/\ell_n$ scale by Proposition 4.2, and the deterministic extrapolation error is still negligible by Lemma 4.1. Thus the log-ratio limit is the centred Gaussian display above. Since $\ell_n/\sqrt{k} \rightarrow 0$, the final passage from log-ratio to relative error is the same exponential Taylor argument used in the proof of

Theorem 4.4.

For estimated weights, $\hat{\omega}_\gamma^\top \widehat{\mathbf{B}}_c \xrightarrow{\mathbb{P}} \omega_\gamma^\top \mathbf{B}_c$ by the assumed convergences, and the estimated-weight extension of [Proposition 4.2](#) supplies the corresponding Slutsky step for the pooled Hill component. With the explicit estimated-weight definition above,

$$\sqrt{k}\{\bar{\gamma}_n(\hat{\omega}_\gamma) - \gamma\} = \sqrt{k}\{\hat{\gamma}_n(\hat{\omega}_\gamma) - \gamma\} - \hat{\omega}_\gamma^\top \widehat{\mathbf{B}}_c \xrightarrow{d} \mathcal{N}(0, \omega_\gamma^\top \mathbf{V}_c \omega_\gamma).$$

The anchor weight remains first-order invisible on the very-extreme scale, as in [Theorem 4.4](#). $\square$

A.14 Proof of Corollary 4.7

The proof of [Theorem 4.4](#) gives the log-scale convergence

$$\frac{\sqrt{k}}{\ell_n} \left\{ \log \hat{\xi}_{\tau'_n}^{\text{pool},*}(\hat{\omega}_\gamma, \hat{\omega}_q) - \log \xi_{\tau'_n} \right\} \xrightarrow{d} \mathcal{N}(\omega_\gamma^\top \mathbf{B}_c, \omega_\gamma^\top \mathbf{V}_c \omega_\gamma),$$

with the same estimated-weight Slutsky step as in the theorem. Under $\omega_\gamma^\top \mathbf{B}_c = 0$ this limit is centred. Under the bias-reduced construction, [Corollary 4.6](#) gives the identical centred log-scale limit with the same variance. Consequently, in either case in the statement,

$$\frac{\sqrt{k}}{\ell_n} \left\{ \log \hat{\zeta}_{\tau'_n} - \log \xi_{\tau'_n} \right\} \xrightarrow{d} \mathcal{N}(0, \omega_\gamma^\top \mathbf{V}_c \omega_\gamma).$$

The consistency of $\widehat{\mathbf{V}}_c$ and $\hat{\omega}_\gamma$ implies

$$\hat{\sigma}_{\gamma,n}^{*2} = \hat{\omega}_\gamma^\top \widehat{\mathbf{V}}_c \hat{\omega}_\gamma \xrightarrow{\mathbb{P}} \omega_\gamma^\top \mathbf{V}_c \omega_\gamma,$$

so self-normalisation gives a standard normal limit. The coverage of [\(4.26\)](#) follows, and the original-scale statement is the image of the same event under the strictly increasing map $x \mapsto \exp(x)$. $\square$

Appendix B

Additional finite-sample diagnostics

This appendix reports the simulation diagnostics referenced in [Chapter 5](#). They are kept out of the main body because they support implementation checks rather than the central Monte Carlo comparison.

DGP	Estimator	Invalid rate	Valid reps
AbsStudent3	amse_oracle	0.0000	5000
AbsStudent3	amse_plugin	0.0000	5000
AbsStudent3	full_sample	0.0000	5000
AbsStudent3	naive	0.0000	5000
AbsStudent3	variance	0.0000	5000
Burr	amse_oracle	0.0000	5000
Burr	amse_plugin	0.0000	5000
Burr	full_sample	0.0000	5000
Burr	naive	0.0000	5000
Burr	variance	0.0000	5000
BurrHeavy	amse_oracle	0.0000	5000
BurrHeavy	amse_plugin	0.0000	5000
BurrHeavy	full_sample	0.0000	5000
BurrHeavy	naive	0.0000	5000
BurrHeavy	variance	0.0000	5000
Frechet	amse_oracle	0.0000	5000
Frechet	amse_plugin	0.0000	5000
Frechet	full_sample	0.0000	5000
Frechet	naive	0.0000	5000
Frechet	variance	0.0000	5000
Pareto	amse_oracle	0.0000	5000
Pareto	amse_plugin	0.0010	4995
Pareto	full_sample	0.0000	5000
Pareto	naive	0.0000	5000
Pareto	variance	0.0000	5000
ParetoHeavy	amse_oracle	0.0000	5000
ParetoHeavy	amse_plugin	0.0004	4998
ParetoHeavy	full_sample	0.0000	5000
ParetoHeavy	naive	0.0000	5000
ParetoHeavy	variance	0.0000	5000

Table B.1: QB domain-validity diagnostic for the intermediate target. Invalid rows correspond to finite-sample pooled or full-sample Hill estimates outside $(0, 1)$; they are not truncated and are excluded from the bias, MSE and coverage summaries.

DGP	Regime	m	Estimator	Neg. rate	Avg. max $ w $
AbsStudent3	balanced	10	amse_oracle	0.000	0.100
AbsStudent3	balanced	10	amse_plugin	0.000	0.100
AbsStudent3	moderate	10	amse_oracle	0.000	0.136
AbsStudent3	moderate	10	amse_plugin	0.000	0.134
AbsStudent3	strong	10	amse_oracle	0.000	0.247
AbsStudent3	strong	10	amse_plugin	0.000	0.246
AbsStudent3	strong_equal_k	10	amse_oracle	1.000	0.468
AbsStudent3	strong_equal_k	10	amse_plugin	1.000	0.316
Burr	balanced	10	amse_oracle	0.000	0.100
Burr	balanced	10	amse_plugin	0.000	0.100
Burr	moderate	10	amse_oracle	0.000	0.134
Burr	moderate	10	amse_plugin	0.000	0.134
Burr	strong	10	amse_oracle	0.000	0.256
Burr	strong	10	amse_plugin	0.001	0.254
Burr	strong_equal_k	10	amse_oracle	1.000	0.348
Burr	strong_equal_k	10	amse_plugin	0.998	0.281
BurrHeavy	balanced	10	amse_oracle	0.000	0.100
BurrHeavy	balanced	10	amse_plugin	0.000	0.100
BurrHeavy	moderate	10	amse_oracle	0.000	0.135
BurrHeavy	moderate	10	amse_plugin	0.000	0.135
BurrHeavy	strong	10	amse_oracle	0.000	0.248
BurrHeavy	strong	10	amse_plugin	0.000	0.247
BurrHeavy	strong_equal_k	10	amse_oracle	1.000	0.451
BurrHeavy	strong_equal_k	10	amse_plugin	1.000	0.463
Frechet	balanced	10	amse_oracle	0.000	0.100
Frechet	balanced	10	amse_plugin	0.000	0.100
Frechet	moderate	10	amse_oracle	0.000	0.134
Frechet	moderate	10	amse_plugin	0.000	0.134
Frechet	strong	10	amse_oracle	0.000	0.257
Frechet	strong	10	amse_plugin	0.007	0.258
Frechet	strong_equal_k	10	amse_oracle	1.000	0.891
Frechet	strong_equal_k	10	amse_plugin	1.000	0.904
Pareto	balanced	10	amse_oracle	0.000	0.100
Pareto	balanced	10	amse_plugin	0.000	0.100
Pareto	moderate	10	amse_oracle	0.000	0.134
Pareto	moderate	10	amse_plugin	0.005	0.148
Pareto	strong	10	amse_oracle	0.000	0.258
Pareto	strong	10	amse_plugin	0.017	0.270
Pareto	strong_equal_k	10	amse_oracle	1.000	1.318
Pareto	strong_equal_k	10	amse_plugin	1.000	1.495
ParetoHeavy	balanced	10	amse_oracle	0.000	0.100
ParetoHeavy	balanced	10	amse_plugin	0.000	0.100
ParetoHeavy	moderate	10	amse_oracle	0.000	0.135
ParetoHeavy	moderate	10	amse_plugin	0.002	0.138
ParetoHeavy	strong	10	amse_oracle	0.000	0.249
ParetoHeavy	strong	10	amse_plugin	0.004	0.260
ParetoHeavy	strong_equal_k	10	amse_oracle	1.000	0.457
ParetoHeavy	strong_equal_k	10	amse_plugin	1.000	0.457

Table B.2: AMSE-weight stability diagnostics. Negative weights are not forbidden by the affine pooling theory, but high negative-weight rates or large maximum absolute weights indicate finite-sample instability.

Bibliography

- Artzner, P., F. Delbaen, J.-M. Eber, and D. Heath (1999). “Coherent measures of risk”. In: *Mathematical Finance* 9.3, pp. 203–228.
- Bellini, F., B. Klar, A. Müller, and E. Rosazza Gianin (2014). “Generalized quantiles as risk measures”. In: *Insurance: Mathematics and Economics* 54, pp. 41–48.
- Daouia, A., S. Girard, and G. Stupfler (2018). “Estimation of tail risk based on extreme expectiles”. In: *Journal of the Royal Statistical Society. Series B* 80.2, pp. 263–292. DOI: [10.1111/rssb.12254](https://doi.org/10.1111/rssb.12254).
- (2019). “Extreme M-quantiles as risk measures: From L^1 to L^p optimization”. In: *Bernoulli* 25.1, pp. 264–309. DOI: [10.3150/17-BEJ987](https://doi.org/10.3150/17-BEJ987).
- (2020). “Tail expectile process and risk assessment”. In: *Bernoulli* 26.1, pp. 531–556. DOI: [10.3150/19-BEJ1137](https://doi.org/10.3150/19-BEJ1137).
- Daouia, A., S. A. Padoan, and G. Stupfler (2024). “Optimal weighted pooling for inference about the tail index and extreme quantiles”. In: *Bernoulli* 30.2, pp. 1287–1312. DOI: [10.3150/23-BEJ1632](https://doi.org/10.3150/23-BEJ1632). arXiv: [2111.03173](https://arxiv.org/abs/2111.03173).
- Davison, A. C., S. A. Padoan, and G. Stupfler (2023). “Tail risk inference via expectiles in heavy-tailed time series”. In: *Journal of Business & Economic Statistics* 41.3, pp. 876–889. DOI: [10.1080/07350015.2022.2078332](https://doi.org/10.1080/07350015.2022.2078332).
- Ehm, W., T. Gneiting, A. Jordan, and F. Krüger (2016). “Of quantiles and expectiles: Consistent scoring functions, Choquet representations and forecast rankings”. In: *Journal of the Royal Statistical Society. Series B* 78.3, pp. 505–562.
- Gneiting, T. (2011). “Making and evaluating point forecasts”. In: *Journal of the American Statistical Association* 106.494, pp. 746–762.
- Haan, L. de and A. Ferreira (2006). *Extreme Value Theory: An Introduction*. Springer Series in Operations Research and Financial Engineering. New York: Springer.
- Hill, B. M. (1975). “A simple general approach to inference about the tail of a distribution”. In: *The Annals of Statistics* 3.5, pp. 1163–1174.
- Newey, W. K. and J. L. Powell (1987). “Asymmetric least squares estimation and testing”. In: *Econometrica* 55.4, pp. 819–847.
- Padoan, S. A. and G. Stupfler (2022). “Joint inference on extreme expectiles for multivariate heavy-tailed distributions”. In: *Bernoulli* 28.2, pp. 1021–1048. DOI: [10.3150/21-BEJ1375](https://doi.org/10.3150/21-BEJ1375).
- Weissman, I. (1978). “Estimation of parameters and large quantiles based on the k largest observations”. In: *Journal of the American Statistical Association* 73.364, pp. 812–815.

- Ziegel, J. F. (2016). “Coherence and elicibility”. In: *Mathematical Finance* 26.4, pp. 901–918.
DOI: [10.1111/mafi.12080](https://doi.org/10.1111/mafi.12080).